



# UNIVERSITÀ DEGLI STUDI DI PADOVA

DEPARTMENT OF MATHEMATICS "TULLIO LEVI-CIVITA"

*MASTER THESIS IN TRANS-MEDITERRANEAN DATA SCIENCE*

## **THE PRICE OF ATTENTION: BEHAVIOURAL AND EEG RESPONSES TO ADVERTISING IN LLM CONVERSATIONS**

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*ACADEMIC YEAR*

2024-2026

# Acknowledgments

I am deeply grateful to all the people who have pushed me throughout this journey toward graduation, especially my father, Alessandro Troiani, my sister, Elisabeth Troiani. Your support during the hardest moments of my academic and professional career has been a constant source of strength and encouragement to never give up. Patricia, thank you for everything you have done throughout this journey, and for the way your love has transformed me. What I feel for you is something words could never fully express.

To all the Telefonica Research Team that helped to make this work possible (Katerina, Ioannis, Sebastian, Angela, Georgios...).

To all the great teachers I have had the privilege of encountering throughout my academic journey, especially Conrado Martinez Parra, who has been one of my greatest role models as a teacher. You not only helped me inside and outside the classroom, but also pushed me to pursue my childhood dream of one day working at CERN. To Carlos Escolano and Lluís Belanche, thank you for transmitting to me your curiosity and hunger for LLMs and deep learning. Finally, to Prof. Rossi, whose depth of knowledge, eagerness to teach, and genuine care for his students have surpassed any expectations I could have had: I applaud you. Without you, this thesis would not have been possible.

To all the friends who have accompanied me throughout this journey called life; to my dear friends from my Bachelor's, especially Alex Herrero, Miquel Torner, and Oriol Miró; and to Daniel Weroniski from my Master's. To all the wonderful people I met at Mango, BSC, CERN, and especially Multiverse Computing; yes, Miguel, Mikel and Xuban, especially you; trauma really does bond people together. And to my beloved friends at CERN, especially my Erasmus group: all of you hold a special place in my heart. You have challenged and reshaped my rigid perceptions of the world again and again. Thank you for all the experiences. I will never forget them.

Lastly, I wanted to thank my younger self for somehow having the hunger and strength to have consistently pushed through trauma, pain and discomfort and have constantly improved again and again. For having had the courage to rebuild a broken life, to break free of limitations, addictions and unhealthy relationships despite all the odds. Thank you for developing that relentless, military pursuit of excellence, always setting greater goals beyond the horizon. Over the past four years, that mindset has taken me further than I could have imagined: becoming a top student, a high-performing engineer, a better friend, son, brother, partner, and athlete.

I hope I can carry that same hunger, curiosity, discipline, and relentless pursuit of excellence into whatever comes next. But I also hope that, as I continue to evolve, I become wiser, more mature, and more humble. I hope I never lose the desire to help others, to contribute to something greater than myself, to change the world and keep learning. Now that i'll be starting a new life as an ML engineer in San Francisco, thank you Walter for having helped me reach up to this point, I promise I will raise the bar once again and help our generation.

To my grandfather, Walter Troiani Pasqualucci: whatever I become, a part of it will always be yours. I will carry the torch forward. You can rest assured I will give everything I've got.

# Abstract

When advertising enters a conversational assistant, its cost is paid by the user: in attention, trust, and overall experience. The central question is not simply whether advertising can be inserted, but when its cost is incurred, how large it is, and whether the way an advertisement enters the conversation changes that cost. This thesis therefore treats advertising not as an isolated interface feature, but as an intervention in the interaction itself, measuring how its presentation and timing affect the user, so that the cost any serving policy imposes on the user can be measured before it is accepted.

It contributes a theory and a controlled experiment. The theory pairs a taxonomy of advertising interventions with an intent-trajectory formalism for defining annotation-free impact measures for advertising. In the experiment,  $N = 54$  participants completed five four-turn tasks with a purpose-built retrieval-based assistant: an implicit mention and an explicit banner, each presented early (turn 2) or late (turn 4), plus an advertisement-free control. A subset of  $n = 18$  also completed the experiment with synchronised 32-channel EEG.

Any advertisement raised perceived manipulation by +1.27 on a seven-point scale, with stronger effects for the banner than for the mention and for early than for late insertion. The banner was noticed more often than the mention; about half of participants did not identify the mention as sponsored. Explicit banners were remembered far better than implicit mentions (81–87% versus 57–59%), and more than half of those who initially failed to notice the mention recognised it when shown again: a cost the user cannot name while it happens. Early insertion reduced credibility by about one third of a point relative to late insertion; trust did not differ on any planned contrast, and in a post hoc localisation it was lower only under the early banner; trust after re-exposure was lower for early than for late insertions. Presentation and timing did not interact on any of the four primary outcomes, and no moderation by personality or by the recorded demographics was detected.

No advertisement effects were detected in the intent trajectories; the proposed theory remains to be validated as a human-free metric for advertising.

In the EEG, aggregated over each condition, posterior  $\alpha$  power was lower for early than for late insertion, a contrast that also differs in conversational depth. At early-banner onset, an exploratory transient slow-power tilt (absolute  $\delta$  and  $\theta$  up, relative  $\alpha$  and  $\beta$  down) was observed, and none after the implicit mention (the two formats were not compared directly). The onset-locked posterior  $\alpha$  response covaried with the change in reported trust ( $\rho = .80$ ,  $n = 18$ ), making it a candidate neural correlate of a trust drop to be tested in replication.

The theory, platform, analysis code, and multimodal datasets are released openly. Together, they provide a foundation for measuring the user-side cost of conversational advertising and for holding serving policies to it. **The price of attention** is part of the cost of monetising conversation; it should be measured and minimised in the design of advertising systems.

*“FORGETTING WHAT IS BEHIND AND REACHING FORWARD TO WHAT IS AHEAD, I PURSUE AS MY GOAL THE PRIZE PROMISED BY GOD’S HEAVENLY CALL IN CHRIST JESUS.”*

— PHILIPPIANS 3:13–14, *CHRISTIAN STANDARD BIBLE*

*“REGARDLESS OF WHAT’S IN FRONT OF YOU, YOU STILL MUST GRIND. I’LL NEVER BE IN THE OLYMPICS, I’LL NEVER BE A PROFESSIONAL ATHLETE, BUT STILL I GRIND. I FAIL AT MOST THINGS I DO, BUT STILL I GRIND. I DON’T WANT TO DO HALF THE SHIT I DO, BUT STILL I GRIND. AND THAT ONE DAY YOU SEE ME DOWN THAT DARK ALLEY, RUNNING AT I AM IN THE MORNING, KNOW ONE THING: I WAS GRINDING. WHEN THE ENDING IS UNKNOWN AND THE DISTANCE IS UNKNOWN, THAT’S WHEN YOU KNOW WHO THE FUCK YOU ARE.”*

— DAVID GOGGINS

## Note on the companion paper

A companion paper, in preparation for submission to a journal, reports this study in condensed form: the advertising taxonomy, the serving policy, the behavioural design, and the confirmatory EEG analysis. Material that appears only here, the genre-intent trajectory theory with its results and discussion, the system design, the implementation details, and the walkthrough of each dataset, is omitted there, which makes this thesis a strict superset of the paper. There are two reasons for those omissions.

1. The system design, the datasets, and the implementation details are too technical, too long, and too specific to this deployment to fit a short-form publication of five to ten pages.
2. The intent theory and the metrics it proposes for LLM advertising are useful but dense, and their results were not strong enough to carry a short paper. As section 8.4 discusses, that outcome is attributable mainly to the advertisement insertion policy and to the genre-prediction model, neither of which this study set out to build.

The companion paper and related forthcoming work will be tracked at my personal project page. My publications can also be tracked through my ORCID and the arXiv author search of myself (Currently empty until the draft is sent).

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## Listing of acronyms

|                     |                                                      |
|---------------------|------------------------------------------------------|
| <b>AI</b> .....     | Artificial Intelligence                              |
| <b>ANN</b> .....    | Approximate Nearest Neighbour                        |
| <b>API</b> .....    | Application Programming Interface                    |
| <b>BFI-10</b> ..... | Big Five Inventory, ten-item form                    |
| <b>BGE</b> .....    | BAAI General Embedding                               |
| <b>CLS</b> .....    | Classification token of a transformer input sequence |
| <b>CRS</b> .....    | Conversational Recommendation System                 |
| <b>EEG</b> .....    | Electroencephalography                               |
| <b>EOG</b> .....    | Electrooculography                                   |
| <b>ERP</b> .....    | Event-Related Potential                              |
| <b>ETL</b> .....    | Extract, Transform, Load                             |
| <b>FAA</b> .....    | Frontal Alpha Asymmetry                              |
| <b>FAISS</b> .....  | Facebook AI Similarity Search                        |
| <b>GEE</b> .....    | Generalised Estimating Equations                     |
| <b>GGUF</b> .....   | GPT-Generated Unified Format                         |
| <b>GPU</b> .....    | Graphics Processing Unit                             |
| <b>HNSW</b> .....   | Hierarchical Navigable Small World                   |
| <b>ICA</b> .....    | Independent Component Analysis                       |
| <b>ICC</b> .....    | Intraclass Correlation Coefficient                   |
| <b>IQR</b> .....    | Interquartile Range                                  |
| <b>IVF</b> .....    | Inverted File index                                  |
| <b>JSONL</b> .....  | JSON Lines                                           |
| <b>KV</b> .....     | Key-Value cache                                      |
| <b>LLM</b> .....    | Large Language Model                                 |

|                    |                                                                       |
|--------------------|-----------------------------------------------------------------------|
| <b>LSL</b> .....   | Lab Streaming Layer                                                   |
| <b>MNE</b> .....   | MNE-Python, the EEG analysis toolbox                                  |
| <b>OCEAN</b> ..... | Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism |
| <b>RAG</b> .....   | Retrieval-Augmented Generation                                        |
| <b>RAGAS</b> ..... | Retrieval-Augmented Generation Assessment                             |
| <b>RRF</b> .....   | Reciprocal Rank Fusion                                                |
| <b>SD</b> .....    | Standard Deviation                                                    |
| <b>SEP</b> .....   | Separator token of a transformer input sequence                       |
| <b>UI</b> .....    | User Interface                                                        |
| <b>VRAM</b> .....  | Video Random-Access Memory                                            |
| <b>XDF</b> .....   | Extensible Data Format                                                |

# 1

## Introduction

### 1.1 MOTIVATION

Just as the internet did in the 1990s, modern pre-trained transformer models have changed the landscape for good: they show capabilities in natural language understanding and generation never seen before, and they have been placed in production and within everybody's reach as a mass medium.

Pioneers such as OpenAI, xAI, Anthropic, and Google offer broad access to models whose training and daily inference demand extensive funding: GPU farms, engineering talent, and power. This raises a fundamental question about how such companies can sustain the cost of providing these services at scale.

The strategies frontier AI companies have adopted are varied and tailored to their goals: free universal inference (OpenAI, Google), subscriptions (Perplexity, Anthropic), and, for most of them, hybrid proposals that mix free inference for data collection, subscriptions, API usage, agentic products, enterprise plans, and cloud distribution. The segment is young enough that the list keeps growing; agentic coding harnesses such as Codex, Claude Code, or Cursor are the latest addition.

How companies like OpenAI or Google can offer what feels like infinite inference for free without going bankrupt is beyond the scope of this thesis and better suited to business and economics research. But one option these companies are exploring to alleviate day-to-day operating costs, and one gaining traction just as it did for the internet, is **advertisement**.

A final argument for the importance of advertising, in the technology industry at large and not only for LLMs, is how much of the revenue of the profitable long-running companies rests on the attention economy. Alphabet, one of the biggest players in AI, reported 402.8B\$ in total revenue, of which approximately 294.7B\$ came from advertising alone, while DeepMind’s share is negligible [2]; of Meta’s 201B\$ in 2025, 196.2B\$ came from advertising, and no significant share is reported from Meta AI [3]. Newer AI companies such as OpenAI or Anthropic report funding rounds, valuations, and large contracts, but avoid disclosing revenue. So, while advertising is not yet an established revenue stream for frontier AI companies, its historical role in financing the adoption of new technologies makes it a particularly relevant avenue as the industry moves toward sustainable business models.

## 1.2 PROBLEM STATEMENT

Just as advertisement was transformed with each change of medium, from newspapers through radio and television to web pages, it now has to be transformed again for a new medium of textual mass communication: LLM chatbots. The word itself comes from the Latin *advertere*, *ad* (“toward”) and *vertere* (“to turn”): to turn one’s attention to something. AI companies are now asking which policies to adopt, and how to present advertising without disrupting the user’s trust and experience, and with them the attention that is the currency of this economy.

Advertisements could be inserted implicitly in the LLM responses via context engineering, steering vectors or fine-tuning, presented explicitly in sponsored banners akin to web page ads, in sponsored searches (like Perplexity)... But the implementations can be seen as instances of user interface designs and definitions of company-wide advertisement policies. AI companies that aim to implement advertisement are faced with a greater challenge beyond UI, which is deciding where they stand on this welfare optimisation problem in the triad of advertisers, end-users and the company itself, often with conflicting objectives (e.g. optimise for the welfare of the company and end-users will be drowned in ads, optimise for advertisers and the users will be shown irrelevant or annoying ads...). The main goal of this thesis is therefore to measure how these critical policy decisions land on the user, so that companies can lean toward the user side of the triad and the everyday person can meet this inevitable future with a better experience.

On the theoretical side, our work will involve a collection and summarization of previous related work with some new conceptualizations and abstract hierarchies in order to help to lay the foundations of advertisement in AI platforms. On the experimental side, this work will run experiments on different formats of ads and different timings, on different conversational guided tasks. Behavioural data will be acquired from a crowd-sourcing platform and similarly neurophysiological EEG data will be ingested

from our laboratory study. The analysis of that data aims to show what each policy costs the user, so that native advertising policies can be designed around that cost.

Our crowdsourcing study will help us establish the statistical foundation of trust and user experience under these monetized environments and the laboratory study will take this further to look for neural correlates from the EEG data to validate the differences in ad formats and timings beyond self-report. Beyond that, the foundations and heuristics laid out here are a starting point for data-rich organisations that want to learn advertisement policies from data.

To summarise, this work does not address auction algorithms, intent clustering, advertisement generation, or the other engineering topics of chapter 2. It asks how these advertisements affect end-users, behaviourally and neurophysiologically; it builds the theoretical abstractions needed to ask that question precisely; and it implements the intent-trajectory metrics and shifts that describe what happens to the conversation. From that, we aim to say what each insertion choice costs the user, so that the people who design, regulate, or use these systems can weigh advertising against user welfare with evidence rather than intuition.

### 1.3 RESEARCH QUESTIONS

The questions below ask what an advertisement placed inside a conversational assistant costs the person using it. They come in four blocks: what the presentation format does, what the moment of insertion does, whether the individual user changes either answer, and what the neurophysiological record adds to what people report. Each question names the measures that answer it, so that chapter 7 can answer every one of them positively or negatively.

#### 1.3.1 ADVERTISEMENT FORMAT

**RQ1:** How does advertisement presentation format affect the user’s overall experience of the conversational assistant?

**RQ2:** Does advertisement presentation format alter the trajectory of the conversation following the advertisement?

#### 1.3.2 TEMPORAL EFFECTS

**RQ3:** How does the moment of advertisement insertion affect the user’s overall experience of the conversational assistant?

**RQ4:** Does the moment of advertisement insertion alter the trajectory of the conversation following the advertisement?

**RQ5:** Does the effect of advertisement presentation format on user experience depend on the moment of insertion?

#### 1.3.3 USER HETEROGENEITY

**RQ6:** Do personality traits moderate the effect of advertisement presentation format on user experience?

**RQ7:** Do personality traits moderate the effect of advertisement insertion timing on user experience?

#### 1.3.4 NEUROPHYSIOLOGICAL MECHANISMS

**RQ8:** How do advertisement presentation format and insertion timing manifest in the user’s neurophysiological response?

**RQ9:** What is the relationship between the user’s neurophysiological response at advertisement onset and the subjective experience subsequently reported by the user?

### 1.4 CONTRIBUTIONS

From an academic viewpoint, the aforementioned contributions of this work can be summarized and categorized in 2 arms:

- Theoretical Work:
  1. **Small Survey on LLM based Advertisement:** We present the most impactful previous work that can serve to unify the field of advertisement in LLMs, including the theoretical works, engineering behind of it and population surveys, so the reader can quickly gather all the context just from reading our introduction and related work section.
  2. **Theoretical Definitions laid out:** We also propose some advancements in the underlying theory defining an advertisement taxonomy, expand the concept of intent and define annotation-free trajectory metrics such as the ad-associated genre shift.
- Experimental Work:
  1. **Unified Experimental Framework for Studying Advertisement in LLM Conversations:** We propose a controlled within-subject experimental design that enables systematic analysis of how ad format, timing, and conversational context jointly influence user experience and conversational dynamics.
  2. **Open-Source Dataset for Conversational Advertising Interventions:** We introduce a structured dataset of multi-turn LLM interactions where ad type, insertion timing, and conversational context are systematically manipulated and paired with subjective, behavioural,

and conversational outcome measures. This can help other works of synthetic dataset generation for ads such as [4] to improve their grounding on real human responses.

3. **A Multimodal Analysis of Cognitive and Attentional Effects of Ad Interventions:** We integrate EEG signals to study the neurophysiological correlates of conversational advertising and relate the response recorded at the advertisement's onset to what participants report about the same advertisement.

# 2

## Related Work

This chapter reviews the literature that motivates and contextualizes the present work. As this dissertation lies at the intersection of conversational AI, advertising, recommender systems, personality psychology, and neurophysiology, we position our contribution with respect to previous research, identifying the advances that have enabled this work as well as the remaining open challenges.

The chapter focuses on the research most directly related to the questions addressed in this dissertation. Basic concepts in language modeling, embeddings, information retrieval, retrieval-augmented generation, recommender systems, and advertising are therefore only introduced where they are necessary to establish the context of the reviewed work. The theoretical concepts developed specifically in this dissertation are introduced in chapter 3.

For clarity, we organize the related literature into six complementary themes: (i) the engineering foundations of conversational advertising and its optimization; (ii) advertisement quality, personalization, and personality-aware conversational systems; (iii) the effects of conversational advertisements on user behaviour and trust; (iv) advertisement detection and transparency; (v) safety, alignment, and ethical challenges in monetized LLMs; and (vi) the intersection of LLMs with EEG and neuromarketing. Together, these research directions motivate the present work and highlight the currently unexplored intersection between conversational advertising and neurophysiological analysis.

The rapid adoption of large language models has fundamentally changed the way users interact with information online. Rather than behaving as traditional search engines, modern LLMs increasingly act as conversational assistants with which users engage through natural dialogue, often attributing human-

like characteristics such as trustworthiness, empathy, or expertise [5]. This transition from keyword-based retrieval to conversational interaction creates entirely new opportunities and challenges for digital advertising. Unlike conventional web advertisements, conversational advertisements can be dynamically generated, personalized, and seamlessly integrated into an ongoing dialogue, making them substantially more persuasive but also considerably harder for users to recognize. Consequently, understanding how these advertisements are engineered, how they affect users, and how they can be deployed responsibly has recently emerged as an important research direction.

Given the novelty of the field, LLM advertising needs work from several angles. The first, presented in the next paragraph, is the *how*: how advertising can realistically be brought into production. The next is its impact and quality, since end-user trust is one of the main constraints of these monetised environments. A third, more ML-oriented dimension is the creation of standardised datasets for training intent classifiers and advertisement-serving policies, which the field still lacks. Zhang et al. [4] address this with synthetic datasets in the order of 50k examples, but open data remain far from the proprietary datasets that large AI companies hold.

The intersection between LLMs and advertising dates to 2023, when [6] and [7] first asked how the traditional framework of keywords and bids could be translated to LLMs, with keyword-level and token-level bids respectively. [8] advanced both engineering and theory by bidding on the inferred intent of user messages (clusters of topics whose granularity the company chooses) and by formalising the optimisation of utility and welfare across the opposing triad of advertisers, users, and companies. This is the core foundation of how it is done and the closest description of how ads in LLMs can run in production.

Moreover, we have more theoretical works like [9], which also expand the taxonomy of LLM advertisement and propose a new perspective on this joint optimisation problem of utility and welfare, similar to the aforementioned [8]. Yun et al. later moved allocation from generated text to neuron-level intervention budgets [10].

Some works like [11] took another angle to explore how good are the ads generated by multimodal LLMs (N=1200), writing ads customized to their OCEAN personality traits and achieved a tie with human marketers and in a second study, latest models of 2025 won by a 59.1% vs 40.9% preference of human made content. When people correctly identified an ad as AI-generated, they penalized it (a 21 percentage point drop). But even penalized, AI ads still beat human ones and 29.4% of people chose the AI ad even though they knew it was AI. This paper alone shows that LLM-generated advertisements need not be lower in quality than human-made ones, contrary to a common first impression. Before this, [12] (N=175) explored in the very beginning how natural ads felt for YouChat and ChatGPT models,

finding that when they naturally fitted the context they were highly accepted, and highly rejected when not, so even with the lower quality of 2024 LLMs and the huge advancements that have been made in the last 2 years, this tendency showed up and the gap is closing every day, especially taking into account how many advancements in post-training have been made lately.

Talking about personality and OCEAN, we would like to mention some foundational work that allowed us to make use of this personality framework. Firstly, the original OCEAN Big Five publication [13] that established the statistical framework. Also the BFI-10 has been foundational to integrate OCEAN in our time-constrained experimental setup [1]. Moreover, [14] presents an extensive discussion about the 3 typical personality clusters that arise from OCEAN and the data analysis best practices, which will potentially serve us for handling this data in the analytical part.

Before the emergence of LLM advertising, many of the underlying ideas originated from Conversational Recommender Systems (CRSs), where dialogue was used to infer user preferences and deliver personalized recommendations [15]. A common theme across this literature is that users should not be treated as a homogeneous population. Zhao et al. [16] conditioned GPT-4o on Big Five (OCEAN) personality traits and showed that personality significantly influences conversational dynamics: the model achieved an overall personality consistency F1-score of 0.7398, highly agreeable users reached agreements in fewer dialogue turns, whereas highly neurotic users were less easily persuaded and required longer conversations. Similarly, Martina et al. [17] introduced DistillRecDial, a conversational recommendation dataset generated through knowledge distillation, where LLMs simulated diverse user personas instead of relying exclusively on costly human conversations. The resulting dataset outperformed previous benchmarks such as ReDial and INSPIRED in dialogue coherence and domain expertise while explicitly modelling personality, preference expression styles, initiative levels, and user diversity. Together, these works demonstrate that modern LLMs are capable of modelling personality and generating realistic synthetic users, opening the door to future personality-aware synthetically augmented datasets and advertisement policy models training data.

Many scholars have explored topics of great interest such as the impact of advertisement on end-users, for example [18] (N=179) showed that by fine-tuning Phi-4 so it learns to sneak implicit advertisements, 49% of participants did not realise they were being advertised to even with “Sponsored” labels; a “halo effect” was found on stronger base models, with the older model (GPT-3.5) perceived as pushy and negative more often than the newer one (GPT-4o); almost nobody clicked a link, and many asked the chatbot to stop the explicit advertisement. This work will serve as the statistical foundation of our analysis where we reuse the rigour of their questionnaires to our analytics and expand it.

Another important point is the difficulty associated with identifying advertisements in this medium, such as [19] and [20], the latter showing that in a specific scenario CTR/persuasion is increased by a factor of 3 (22% to 61%) when using chatbot to promote content rather than the regular search engine ads, while only 18% of users noticed. Even when the sponsored labels came out, persuasion only dropped from 61 to 55 which was shown statistically insignificant. When told to go subtle, detection from users dropped to 9.5%! The works from Heineking et al for creating LLM ad-blockers suggest that overt native ads in RAG are detectable up to 99% (precision and recall) at context level, but covert ads recall dropped as low as 63%, and conventional machine learning models were extremely brittle, so BERT variants were the only option which is not ideal for on-device performance, and that newer llms keep getting better at being undetectable, so it is a cat-and-mouse scenario. Previous works like [21] found similar results creating their own benchmark dataset, that implicit/covert advertisements can in fact be detected, but for their SOTA LLM's the F1-scores were low under 50%. With specialized fine-tuning sentence transformers on the aforementioned dataset, they achieved F1-scores of 97% using MPNet as the base, although the "cat and mouse game" appears again as newer LLM's seem to keep getting better at small long-term nudges and covert advertisement.

This brings us again to the next logical point which are then the hidden risks of llm advertisement. When navigating conflicts of interests, [22] showed that in monetized environments models refused to disclose sponsorship (Opus 4.5 did it 98% of the time!), when told to recommend predatory loans more than 60% of the time on average all models did, they show bias against poor looking people, hid prices when it is convenient for the sponsored product and re-surface sponsored products multiple times even when you chose non-sponsored products (Grok 100% of the time, GPT-5 88% of the time), showcasing that just by changing prompts, LLM's are not in the best interest of the end user but rather optimizing for utility of the company. Then [23] goes a step beyond showing to the extent LLMs are capable of obeying the monetary incentives, disregarding morals, the "fake friend dilemma". Once advertisement is blended into conversations it is difficult to differentiate, and the most vulnerable users are the most susceptible to this. The call to action of both papers is to enforce guardrails to prevent this fake friend behaviour. Moreover, the more people know about this, the more law enforcements and more rejection will grow from the public eye, forcing companies to adopt ethical advertisement policies or either to risk people's trust.

There is some previous work related to the intersection between LLM and EEG or neural correlates analysis, from the most influential we would like to mention that nudge this future where EEG tracking earbuds and other devices become more widely industrialised. [24] trained a satisfied/dissatisfied EEG classifier (81% F1) so that users could steer an iterative image-generation loop (GPT-4 with Stable Diffusion) without explicit input, raising final image satisfaction from 65% to 92%. [25] drew wide attention

to “cognitive debt”: across four months of essay writing, the ChatGPT group showed the weakest EEG connectivity, the poorest recall and sense of ownership of its own text, and essays that were more homogeneous in n-gram terms than those of the search-engine and brain-only groups. Neurochat [26] proved that the quality of conversations could be improved iteratively to the satisfaction of the user using only 4 dry electrodes in a headband and the engagement index, lowering the barrier to entry in this area. Finally we have the works of [27] which shows that the frozen hidden states of the last layers of LLMs (24/28 of Layer 24/28 of Qwen 2.5 7B) encode person-specific neural signatures. A separate linear probe for each person dramatically outperforms a one-size-fits-all population probe; for high-gamma power, person-specific  $\rho = 0.183$  vs population  $\rho = 0.020$  (a  $9\times$  improvement). They also found it is replicable across different architectures (like Llama 3), so this gives the finding that individual neural variation has structured geometry in LLM activation space, these individual neural directions are discussed to be grounded on how the individual’s brain processes language. All of this work gives exciting directions of research on how this could be applied directly to LLM advertisement.

The third intersection, EEG and advertising, is the field of neuromarketing, which established the baseline for combining neurophysiological signals with advertising stimuli. The comprehensive survey by Quiles Pérez et al. [28] catalogues the objectives, biosignals, and processing pipelines used across the literature, finding that EEG is the most widespread biosignal (used in nearly 50% of studies), that audio-visual stimuli dominate, and that ML/DL methods have largely replaced purely statistical approaches. Following this paradigm, Afshar and Azimi [29] used the NeuMa dataset, 42 participants viewing 144 products in a simulated supermarket browsing task, to compare classical ML models (SVM, Random Forest, XGBoost) against graph neural networks trained on brain connectivity features for predicting “Buy” vs. “NoBuy” decisions. While classical models achieved higher overall accuracy (79%), they were biased toward the majority non-purchase class, which makes them uninformative for the minority decision; GNNs, by modelling the interactions between EEG electrodes as a graph, achieved better sensitivity for actual purchase decisions (37%) despite lower aggregate accuracy. This thesis borrows from that field only its two best-established spectral markers, frontal-midline  $\theta$  and posterior  $\alpha$ , and none of its decoding machinery; purchase prediction is not the question here, and §4.2.4 explains why the two markers were chosen.

To the best of our knowledge, no research to date sits at the intersection of LLMs, advertising, and EEG ; the gap this work aims to fill, laying a foundation for future studies at this three-way junction, so we aim not only to accomplish our goals but also lay a foundation for the analysis of data for future research on this same topic.

In one sentence, the state of the field is: AI-generated advertisements can now outperform human-

created ones, while users often fail to recognise them as advertising, disclosure labels provide only limited mitigation, and models may readily act against the user’s interests when doing so serves the interests of their corporate owners. Our work therefore takes a user-centric perspective: it asks how people experience advertising in LLMs, so that evidence about what it costs them, rather than yield, can steer its development as LLM advertising becomes, in the long term, an inevitable part of these systems. Helping companies improve their policies follows from that.

# 3

## Theoretical Foundations

This chapter records the formal vocabulary used in the rest of the thesis: intent, company-wide policies  $\Pi$ , advertisement serving policies  $\pi$ , genre trajectories in conversations with their associated metrics (number of shifts, persistence, entropy; Section 3.1), a two-level taxonomy of a conversational advertisement instance (Section 3.2), and a split between platform policy and online serving (Section 3.3). The experimental design in chapter 6 realises two instances from that taxonomy and a null control.

### 3.1 INTENT AND ATTENTION SHIFTS

One abstraction that has gained considerable traction in NLP, retrieval systems, and, consequently, LLM-based advertising is the concept of *intent*. When a user queries an interface (e.g., a browser or chatbot), the query may reflect an underlying intent  $t \in \mathcal{T}$ : the concrete goal the user is trying to accomplish with that message, which extends beyond the literal semantics of the text. For example, the message “Which airlines fly Milan–Lisbon direct in October?” realises the intent *book a flight to Lisbon*. This intent may itself arise from a latent need  $n \in \mathcal{N}$ , the motive behind the goal, which is generally unobservable to the interface (the same user may want a first-class seat because they recently received a bonus and want to feel rewarded). The thesis works at the level of intents; the relationship between needs and intents is beyond its scope.

In a browser, intent can often be approximated from keywords and other contextual signals, whereas conversational systems introduce a more complex and dynamic setting. The space of possible intents is effectively unbounded due to the many possible nuances of language and context. Consequently, working directly with intent for advertisement allocation, as noted in [8], is intractable. This motivates

the introduction of *genres*, which provide a finite, coarse-grained partition of the intent space via a map  $\kappa$  from intents to genres:

$$\mathcal{G} = \{g^{(1)}, \dots, g^{(|\mathcal{G}|)}\}, \quad \kappa : \mathcal{T} \rightarrow \mathcal{G}.$$

A genre is therefore a coarse class of activities that many different intents fall into. Two genres used as a running example throughout this section are *Organising a trip* (abbreviated TRIP), which groups intents such as *book a flight to Lisbon*, *find a hotel near the old town*, or *check how many days Sintra needs*, and *Shopping* (abbreviated SHOP), which groups intents such as *find a compact camera under 400 euros* or *compare the battery life of two models*. Both are activities the user is carrying out through the conversation, so they are homogeneous in kind; the map  $\kappa$  sends each fine-grained intent to exactly one of them.

Let us start by defining a multi-turn conversation  $C$  of  $T$  turns from the user’s point of view. Each user message is an utterance  $u \in \mathcal{U}$ , so a conversation is an ordered sequence of utterances:

$$C = (u_1, u_2, \dots, u_T).$$

An utterance  $u_k$  is a linguistic realisation of the intent  $t_k$  held at turn  $k$ ,

$$u_k \sim \mathcal{U}(t_k).$$

The running example is the five-turn conversation

- $u_1$  : “Which airlines fly Milan–Lisbon direct in October?”
- $u_2$  : “Is a hotel in Alfama a good idea for a first visit?”
- $u_3$  : “How many days do I need for Lisbon and Sintra?”
- $u_4$  : “I also need a compact camera for the trip, anything under 400 euros?”
- $u_5$  : “Does that one have a decent battery life?”

whose first three intents belong to TRIP and whose last two belong to SHOP. The trajectory of optional advertisements  $a_k$  introduced between utterances in a conversation  $C$  is defined as

$$\mathbf{A} = (a_1, a_2, \dots, a_{T-1}), \quad a_k \in \mathcal{A} \cup \{\emptyset\} \quad \text{where} \quad u_k \rightarrow a_k \rightarrow u_{k+1}.$$

Here  $\mathcal{A}$  is the catalogue of advertisements, a set, and  $\mathbf{A}$  is the ordered sequence of insertions, one slot between consecutive utterances; an insertion after the final utterance  $u_T$  is written  $a_T$  and occupies no slot.

It has been proposed that the usage of intent and genres allow for response-decoupled and bidding-decoupled ad allocations, which are the most ethical and interesting for companies wanting to push advertisements in their LLM platforms [8]. Models like the open-source ThrashBERT (based on [30]) try to classify text into extremely coarse genres, which can pave the way in the production of engineering of LLM monetized environments. These models can either be trained to produce soft-labels or hard-labels, which are more common in practice due to the labelling effort, this is why our work will be more centered in the hard label side. A trained genre model  $f_{\text{genre}}$

$$f_{\text{genre}} : \mathcal{U} \rightarrow \mathcal{G}$$

performs inference by mapping an utterance  $u_k$  to a genre  $\hat{g}_k = f_{\text{genre}}(u_k) \in \mathcal{G}$ . We distinguish the temporal index  $k$  from the genre index  $i$ , with  $g^{(i)}$  denoting the  $i$ -th genre in the finite genre space  $\mathcal{G}$  and  $k$  denoting the temporal position of an utterance.

**DEFINITION 1 (GENRE TRAJECTORY).** Given a conversation  $C = (u_1, \dots, u_T)$  and a genre classifier  $f_{\text{genre}}$ , the predicted genre trajectory is the discrete ordered sequence \*

$$\hat{\mathbf{G}} = (\hat{g}_1, \dots, \hat{g}_T), \quad \text{where } \hat{g}_k \in \mathcal{G} \quad \wedge \quad \hat{g}_k = f_{\text{genre}}(u_k). \quad (3.1)$$

The trajectory represents the observable, coarse-grained evolution of the user’s inferred conversational intent over the course of the multi-turn conversation  $C$ . For the running example,  $\hat{\mathbf{G}} = (\text{TRIP}, \text{TRIP}, \text{TRIP}, \text{SHOP}, \text{SHOP})$ . Note that this concept can also be applied symmetrically to intent, although it is not observable, similar to observations and states in Hidden Markov Models.

**DEFINITION 2 (GENRE TRANSITIONS AND SHIFTS).** A genre transition is the ordered pair of consecutive predicted genres

$$\tau_k = (\hat{g}_k, \hat{g}_{k+1}), \quad k \in [1, T-1].$$

A transition constitutes a *genre shift* when the predicted genre changes between consecutive utterances. We therefore define the binary genre-shift indicator and its summation in the whole trajectory as:

$$\delta_k = \mathbb{I}[\hat{g}_{k+1} \neq \hat{g}_k], \quad k \in [1, T-1], \quad N_{\text{shift}} = \sum_{k=1}^{T-1} \delta_k. \quad (3.2)$$

---

\*No Markov assumption is imposed; the transition frequencies simply describe observed local genre dynamics. Markov or higher-order models are left for theoretical future work.

Hence,  $\delta_k = 0$  corresponds to a self-transition  $\hat{g}_{k+1} = \hat{g}_k$ , whereas  $\delta_k = 1$  corresponds to a genre-changing transition  $\hat{g}_{k+1} \neq \hat{g}_k$ . In the running example the only shift is  $\tau_3 = (\text{TRIP}, \text{SHOP})$ , so  $\delta = (0, 0, 1, 0)$  and  $N_{\text{shift}} = 1$ . Importantly, note that a genre shift is an observable change in the inferred genre and does not by itself imply that an advertisement caused the change.

**DEFINITION 3 (GENRE PERSISTENCE).** Number of consecutive user utterances for which the same predicted genre is maintained. For a maximal contiguous run of genre  $g^{(i)}$ , its persistence is defined as

$$R(g^{(i)}) = \text{length of the maximal contiguous run of } g^{(i)}. \quad (3.3)$$

The running-example trajectory  $(\text{TRIP}, \text{TRIP}, \text{TRIP}, \text{SHOP}, \text{SHOP})$  contains a  $\text{TRIP}$  run of length 3 and a  $\text{SHOP}$  run of length 2, so  $R(\text{TRIP}) = 3$  and  $R(\text{SHOP}) = 2$ . A genre may occur in several separate runs;  $R(g^{(i)})$  is the length of its longest one. At the conversation level,  $R_{\max} = \max_i R(g^{(i)})$  is the longest run of any genre, and it is this scalar that Chapters 4 and 7 abbreviate to  $R$ . Genre persistence therefore provides a local measure of conversational stability.

**DEFINITION 4 (GENRE FREQUENCY).** For a genre  $g^{(i)} \in \mathcal{G}$ , its empirical frequency within a conversation is defined as

$$p_i = \frac{1}{T} \sum_{k=1}^T \mathbb{I}[\hat{g}_k = g^{(i)}], \quad \text{where} \quad \sum_{i=1}^{|\mathcal{G}|} p_i = 1. \quad (3.4)$$

The empirical number of transitions from genre  $g^{(i)}$  to genre  $g^{(j)}$  is

$$N_{ij} = \sum_{k=1}^{T-1} \mathbb{I}[\hat{g}_k = g^{(i)} \wedge \hat{g}_{k+1} = g^{(j)}]. \quad (3.5)$$

In the running example  $p_{\text{TRIP}} = 3/5$ ,  $p_{\text{SHOP}} = 2/5$ , and the only non-zero off-diagonal count is  $N_{\text{TRIP}, \text{SHOP}} = 1$ .

**DEFINITION 5 (GENRE DIVERSITY AND ENTROPY).** The diversity of a  $\hat{\mathbf{G}}$  measures the number of distinct genres visited throughout a conversation. For a trajectory  $\hat{\mathbf{G}} = (\hat{g}_1, \dots, \hat{g}_T)$ , genre diversity is defined as

$$D(\hat{\mathbf{G}}) = |\{\hat{g}_k \mid k = 1, \dots, T\}|, \quad 1 \leq D(\hat{\mathbf{G}}) \leq |\mathcal{G}|. \quad (3.6)$$

While diversity captures the breadth of genres encountered, it does not account for how conversational turns are distributed among those genres. Therefore, let the corresponding genre entropy be

defined as the Shannon entropy of this empirical distribution, which is minimized when the trajectory is concentrated in a single genre and maximized when the observed genres are uniformly distributed.

$$H(\hat{\mathbf{G}}) = - \sum_{i=1}^{|\mathcal{G}|} p_i \log p_i. \quad (3.7)$$

The running example visits two genres,  $D(\hat{\mathbf{G}}) = 2$ , with entropy  $H(\hat{\mathbf{G}}) = -(0.6 \log 0.6 + 0.4 \log 0.4) \approx 0.67$  nats; a conversation that stayed in TRIP for all five turns would have  $D = 1$  and  $H = 0$ .

**DEFINITION 6 (AD-ASSOCIATED GENRE SHIFT).** Let  $I_k = \mathbb{I}[a_k \neq \emptyset]$  denote whether an advertisement is presented at turn  $k$ . An *ad-associated genre shift* occurs when an advertisement is presented at turn  $k$  and the genre of the subsequent user utterance differs from that of the pre-advertisement utterance. For  $k < T$ , we define

$$\delta_k^{(a)} = I_k \delta_k = I_k \mathbb{I}[\hat{g}_{k+1} \neq \hat{g}_k]. \quad (3.8)$$

Thus,  $\delta_k^{(a)} = 1$  indicates that a genre shift is observed immediately after advertisement  $a_k$ . This temporal association does not, by itself, imply that the advertisement caused the observed genre shift.

A relevant subcase occurs when the observed shift leads to the genre associated with the advertisement. Let  $g_k^{(a)}$  be the genre associated with advertisement  $a_k$  on turn  $k$ . We define a *genre-aligned ad-associated genre shift* as

$$\tilde{\delta}_k^{(a)} = \delta_k^{(a)} \mathbb{I}[\hat{g}_{k+1} = g_k^{(a)}]. \quad (3.9)$$

Hence,  $\tilde{\delta}_k^{(a)} = 1$  identifies an ad-associated genre shift in which the subsequent user utterance adopts the genre associated with the advertisement. Suppose that in the running example a compact camera was advertised after  $u_3$ , i.e.  $a_3 \neq \emptyset$  with  $g_3^{(a)} = \text{SHOP}$ , and no other slot was filled. Then  $I = (0, 0, 1, 0)$ , the shift at  $k = 3$  is ad-associated,  $\delta_3^{(a)} = 1$ , and it is also genre-aligned,  $\tilde{\delta}_3^{(a)} = 1$ , because  $u_4$  moved to SHOP. Had the camera been advertised after  $u_1$  instead,  $\delta_1^{(a)} = 0$ : the user stayed in TRIP. By construction,

$$\tilde{\delta}_k^{(a)} \leq \delta_k^{(a)} \leq \delta_k.$$

Finally, we reserve the term **attention shift** for an ad-associated genre shift sub-case that can be causally attributed to the advertisement rather than to the natural progression of the conversation. Establishing such causal attribution requires additional analysis and is outside the scope of the present thesis.

### 3.2 TAXONOMY OF CONVERSATIONAL ADVERTISING

A conversational advertisement can be described independently of the policy that selected and delivered it. The taxonomy characterises the served artefact  $a_k$ : the advertisement encountered after utterance  $u_k$ . The policy in Section 3.3 determines eligibility, implementation, and delivery. The same artefact may be shown at a different turn or to a different user without changing its taxonomic description. We describe  $a_k$  by seven properties arranged in four layers: an influence target  $\iota$ , a style pair  $(\alpha, \varepsilon)$ , a surface triple  $(\sigma, \lambda, \theta)$ , and a realised personalisation  $\varphi$ :

$$a_k = \langle \iota, (\alpha, \varepsilon), (\sigma, \lambda, \theta), \varphi \rangle. \quad (3.10)$$

Each symbol answers one question about the served advertisement, listed in Table 3.1, which is the reference definition of the taxonomy. The *influence target*  $\iota$  is the commercial variable the advertisement tries to move (e.g. exposure to a product, or the framing of information) [9]. The *style pair* describes the wording of the message and follows Heineking et al. and the marketing literature they synthesise [19]: the *appeal*  $\alpha$  is the persuasive logic (rational, i.e. informational, versus emotional), and the *explicitness*  $\varepsilon$  is how directly the wording announces that it is promoting. We call wording *overt* ( $\varepsilon = \text{overt}$ ) when it openly recommends or praises the product, for instance “X is the best choice, you should get it”, and *covert* ( $\varepsilon = \text{covert}$ ) when the product is mentioned in neutral, informational language that reads like the rest of the answer, for instance “X weighs 300 g and lasts about ten hours”. The *surface triple* describes the container rather than the words. The *channel*  $\sigma$  is the communicative medium (text, image, audio); the *presentation*  $\lambda$  states whether, in that channel, the promotion is woven into the organic reply (*implicit*) or shown as a visually distinct unit such as a banner (*explicit*); and the *realised disclosure*  $\theta$  is the commercial information actually made available to the user, from nothing to the sponsor’s identity and the targeting rationale. Finally, the *realised personalisation*  $\varphi$  records which user evidence was used in the served text; it belongs to  $a_k$  because it describes the delivered artefact, whereas the data a system is *permitted* to use belong to the policy of Section 3.3 [15, 11].

Explicitness  $\varepsilon$  and presentation  $\lambda$  share everyday vocabulary but are distinct dimensions:  $\varepsilon$  is about the words,  $\lambda$  about the container, and all four combinations occur. An implicit advertisement can be rhetorically overt (a sentence inside the reply that urges the user to buy), and an explicit banner can be covert in wording and carry only a weak label [18]. In this thesis, *implicit* and *explicit* always refer to  $\lambda$ , and *covert* and *overt* always refer to  $\varepsilon$ .

**Table 3.1:** Morphological taxonomy of the conversational advertisement instance  $a_k$ .

| Layer      | Dimension                          | Question                                                                               | Levels                                                                                                    |
|------------|------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Influence  | $\iota$ influence target           | What commercial variable is targeted?                                                  | Product exposure; information framing; behavioural redirection; preference shaping [9]                    |
| Style      | $\alpha$ appeal                    | What persuasive logic does the message use?                                            | Rational/informational; emotional/transformational [31, 32, 19]                                           |
|            | $\varepsilon$ explicitness         | How directly does the wording announce the promotion?                                  | Covert; overt [19]                                                                                        |
| Surface    | $\sigma$ channel                   | Through which medium does the advertisement reach the user?                            | Text; image/video; audio/voice; multi-modal... [9, 15]                                                    |
|            | $\lambda$ presentation             | In that channel, is the promotion integrated natively or in a distinct, separate unit? | Implicit; explicit                                                                                        |
|            | $\theta$ realised disclosure       | What commercial information is made available to the user?                             | None; on request; commercial nature disclosed; sponsor identified; targeting rationale disclosed [18, 33] |
| Adaptation | $\varphi$ realised personalisation | What user evidence is used or encoded in the served advertisement?                     | None; last utterance; session memory; persistent memory; traits & demographics [15, 11, 33]               |

The present study realises two advertisement instances and a null control. Influence, style, channel, and personalisation are held fixed:  $\iota$  is product exposure,  $\alpha$  is informational,  $\varepsilon$  is covert,  $\sigma$  is text, and  $\varphi$  is session memory. The two advertisements differ only in the surface pair  $(\lambda, \theta)$ . Let us define the following 3 advertisement instances used throughout the thesis for brevity:

$$a^{\text{imp}} = \langle \iota, (\alpha, \varepsilon), (\sigma, \text{implicit, on-request}), \varphi \rangle, \quad (3.11)$$

$$a^{\text{exp}} = \langle \iota, (\alpha, \varepsilon), (\sigma, \text{explicit, nature}), \varphi \rangle, \quad (3.12)$$

$$a^\emptyset = \emptyset. \quad (3.13)$$

### 3.3 CONVERSATIONAL ADVERTISING POLICY

The taxonomy above describes the realised properties of an advertisement instance, whereas an advertising policy encompasses the broader organisational choices governing how advertisements are introduced and delivered for inference providers. The same advertisement may be implemented through different

technical mechanisms, presented at different turns or opportunities, or served to different users without changing its taxonomic description. To avoid confusion, we write uppercase  $\Pi$  for the platform-level advertising policy of the company and lowercase  $\pi$  for the online serving policy that decides, turn by turn, what is shown. Without claiming an exhaustive policy theory, we propose the provisional organising representation

$$\Pi = \langle \Gamma, \mu, \pi \rangle, \quad (3.14)$$

whose components are:

- **Governance constraints and objectives  $\Gamma$ :** the relatively stable organisational boundaries within which advertising operates, including disclosure requirements, privacy restrictions, safety requirements, advertiser eligibility, user eligibility, intervention mechanisms allowed and most importantly the relative importance assigned to user, advertiser, and platform welfare.
- **Intervention architecture  $\mu$ :** the execution layer through which an advertising decision is realised in the chatbot response. It does not decide whether, which, or to whom an advertisement should be served. Instead, it determines how an advertisement selected by the serving policy is incorporated into the system's output. Examples include inserting a fixed creative into an independently generated response, introducing commercial information through retrieval or prompt context, modifying generation or token selection, and intervening through internal representations or neurons [6, 7, 8, 10].
- **Online serving policy  $\pi$ :** the decision rule that runs at every slot of the conversation. Given what is known at turn  $k$  (the utterances so far, the predicted genre  $\hat{g}_k$ , session memory, the candidate products),  $\pi$  decides whether to show an advertisement at all and, if so, which instance  $a_k \in \mathcal{A}$  to show, to whom, and when; formally  $\pi$  maps that state to an action in  $\mathcal{A} \cup \{\emptyset\}$ . It may be rule-based, optimisation-based, or learned from data. Two simple examples fix the idea of serving policy:
  1. *Fixed-slot heuristic.* Serve exactly one advertisement per conversation at a pre-assigned turn  $k^*$ , take the product with the highest retrieval score for the current session, and stay silent at every other slot. This is the serving policy of the experiment in chapter 6:  $k^* = 2$  in the early condition,  $k^* = 4$  in the late condition, and  $a_k = \emptyset$  everywhere for the control.
  2. *Learned policy.* Treat each slot as a decision of a contextual bandit or reinforcement-learning agent: the state is  $(\hat{g}_k, \text{turns since the last advertisement, session memory})$ , the action is an element of  $\mathcal{A} \cup \{\emptyset\}$ , and the reward combines advertiser value (a click or purchase) with a penalty for user cost (a drop in reported trust or an abandoned conversation), weighted as  $\Gamma$  prescribes. Such a policy learns, for instance, to advertise only after a genre shift into SHOP and to stay silent otherwise.

In both cases  $\pi$  operates within the constraints established by  $\Gamma$ , and its selected action is executed through  $\mu$ .

The precise boundary between  $\mu$  and  $\pi$  may depend on the system architecture. For example, a serving policy may operate through a fixed intervention mechanism or select among multiple mechanisms at runtime. Nevertheless, separating these components remains analytically useful:  $\Gamma$  specifies the constraints and objectives under which advertising operates,  $\mu$  specifies how commercial intervention is technically realised, and  $\pi$  specifies the decisions made during serving and can even choose dimensions of the realized ad such as  $\varphi, \theta$ . Together, these components provide a practical checklist for describing and comparing conversational advertising systems without claiming that their boundaries are always mutually exclusive or exhaustive.

# 4

## Datasets

For the purpose of this work, we have ingested an unstructured dataset in order to have a database of potential products to advertise during the experiment conditions. However, its important to note that the experiments runs resulted in multiple datasets, the experimental data, spanning 3 data formats, which will lay the foundation not only for our data analysis but potentially for advertisement policy training or synthetic augmentation even in future work.

### 4.1 PRODUCT CATALOGUE

#### 4.1.1 DESCRIPTION

This dataset called Amazon Products 2023 is an unstructured dataset transformed and traced back from the huge and open-source Amazon Reviews 2023 dataset [34, 35], highly used in Recommender Systems literature, comprises over 100k products. Put simply, every row/entity is a product/item and it contains many columns/variables such as an identifier (parent\_asin), title, description, main category... Following up the first five examples of the dataset are shown so the reader can familiarize with it 4.1. This dataset will be used as the source of candidates for advertisement by the advertisement insertion algorithm.

The Amazon Products 2023 dataset was selected for three main reasons. First, it provides a large and heterogeneous catalogue covering dozens of product categories, which increases the diversity of potential advertisement candidates. Second, unlike synthetic product collections, the products correspond

**Table 4.1:** Summary statistics of the preprocessed Amazon Products 2023 catalogue.

| Statistic                                       | Value                        |
|-------------------------------------------------|------------------------------|
| Products                                        | 117,343                      |
| <b>Unique product IDs</b>                       | 117,343                      |
| Duplicates removed                              | 0                            |
| <b>Normalized categories</b>                    | 39                           |
| Missing/empty descriptions                      | 0                            |
| Missing/empty titles                            | 0                            |
| Missing price                                   | 35,984                       |
| <b>Average description length (characters)</b>  | 1,669.0                      |
| <b>Median description length (characters)</b>   | 1,547.0                      |
| Minimum description length (characters)         | 17                           |
| Maximum description length (characters)         | 18,084                       |
| <b>Standard deviation of description length</b> | 1,008.1                      |
| <b>Average title length (characters)</b>        | 119.5                        |
| <b>Median title length (characters)</b>         | 120.0                        |
| Average price (\$)                              | 55.40                        |
| Median price (\$)                               | 20.48                        |
| Price range (\$)                                | \$0.01 – \$49,484.08         |
| Average rating                                  | 4.07                         |
| Median rating                                   | 4.40                         |
| Items with rating (Amazon rows)                 | 117,243                      |
| <b>Embedding dimension</b>                      | 1,024                        |
| <b>FAISS index size</b>                         | 459 MB                       |
| Index type                                      | IndexFlatIP (Inner Product)  |
| Dataset source                                  | milistv/AMAZON-Products-2023 |
| License                                         | MIT                          |

to real-world commercial items with realistic titles and descriptions, making generated advertisements more natural. Finally, the dataset has been extensively adopted in recommender systems research, facilitating reproducibility and comparison with future work.

This dataset will serve as the pool of products that our advertisement backbone will navigate to retrieve potential products to advertise. The simple advertisement policy used for the experiment is detailed in the section 5.2 and 6.2. For any good recommendation system to succeed, diversity is key; so another reason this dataset was chosen is due to the big amount of categories it contains (+20) 4.2.

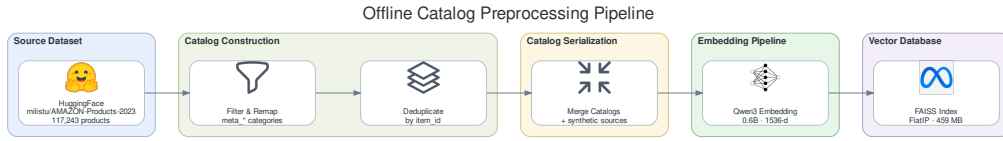
**Table 4.2:** Distribution of products across Amazon categories.

| Category                  | Products       | (%)           |
|---------------------------|----------------|---------------|
| Amazon Fashion            | 470            | 0.40          |
| Appliances                | 573            | 0.49          |
| Arts, Crafts & Sewing     | 2,948          | 2.51          |
| Automotive                | 7,161          | 6.11          |
| Baby Products             | 526            | 0.45          |
| Beauty & Personal Care    | 1,402          | 1.20          |
| Books                     | 2              | 0.00          |
| CDs & Vinyl               | 1,319          | 1.13          |
| Cell Phones & Accessories | 5,062          | 4.32          |
| Clothing, Shoes & Jewelry | 41,777         | 35.63         |
| Digital Music             | 56             | 0.05          |
| Electronics               | 7,681          | 6.55          |
| Gift Cards                | 8              | 0.01          |
| Grocery & Gourmet Food    | 96             | 0.08          |
| Handmade Products         | 1,018          | 0.87          |
| Health & Household        | 4,760          | 4.06          |
| Health & Personal Care    | 93             | 0.08          |
| Home & Kitchen            | 17,326         | 14.78         |
| Industrial & Scientific   | 1,216          | 1.04          |
| Magazine Subscriptions    | 3              | 0.00          |
| Musical Instruments       | 639            | 0.55          |
| Office Products           | 3,545          | 3.02          |
| Patio, Lawn & Garden      | 3,075          | 2.62          |
| Pet Supplies              | 2,742          | 2.34          |
| Software                  | 157            | 0.13          |
| Sports & Outdoors         | 6,343          | 5.41          |
| Tools & Home Improvement  | 4,776          | 4.07          |
| Toys & Games              | 1,367          | 1.17          |
| Unknown                   | 541            | 0.46          |
| Video Games               | 561            | 0.48          |
| <b>Total</b>              | <b>117,243</b> | <b>100.00</b> |

Despite its size, the dataset presents an imbalanced category distribution. Categories such as Clothing, Home & Kitchen, and Electronics dominate the catalogue, whereas others (e.g., Books or Gift Cards) contain only a handful of products. Consequently, the retrieved advertisements are naturally biased toward the largest categories. To mitigate this, the category will be fixed per task in the running study.

#### 4.1.2 PREPROCESSING

In this section we will briefly discuss the transformations applied to the data to make it retrievable by our advertisement serving policy  $\pi$ , being a deterministic static offline process which does not need to be relaunched per experiment, ensuring the same catalogue can be regenerated and traced back from the original dataset. The pipeline mainly consists of a filtering to avoid deduplication and the merging of different formats onto a single one, and then applying an embedding model to that textual data offline in order to create the embeddings that will be stored in the vector database persistently. Once the process is ran, it can be reused many times without having to preprocess the data again, making it extremely important to meet the latency requirements described in 5.1.



**Figure 4.1:** Offline product-catalogue transformation into the FAISS index.

#### RETRIEVAL RATIONALE

The objective of the retrieval system is to identify products that are semantically relevant to the current conversational context rather than those sharing only lexical similarities. For example, a user requesting *"wireless headphones for travelling"* should retrieve products related to active noise cancellation even if the exact wording does not appear in the product description. Consequently, we commit to the school of thought of semantic dense vectorial representations. Through this representation our advertisement insertion algorithm (section 5.3) will be able to leverage items more semantically rather than lexically.

Let a product  $i$  be described by its category  $C_i$ , title  $T_i$ , feature list  $F_i$ , and textual description  $Desc_i$ .

A single textual document is constructed by concatenating these fields,

$$d_i = C_i \oplus T_i \oplus F_i^{(1:10)} \oplus Desc_i, \quad (4.1)$$

where  $\oplus$  denotes string concatenation and only the first ten product features are retained to limit excessively long documents while preserving the most informative attributes. The catalogue is  $\mathcal{D} = \{d_1, \dots, d_N\}$ , the same objects as in subsection 5.2.1.

Each document  $d_i$  is then mapped onto a dense embedding (1024-dimensional in our specific case) using a bi-encoder  $f_{\text{embed}}$ :

$$\mathbf{e}_{d_i} = f_{\text{embed}}(d_i), \quad \mathbf{e}_{d_i} \in \mathbb{R}^{1024}, \quad (4.2)$$

Similarly, given the conversational context  $q$ , composed of the most recent user messages, the retrieval query is embedded using the same encoder (in deployment,  $q$  is the HyDE expansion of section 5.3 rather than the raw utterance),

$$\mathbf{e}_q = f_{\text{embed}}(q). \quad (4.3)$$

Since both products and user queries are projected into the same embedding space, semantic retrieval can be formulated as a nearest-neighbour search. Because all embeddings are L2-normalized prior to indexing, the inner product employed by the FAISS `IndexFlatIP` index is equivalent to cosine similarity,

$$s_{\text{bi}}(q, d_i) = \mathbf{e}_q^\top \mathbf{e}_{d_i}, \quad (4.4)$$

where larger similarity values indicate that the product better matches the semantic intent expressed by the user. The retrieval stage therefore returns the products maximizing for each successive candidate:

$$\arg \max_i s_{\text{bi}}(q, d_i). \quad (4.5)$$

#### 4.1.3 VECTOR DATABASE

As any other type of processing pipeline following the Extract-Transform-Load (ETL) paradigm, the last step is to define the loading sink appropriately to meet our requirements and needs, to weed out the endless possibilities (relational databases like MySQL, blob storages like S3...).

Given the characteristics of the catalogue, approximately 117,000 products occupying less than 500 MB after indexing, the entire collection can be stored comfortably in main memory. This allows the retrieval system to perform similarity search with very low latency without requiring constant disk access, distributed storage or external vector database services.

For this purpose, we employ the Facebook AI Similarity Search (FAISS) library and its `IndexFlatIP` index [36]. During preprocessing, every product description is encoded into a 1,536-dimensional embedding using the Qwen3-Embedding-0.6B model [37] as mentioned before 4.1.2. More on this by the retrieval pipeline described in Chapter 5.

From an algorithmic perspective, FAISS offers a broad family of indexing strategies that trade retrieval speed for search accuracy. Popular approximate nearest neighbour (ANN) indices, such as Inverted File (IVF), Hierarchical Navigable Small World (HNSW), and Product Quantization (PQ), reduce the number of vector comparisons performed during retrieval, enabling scalable search over collections containing millions or even billions of embeddings. In practice, these methods achieve sublinear retrieval times (e.g., approximately  $\mathcal{O}(\log N)$  for HNSW or  $\mathcal{O}(N/k)$  for IVF, where  $k$  denotes the number of partitions searched), at the expense of additional preprocessing, hyperparameter tuning, non-determinism and a small approximation error. Consequently, the retrieved neighbours are not always the exact most similar vectors.

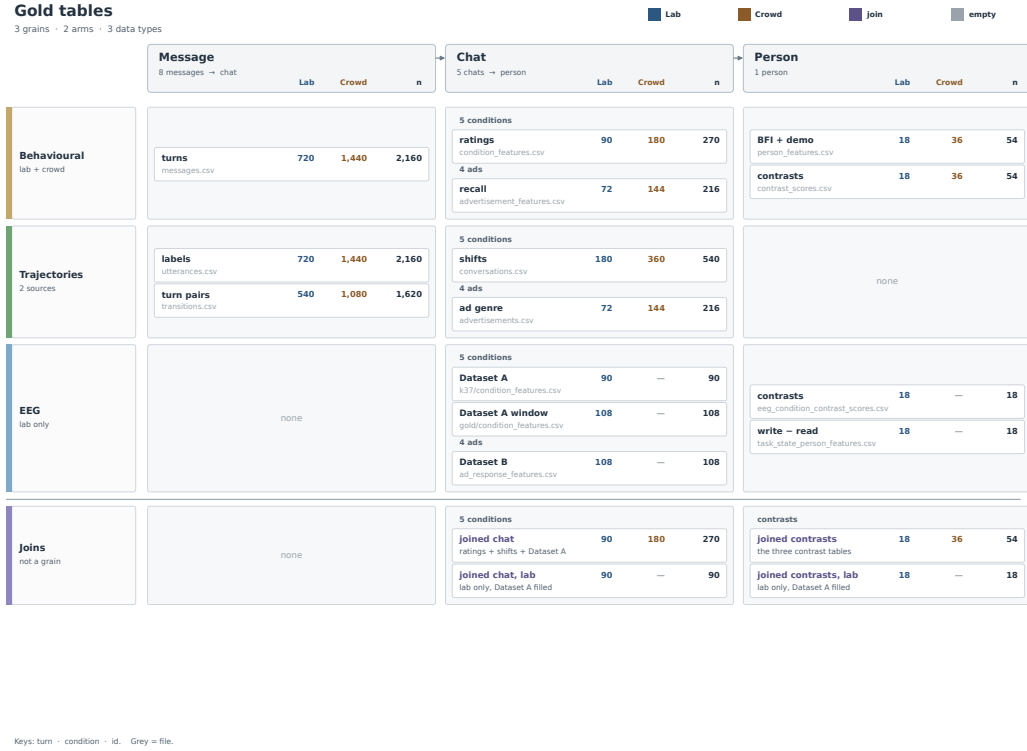
In contrast, the `IndexFlatIP` index adopted in this work performs an exhaustive comparison between the query embedding and every product embedding stored in the catalogue. The retrieval complexity is  $\mathcal{O}(Nd)$ , where  $N$  is the number of indexed products and  $d$  is the embedding dimensionality. Since the embedding dimension is fixed ( $d = 1536$ ), the complexity grows linearly with the size of the catalogue, i.e.,  $\mathcal{O}(N)$ . Although exhaustive search would become impractical for very large collections, our catalogue contains only approximately  $1.17 \times 10^5$  products, allowing exact nearest-neighbour retrieval to be performed in less than 50 ms on our commodity hardware. Consequently, the additional complexity and approximation introduced by ANN indices would provide little practical benefit for the scale of this work, while the deterministic behaviour of `IndexFlatIP` guarantees fully reproducible retrieval results, making it a great solution for the requirements.

## 4.2 EXPERIMENTAL DATA

This section describes the multimodal, mostly unstructured data ingested from the experimental runs, and the preprocessing steps that yield the analytical tables used later in the thesis. The lineage is deliberately staged in the same way as a data lake: immutable **Bronze** (raw logs and recordings), rebuildable **Silver** (parsed, aligned, and labelled intermediates), and **Gold** (the final, analysis-ready tables; every number reported in Chapter 7 is computed from a Gold table and from nothing else). Provenance diagrams for the trajectory and EEG families appear in the subsections below. From a single finished run  $r$  the ingested record is:

$$\mathcal{D}_r = (\mathcal{E}_r, \mathcal{X}_r), \quad \mathcal{X}_r \in \mathcal{X} \cup \{\emptyset\}, \quad (4.6)$$

where  $\mathcal{E}_r$  is the event stream (JSONL) and  $\mathcal{X}_r$  is the optional XDF recording with signals and LSL markers, all aligned under the same clock. Crowdsourced runs have  $\mathcal{X}_r = \emptyset$ ; laboratory runs carry both. Three analytical families are then derived from this pair: the behavioural battery, the genre trajectories, and the EEG features. More on the system that produced the logs is in chapter 5. Figure 4.2 is the big picture summary of this section: three data streams (behavioural, trajectories, EEG) at three grains (message, chat, person), split by arm, with the joined views the association analyses read.



**Figure 4.2:** Gold tables by stream and grain. Numbers are rows per arm; grey cells are empty by design (EEG is laboratory-only).

#### 4.2.1 DATA INGESTION

The finished primary set is  $N = 54$  participants:  $L = 18$  laboratory and  $C = 36$  crowdsourced. Each completed five conditions (implicit or explicit advertisement at turn 2 or turn 4, plus a no-ad control), so the conversation grain is  $54 \times 5 = 270$  conversations and  $270 \times 4 = 1,080$  user utterances. The join key to every other table in the project is `experiment_id`, together with `condition` at the participant  $\times$  condition grain.

Folder tags remove a session from this set: `unfinished` (restarted mid-way), `synthetic`, `crowdfail`, and the early-UI beta testers (kept only as a sensitivity cohort). Sessions tagged `unfocused` remain *in* the finished set when the instruments were completed; inattention is not an exclusion rule. One laboratory folder (`lab_subject_4_crowdfail`) is out of both the behavioural and the EEG Gold tables.

#### 4.2.2 BEHAVIOURAL DATA

The behavioural battery is scored from the same Bronze JSONL as the trajectories, at the participant  $\times$  condition grain; its Gold tables were frozen on 8 September 2026 and are the only source of behavioural

numbers in chapter 7 and chapter 8. The pass follows the same Bronze / Silver / Gold contract as the other two families. Both arms completed every instrument, so this family runs at  $N = 54$ . How items become outcomes is fixed in subsection 6.2.2; this section describes the data format and transformations to gain intuition.

**BRONZE.** Bronze is the tracked laboratory and crowd session log; where a folder holds both an events file and an export-tagged file, the export is read. A session file holds the 22 post-condition items after each of the five conversations, the cued-recall block after the fifth, the BFI-10, the demographics form, and the interaction log of messages, timings, and served advertisements. The roster is the finished set of subsection 4.2.1, with the same folder-tag exclusions and unfocused sessions kept.

**SILVER.** Silver parses that JSONL in memory into a flat participant  $\times$  condition item table: the 22 post-condition items on every condition and the three cued-recall items on each advertisement condition. Reverse-coded items are inverted once, as  $8 - x$ , and never again when outcome means are formed. The join key is `experiment_id`, with `condition` at this grain. No later chapter reads an intermediate item file.

**GOLD.** Gold holds the views of Table 4.3, the behavioural row of Figure 4.2. Ratings are one row per participant  $\times$  condition ( $54 \times 5 = 270$ ) with the four primary outcomes (trust, credibility, perceived manipulation, notice), the four secondary qualities (helpfulness, convincingness, relevance, neutrality), and the raw items beside them. Person features are one row per participant with the five BFI-10 traits, the demographics, and the arm. Contrast scores are one row per participant holding one person-level difference  $D_i$  per planned contrast (6.1). Recall is one row per served advertisement ( $54 \times 4 = 216$ ): format, timing, the served product, cued memory, and trust after re-exposure;  $a^\emptyset$  has no row because there is no product to re-show. The advertised genre  $g^{(a)}$  lives on the trajectory advertisement table (§4.2.3). A message-grain process table (2,160 turns with length and latency) sits beside the ratings and is not an outcome table. The association analyses read joined views of these grains, ratings and contrasts joined to the trajectory and EEG tables, with EEG columns filled only on laboratory rows.

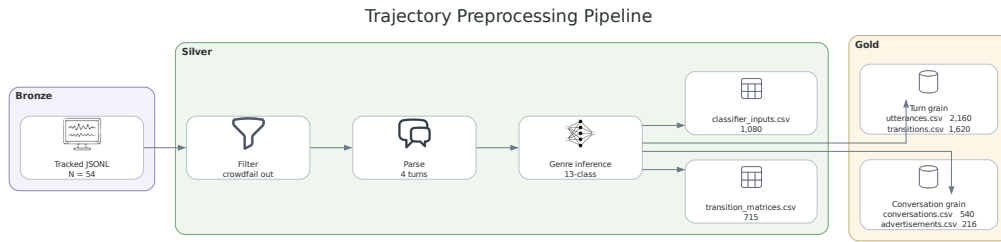
**Table 4.3:** Gold behavioural views. Laboratory rows are the ones in which EEG is filled.

| View                          | Grain                              | Rows |
|-------------------------------|------------------------------------|------|
| Ratings                       | participant $\times$ condition     | 270  |
| BFI + demographics            | participant                        | 54   |
| Planned contrasts             | participant                        | 54   |
| Recall                        | participant $\times$ advertisement | 216  |
| Joined conditions, laboratory | participant $\times$ condition     | 90   |
| Joined contrasts, laboratory  | participant                        | 18   |

A single builder writes these tables, and an independent analysis of the same Bronze by a second analyst was used for cross-checking and is not quoted.

#### 4.2.3 TRAJECTORY DATA

Complementary to the behavioural battery, this family looks only at the conversation as a sequence of intents. Preprocessing is lighter than for EEG: a roster filter, a rearrangement of the four-turn dialogues, and an offline pass of the same 13-class genre classifier  $f_{\text{genre}}$  that the live system used to route retrieval. The pass uses the same Bronze / Silver / Gold contract as the EEG lake, shown in Figure 4.3: Bronze is the tracked JSONL; Silver is the in-memory filter, parse, and genre inference, plus two intermediate tables that analysis does *not* test on; Gold is four datasets at two grains, turn and conversation.



**Figure 4.3:** Trajectory preprocessing from session logs to turn- and conversation-grain Gold tables.

Every Gold table with genre labels carries a `genre_source` column with two values, stored as rows rather than as parallel columns. `utterance` is  $f_{\text{genre}}$  applied to the bare message and is the primary reading. `contextual` concatenates the task prompt, recent history, and the current message, and is the reading the deployed system computed; it is retained as a sensitivity table. The two disagree on most

turns, so every count must state which source it came from. The spine is `conversation_id`; joins to EEG and to the behavioural tables use `experiment_id` and `condition`. Both arms have conversations, so this family runs at  $N = 54$ .

The Silver tables are not inferential. `classifier_inputs.csv` stores the exact contextual string that was fed to  $f_{\text{genre}}$ , so a number can be interpreted without returning to the JSONL. `transition_matrices.csv` is a faceted edge list (overall, by advertisement presence, and by condition); heatmaps used in later chapters are rebuilt from the Gold transition table rather than queried from this kernel.

#### UTTERANCE LEVEL

The core Gold table is `utterances.csv`:  $1,080 \text{ user messages} \times \text{two genre sources} = 2,160$  rows. A genre label lives here once. Alongside the message text it carries the incoming transition ( $\hat{g}_{k-1}$ , the hard shift  $\delta_{k-1}$ , total-variation distance, and whether the step sits before or after the advertisement), blank at turn 1, so the common questions need no join. Table 4.4 shows the first five rows of each Gold table under the primary source, written in the notation of chapter 3. Classifier posteriors are stored on the utterance table but are not shown: confirmatory tests use the hard label  $\hat{g}_k = f_{\text{genre}}(u_k)$ .

`transitions.csv` materialises the same chain as one row per step: three transitions per conversation  $\times$  two sources = 1,620 rows. Each row is  $\tau_k = (\hat{g}_k, \hat{g}_{k+1})$  with  $\delta_k$  and a position flag (`pre_ad`, `crosses_ad`, `post_ad`). Tests that compare the step that crosses an early advertisement against the same step in the no-ad control read this table, after selecting one `genre_source`. Late advertisements have no following utterance, so they cannot appear as a crossing transition.

#### CONVERSATION LEVEL

`conversations.csv` holds only conversation-level aggregates ( $54 \times 5 \times 2 = 540$  rows): the four-label trajectory  $\hat{\mathbf{G}}$ ,  $N_{\text{shift}}$ , diversity  $D(\hat{\mathbf{G}})$ , entropy  $H(\hat{\mathbf{G}})$ , maximum persistence  $R_{\text{max}}$ , and the advertisement-associated shift  $\delta^{(a)}$  when a following turn exists (undefined after a late insertion). It does not duplicate the per-turn rows; anything that needs a single  $\hat{g}_k$  reads the utterance table.

`advertisements.csv` is the second conversation-grain Gold table:  $54 \times 4 = 216$  advertised conversations (the no-ad control has no product). Each retrieved product title is classified with the same  $f_{\text{genre}}$ , so the advertisement genre  $g^{(a)}$  lives in the same 13-class space as the utterances. This table is *not* long on `genre_source`: the product is classified once. It is what the genre-aligned shift  $\tilde{\delta}^{(a)}$  reads.

**Table 4.4:** Example rows from the four trajectory Gold tables under the primary bare-utterance context.  $\delta^{(a)}$  is undefined after a late insertion

| utterances.csv     |     |                 |             |                |  |
|--------------------|-----|-----------------|-------------|----------------|--|
| $a$                | $k$ | $\hat{g}_{k-1}$ | $\hat{g}_k$ | $\delta_{k-1}$ |  |
| $a_4^{\text{exp}}$ | 1   | —               | rel         | —              |  |
| $a_4^{\text{exp}}$ | 3   | guid            | acad        | I              |  |
| $a_4^{\text{exp}}$ | 4   | acad            | acad        | o              |  |

| transitions.csv    |     |             |                 |            |          |
|--------------------|-----|-------------|-----------------|------------|----------|
| $a$                | $k$ | $\hat{g}_k$ | $\hat{g}_{k+1}$ | $\delta_k$ | Position |
| $a_4^{\text{exp}}$ | 1   | rel         | guid            | I          | pre      |
| $a_2^{\text{exp}}$ | 2   | creat       | acad            | I          | crosses  |

| conversations.csv  |                           |                    |     |      |                  |                |
|--------------------|---------------------------|--------------------|-----|------|------------------|----------------|
| $a$                | $\hat{G}$                 | $N_{\text{shift}}$ | $D$ | $H$  | $R_{\text{max}}$ | $\delta^{(a)}$ |
| $a_4^{\text{exp}}$ | (rel, guid, acad, acad)   | 2                  | 3   | 1.04 | 2                | —              |
| $a_2^{\text{imp}}$ | (guid, acad, guid, guid)  | 2                  | 2   | 0.56 | 2                | I              |
| $a^\emptyset$      | (guid, other, acad, acad) | 2                  | 3   | 1.04 | 2                | —              |

| advertisements.csv |                          |           |
|--------------------|--------------------------|-----------|
| $a$                | Product (truncated)      | $g^{(a)}$ |
| $a_4^{\text{exp}}$ | ANCHEER resistance bands | guid      |
| $a_4^{\text{imp}}$ | Organic sea-kelp extract | guid      |
| $a_2^{\text{imp}}$ | Lenovo Slim 7 laptop     | guid      |

#### 4.2.4 EEG DATA

##### DESCRIPTION

Again, each laboratory session produces a pair XDF  $\mathcal{X}_r$  and  $\mathcal{E}_r$ . The data lake is the same Bronze / Silver / Gold contract as the trajectories (Figure 4.4): Bronze inventories recordings and settles the cohort; Silver builds the event clock and the cleaned continuous signal; Gold cuts windows (each user will carry  $K_i$  epochs) and writes two final tables to be analyzed directly. Nineteen sessions were recorded; Subject 4 ran the crowd protocol and stops at Bronze, so every later object is  $n = 18$ . Tests sit outside the lake and operate on people, not epochs [38].

In a high-level fashion, the preprocessing takes the raw signals as input and markers and outputs two analysis-ready datasets, Dataset A (condition aggregation) and Dataset B (onset-locked contrast). Dataset A is composed of one median spectrum (16 features such as the absolute and relative bands, engagement indices...) per participant  $\times$  condition,  $Y_A \in \mathbb{R}^{18 \times 5 \times 16}$ , with the confirmatory cell the median of the 37 epochs nearest visual onset so that short and long conversations are balanced and contribute the same number of epochs. Dataset B is the advertisement contrast: around the moment it becomes visible we cut 4 s before and 4 s after (epochs), take the spectrum of each, and store the difference (after – before). That is done for the four insertions and for two timing-matched replies under  $a^0$  to compare against, so  $Y_B \in \mathbb{R}^{18 \times 6 \times 16}$ . Same cleaning, same epoch size, and the same sixteen spectral features; different epoch cuts, so no shared rows. Person-level contrasts  $D_i^A$  and  $D_i^B$  are formed from these tables in chapter 6. More details appear in the following sections.

**Table 4.5:** Gold EEG tables for condition aggregation and onset-locked responses ( $n = 18$ ).

| Dataset                  | Table                    | Grain                              | Shape                   |
|--------------------------|--------------------------|------------------------------------|-------------------------|
| A, condition aggregation | condition_features.csv   | participant $\times$ condition     | $18 \times 5 \times 16$ |
| B, onset-locked          | ad_response_features.csv | participant $\times$ advertisement | $18 \times 6 \times 16$ |

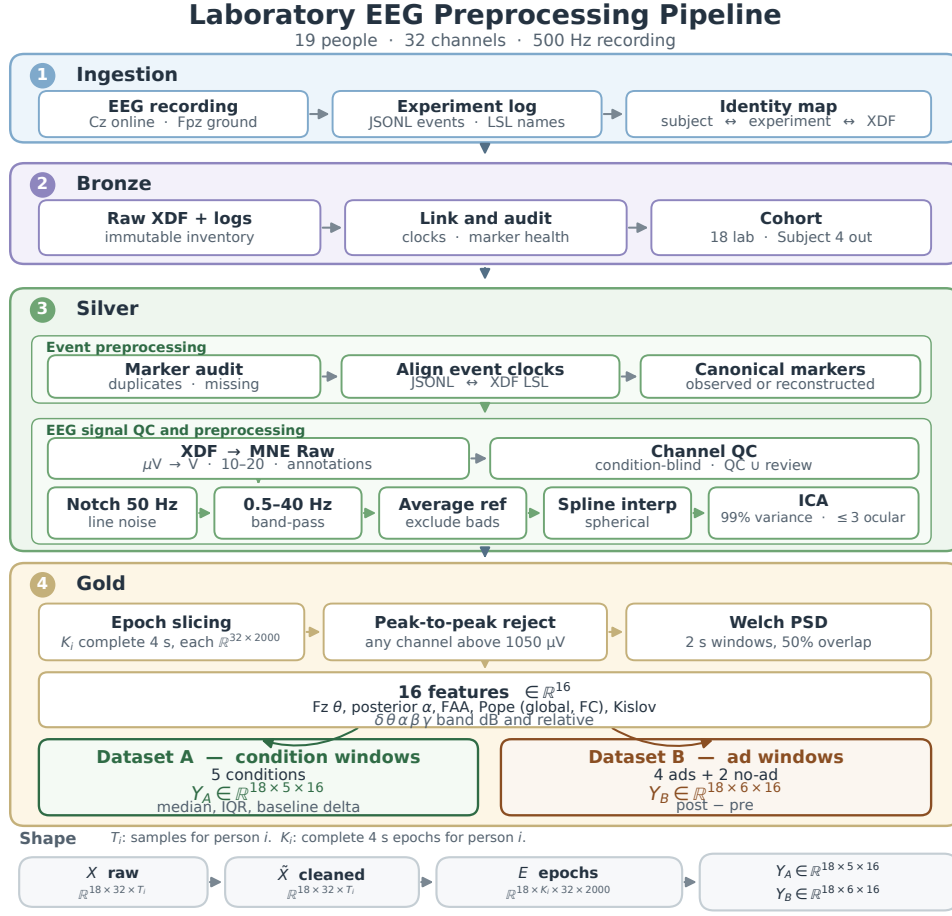
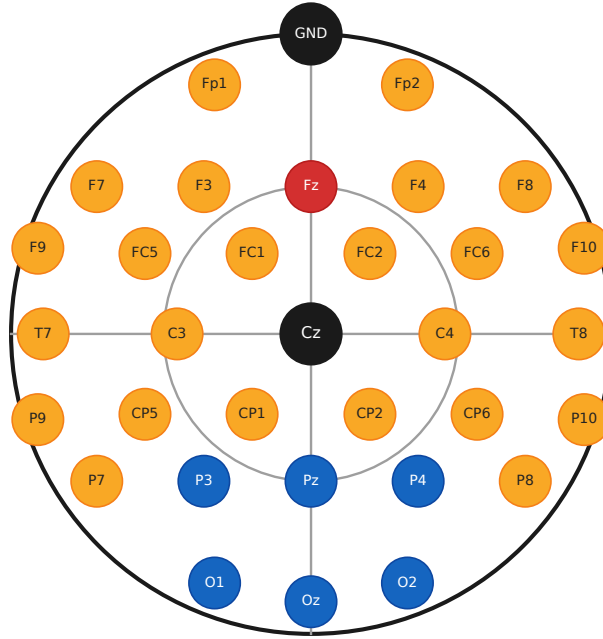


Figure 4.4: EEG lineage from raw recording and event log to condition-aggregation and onset-locked Gold tables.

## ACQUISITION

Stimuli were presented to the users on a 27-inch monitor at a fixed viewing position. EEG was recorded with a gel-based 32-channel actiCHamp system (Brain Products GmbH) at 500 Hz. Electrodes follow the 10 – –20 montage in Figure 4.5, ground Fpz, online reference Cz. Impedances were brought below 10 k $\Omega$  and checked before and during the session. A session is one continuous recording of about an hour. Recording volume, the condition-blind quality profile, and the ICA thresholds are in Appendix D.



**Figure 4.5:** 32-channel actiCHamp 10–20 layout. Red: Fz  $\theta$ ; blue: posterior  $\alpha$ ; yellow: other recorded channels.

## PREPROCESSING

Each zone refines the one before it and never writes back to it, which means the cohort is settled before any signal is filtered, and the signal is filtered before any experimental window is cut. The rest of this subsection follows the zones in order and says what each transform is for; the exact thresholds behind them are in Appendix D.

**INGESTION AND BRONZE.** A session arrives as the two files, both carrying Lab Streaming Layer (LSL) markers on a common clock, and an identity map records which subject, which run, and which file belong together. Bronze does nothing to the signal. It lists the recordings, links each XDF to its log, checks that the expected markers are present and not duplicated, and labels the recording; it computes no spectrum. The cohort is settled here rather than later. Nineteen sessions were recorded, and one of them, Subject 4, ran the crowd protocol by mistake; that session stays in Bronze for the record and is excluded from everything downstream, so every object after this zone contains the same eighteen people as the finished laboratory sample.

**SILVER: THE EVENT CLOCK.** Cutting a window later requires knowing when each phase of the protocol began, and the two clocks do not always agree, because an LSL marker occasionally fails to fire even though the corresponding event is in the log. The two sources were therefore audited against each other for duplicates, gaps, and extra events, and then aligned into a single canonical timeline. A canonical marker is observed when the LSL event fired and reconstructed from the log when it did not. Those canonical times are the only clock Gold is allowed to cut on. They mark interface events, such as a condition starting or a reply appearing on screen, and not brain events; nothing in this zone claims to know when the participant perceived anything.

**SILVER: FROM A RAW FILE TO A CLEAN RECORDING.** Each XDF is read into an MNE-Python Raw object, with channels as rows, values in volts, the 10 — 20 names of Figure 4.5, and the original annotations preserved [39]. Nothing is filtered or resampled at read time, so what sits in memory is still the recorded signal. Every recording was then profiled on twelve 30 s windows spread across the session, summarising each channel by its amplitude, its flatness, its mains contamination, and how well it agrees with the rest of the montage. The windows were placed without consulting the event log, so the profile is condition-blind by construction and cannot end up cleaning one experimental condition more carefully than another. It only writes warnings: no recording was ever excluded by an automatic amplitude rule, and its purpose is to direct human review. Three channels failed that review and were repaired, P4 on Subjects 1 and 9 and C4 on Subject 10.

Cleaning then applies four transforms in a fixed order, each aimed at a different kind of contamination. A narrow notch at 50 Hz removes the mains hum picked up from the building supply, which lies above the 30 — 40 Hz gamma band used here but would otherwise leak into its upper edge through the low-pass transition. A 0.5 — 40 Hz band-pass drops the slow drift caused by sweat and small electrode movements, together with the high-frequency content dominated by muscle activity and equipment noise [40]; it also fixes the interval over which every later spectrum is integrated. Both are zero-phase FIR filters (the MNE-Python defaults: windowed-sinc design, Hamming window, forward-backward application), so no phase shift is introduced; the filters are applied to the continuous recording before any window is cut, and their transition bands span a few hundred milliseconds, small against the 4 s windows of §4.2.4. Since an EEG value is a potential difference, and therefore only means something relative to a reference, the online Cz reference is then replaced by the average over the channels not marked bad,

$$\tilde{x}_c(t) = x_c(t) - \frac{1}{|C_{\text{good}}|} \sum_{c' \in C_{\text{good}}} x_{c'}(t), \quad (4.7)$$

so that no single site is privileged. Last, the channels marked bad are re-estimated from their neighbours

by spherical-spline interpolation, which fits a smooth surface over the scalp instead of leaving a hole in the montage [41]. Interpolated traces keep their bad label, so no later stage can mistake them for recorded data.

Eye movements are the one artefact this chain cannot filter away, because a blink is both far larger than cortical activity and overlapping with the frequencies of interest. Independent component analysis is therefore part of the primary pipeline: FastICA decomposes the recording into components, and only ocular ones are removed [42]. The decomposition is fitted on a high-passed, decimated copy of the recording, so that residual slow drift does not dominate it, and the resulting unmixing is applied to the analysis band. A component counts as ocular only if it both correlates with a frontal ocular proxy and has its topography concentrated at the front of the head; at most three may be removed from any recording, and every automatic exclusion was confirmed by visual inspection of the component map and its time course. This montage has no dedicated EOG channel, so the frontal electrodes stand in for one. Because removing components can in principle impose structure as well as remove it, the identical pipeline without ICA is kept as a mandatory sensitivity branch.

Two habits of this zone matter for reproducibility. Cleaning always runs on the full 32-channel montage, so no later choice of electrodes can influence the reference or the decomposition. And cleaned recordings are never written to disk: Gold calls the cleaner in memory whenever it needs a signal, so there is no intermediate file that could fall out of step with the policy that produced it. What leaves Silver is one cleaned continuous recording per participant,  $\tilde{X} \in \mathbb{R}^{18 \times 32 \times T_i}$ , as long as participant  $i$  sat in the laboratory; removing components changes its values but not its shape. The overall flow can easily be understood by the following shape evolution:

$$X \in \mathbb{R}^{18 \times 32 \times T_i} \rightarrow \tilde{X} \in \mathbb{R}^{18 \times 32 \times T_i} \rightarrow E \in \mathbb{R}^{18 \times K_i \times 32 \times 2000} \rightarrow Y_A \in \mathbb{R}^{18 \times 5 \times 16}, Y_B \in \mathbb{R}^{18 \times 6 \times 16}. \quad (4.8)$$

The first arrow is the cleaning just described, which changes values but neither drops channels nor shortens the recording. The second cuts the continuous signal into  $K_i$  epochs of 4 s, a number that differs by participant because it depends on how long their conditions ran; for Dataset A ( $Y_A$ ) it is the count of complete 4 s tiles, while for Dataset B ( $Y_B$ ) it is 12 when every locked pair survives. The third reduces each epoch to sixteen spectral measures and summarises them per participant. The next subsection covers the cut, and the two after it cover the two aggregations.

## EPOCHING

**AN INDUCED SPECTRUM.** The two datasets measure how much power sits in each frequency band inside a 4 s window, and not the shape of the voltage waveform following a stimulus [38]. The distinction decides everything that follows. An event-related potential averages the raw waveform over many repetitions of a sharply timed event, which buys resolution on the scale of tens of milliseconds but collapses when the timing is uncertain, because misaligned repetitions cancel each other out. An induced spectrum throws away phase and asks only how much energy each band carries, so it survives jitter in exchange for not resolving fast components.

This design needs the tolerant estimator, because an advertisement in a chat interface is not merely a flash: it is painted while the page rerenders, and the moment it becomes visible is uncertain to about 0.23 s for explicit banners and 0.43 s approximately for implicit mentions. Those figures are 95th-percentile leave-one-out estimates of the reconstruction error, obtained by predicting onsets whose display marker had in fact fired and comparing the two. Where it did not, visual onset is taken as the assistant reply plus 0.49 s for an explicit banner, or the injection event plus 1.57 s for an implicit mention, which lands before the reply has finished streaming; 30 of the 36 implicit onsets are reconstructed in this way. An uncertainty of a quarter of a second is already wider than an entire early evoked component, so a window of a few hundred milliseconds around onset would not be defensible here.

**REJECTING EPOCHS AND ESTIMATING THE SPECTRUM.** A complete epoch is  $E_{i\ell} \in \mathbb{R}^{32 \times 2000}$ , that is, 32 channels by 4 s at 500 Hz. It is rejected if any channel amplitude swings more than 1,050  $\mu\text{V}$  peak-to-peak, which is far above physiological range and indicates movement or a lost contact, or if any channel has a standard deviation below 0.5  $\mu\text{V}$ , which indicates a dead one. Surviving epochs are never averaged in the time domain. Each one is turned into a power spectrum with Welch’s method, which cuts the epoch into overlapping segments, estimates a spectrum for each, and averages them, buying a less noisy estimate at the cost of frequency resolution [43]. With 2 s segments overlapping by half, a 4 s epoch gives three segments and a resolution of  $\Delta f = 0.5$  Hz. Writing  $P_{i\ell}(f)$  for that estimate, absolute and relative power in band  $b$  are

$$y_{i\ell,b}^{\text{abs}} = 10 \log_{10} \int_b P_{i\ell}(f) df, \quad y_{i\ell,b}^{\text{rel}} = \frac{\int_b P_{i\ell}(f) df}{\int_{0.5}^{40} P_{i\ell}(f) df}, \quad (4.9)$$

with the band integral in  $\mu\text{V}^2$ . The absolute measure is on a decibel scale, so a difference between two of them is a ratio; the relative measure is the share of the 0.5 — 40 Hz total that falls in the band, so it is a dimensionless proportion. The five bands are the canonical ones,  $\delta$  0.5 — 4 Hz,  $\theta$  4 — 8 Hz,  $\alpha$

$8 - 13$  Hz,  $\beta$   $13 - 30$  Hz, and  $\gamma$   $30 - 40$  Hz [44, 45]. From them plus the 6 additional features described in D.2, one epoch becomes sixteen real numbers,  $E_{i\ell} \mapsto y_{i\ell} \in \mathbb{R}^{16}$ : the five absolute powers averaged over the whole head, the five matching relative powers, two regional measures, and asymmetry and engagement ratios borrowed from the consumer-neuroscience literature. Appendix D defines each one. Only the two regional measures, Fz theta and posterior alpha, are tested confirmatorily (chapter 6); the other fourteen travel in  $Y_A$  and  $Y_B$  without forming a second confirmatory family.

**WHY FOUR SECONDS.** Four seconds is both the conventional width for this kind of estimate and the one that fits this protocol. Kislov et al. scored banner EEG with a 4 s post-advertisement transform [46], and Lopez-Cardona et al. extracted induced spectral features from fixed-length response windows in the same kind of 32-channel laboratory setting [40]. It is also the shortest window that supports the spectral settings above, since it is three 2 s segments at half overlap that produce  $\Delta f = 0.5$  Hz. And it is long enough to contain the assistant reply in nearly every case while staying clear of the next event, which arrives about 53 s later after an early advertisement and about 79 s later after a late one. The two neighbouring choices fail in opposite directions. A 2 s window is as long as the onset uncertainty itself and often ends before an implicit reply has finished streaming, so it may not contain the stimulus at all. An 8 s window folds the reply together with the reading that follows it, so a transient change is averaged away against the surrounding seconds, producing a lot of aggregation. The width was fixed on these grounds before any contrast was estimated and was not revised afterwards; 2, 8, 16, and 32 s appear in section 7.4 as sensitivity against that fixed choice, not as competing primary analyses.

#### CONDITION AGGREGATION

Dataset A is `condition_features.csv`. Canonical markers delimit one baseline window and the five condition windows, each from its start marker to the written conclusion. A window of duration  $D$  seconds yields  $\lfloor D/4 \rfloor$  contiguous non-overlapping epochs; leftover  $< 4$  s is dropped. A window is kept if it retains at least five epochs and at least 80% of the epochs it contains. Conversations in this protocol vary in length (2.5 – 16.8 min; 37 – 251 complete 4 s epochs, median 95). To keep short and long conversations from contributing unequal numbers of epochs, the confirmatory Dataset A cell is the median of the 37 retained epochs whose midpoints lie nearest that condition’s visual onset (for  $a^0$ , the matched turn-2 and turn-4 replies).  $k = 37$  is the shortest eligible window, so every advertisement condition contributes exactly 37 epochs and  $n = 18$  is retained. The fixed count is a standardisation: a whole-window median would summarise one participant over 37 epochs and another over more than 200, so cells would differ in precision and in how much of the conversation they average, and a between-condition difference could then reflect conversation length rather than the advertisement. Taking the

same number of epochs nearest onset for everyone removes that asymmetry. The value 37 is dictated by the shortest window, and no other  $k$  was tried before the confirmatory tests were read. The stored Gold table keeps the whole-window median; the confirmatory tests use this condition aggregation. Write  $N_{ic}(37)$  for that set of 37 epochs. That median is the appropriate summary for log-transformed power [47],

$$Y_{A,ic} = \text{med}_{\ell \in N_{ic}(37)} y_{i\ell}, \quad Y_A \in \mathbb{R}^{18 \times 5 \times 16}. \quad (4.10)$$

IQR and condition-minus-baseline differences are stored, not tested: eye state was uncontrolled at baseline, and the baseline cancels in a paired contrast. Dropping it leaves implicit/explicit  $\times$  early/late and  $a^\emptyset$ . Under primary ICA, 9,449 of 9,468 complete Dataset A epochs were retained (99.8%), and every eligible window passed the rule. The join is `experiment_id` and `condition`; the extra baseline row does not join. Table 4.6 shows one participant’s five confirmatory Dataset A rows.

#### ONSET-LOCKED CONTRAST

Dataset B is `ad_response_features.csv`. Here  $t = 0$  is visibility: the observed display marker, or the reconstruction above. Matched  $a^\emptyset$  locks the same-turn `assistant_reply`. Around onset  $t_{ik}$ , the windows are:

$$E_{ik}^{\text{pre}} = [t_{ik} - 4, t_{ik}), \quad E_{ik}^{\text{post}} = [t_{ik}, t_{ik} + 4). \quad (4.11)$$

Both windows must fall inside the parent condition. The window *is* the epoch, so there is no tiling. A pair is kept only if both sides pass the same rejection rule. The stored cell is

$$Y_{B,ik} = y_{ik}^{\text{post}} - y_{ik}^{\text{pre}}, \quad Y_B \in \mathbb{R}^{18 \times 6 \times 16}. \quad (4.12)$$

All 216 Dataset B epochs were retained (108 pairs). There is no median over many epochs: a single 4 s window is the observation. The lower block of Table 4.6 is the same participant’s six Dataset B rows: four insertions and the two timing-matched  $a^\emptyset$  replies.

**Table 4.6:** Example Gold EEG rows for one laboratory participant; two of sixteen spectral measures are shown.

| <b>condition_features.csv</b> |       |                  |                     |
|-------------------------------|-------|------------------|---------------------|
| $a$                           | $K_c$ | Fz $\theta$ (dB) | post. $\alpha$ (dB) |
| $a_4^{\text{imp}}$            | 238   | 8.90             | 7.97                |
| $a_2^{\text{exp}}$            | 162   | 8.25             | 8.32                |
| $a_2^{\text{imp}}$            | 70    | 8.80             | 8.83                |
| $a_4^{\text{exp}}$            | 54    | 8.55             | 8.86                |
| $a^\emptyset$                 | 109   | 9.18             | 8.89                |

| <b>ad_response_features.csv</b> |        |                           |                              |
|---------------------------------|--------|---------------------------|------------------------------|
| $a$                             | Onset  | $\Delta$ Fz $\theta$ (dB) | $\Delta$ post. $\alpha$ (dB) |
| $a_4^{\text{imp}}$              | recon. | -2.20                     | -2.61                        |
| $a_2^{\text{exp}}$              | obs.   | +2.20                     | +0.77                        |
| $a_2^{\text{imp}}$              | recon. | +3.60                     | +0.73                        |
| $a_4^{\text{exp}}$              | obs.   | -0.43                     | -5.60                        |
| $a^\emptyset$ early             | recon. | +2.32                     | +3.65                        |
| $a^\emptyset$ late              | recon. | +4.15                     | +2.61                        |

# 5

## System Design and Engineering

Beyond the theoretical and experiment research advancements that this thesis offers, one of its biggest strengths is to serve as a case study and all the learnings behind the system design, the engineering to make it happen and how could it be improved in future iterations. This chapter will cover the requirements, hidden trade-offs, the decisions made, engineering efforts such as the retrieval pipeline or the EEG integration and acknowledgement of limitations for future work.

### 5.1 REQUIREMENTS

To be able to have a good enough platform to collect data from the subjects without them being disturbed by technical details and unrealistic and awkward user experience, which could influence the data analysis greatly, the minimal set of functional requirements is described below, side by side with the tools we used to achieve those requirements and the rationale.

#### 5.1.1 FUNCTIONAL REQUIREMENTS

- **Personal Data Privacy:** The information that is disclosed in the conversations and questionnaires is highly sensitive and some information even belongs to the PII category, so this is one reason why we preferred a local deployment of an LLM through Ollama rather than regular APIs and a thorough anonymisation of the users, logging no data that could identify them and posteriorly changing the timestamps so it cannot be known.
- **200k Item Vector Database:** In order to support the product catalogue we need a small-medium sized vector database, so given this small size, in-memory databases such as FAISS are more than enough and give better latencies than more complex hierarchies leveraging persistent disk.

- **Stateful Experiment:** State of the experiment UI must be recorded and stored persistently, thus we deployed an instance of streamlit in the ATLAS Telefonica server with periodic logging of the events into jsonl files associated with each run rather than subject, again for privacy reasons.
- **Multimodal Ingestion Integration:** The system must be able to ingest EEG signals in real-time alongside the logs with synchronized clocks, otherwise EEG data analysis quality degrades. To achieve this we used as described earlier a 32-channel actiCHamp system connected to the laboratory laptop which also received LSL markers from the ATLAS server in the form of network requests, given they are different machines, to guarantee both use the same clock: The laboratory laptop's clock.
- **Crowd Sourcing Platform Integration:** Not only laboratory, but also behavioural runs from the crowd-sourcing platform must be allowed, which supposed some extra steps in the workflow to integrate it with prolific worker ids and with attention questionnaires to avoid degradation of data quality in unfocused or inattentive users.

### 5.1.2 NON-FUNCTIONAL REQUIREMENTS

In terms of non-functional requirements, there are 4 different angles that need to be covered for the success of the system:

- **LLM perceived latency:** The latency of the time to first token must be low in order to avoid disruption of the emotional state of the subjects or to affect behavioural metrics such as trust or helpfulness. Here the GPU choice, LLM and ad retrieval algorithm will greatly influence this metric. Nielsen's three limits still apply: 0.1 s feels instantaneous, 1 s keeps the user's thought intact, and 10 s is about as long as attention will sit on a silent wait before it drifts [48]. The participant UI streams tokens, so the relevant wait is time to first token (TTFT) rather than full completion. Across the finished  $N = 54$ , a completed streamed reply took a median 2.69 s under  $a^0$  and 2.89 s in advertised conditions (IQR about 2.2 – 4.3 s), which with streaming sits inside the 1 – 10 s band. On the one advertised turn, retrieval runs before the first token (median 3.03 s), so that silent wait is about 6 s: noticeable, still below 10 s, and the only ads-versus-no-ads difference in server wait. Warm-up of both LLM and databases was employed to mitigate cold-start latencies and avoid surpassing 10 seconds.
- **Quality of Service (QoS):** Even though it is a local deployment, the quality of the responses must be realistic and high enough compared to production grade LLMs so subjects can engage properly and form an opinion without being degraded by the quality of the responses. Given our constraints, Qwen 3.6 35B-A3B was the best candidate up to that date that surpassed expectations both in TTFT, quality, inference speed even at its size.
- **LLM Parallel Throughput:** Although not a hard requirement, the more users can be connected simultaneously, the more users can be interacting at the same time with the system. vLLM was the only inference engine that allowed us out-of-the-box batching and other optimizations, but

due to the lack of support of this specific model in the past, we were not able to implement it, thus preferring serialized Ollama experiments. The worst case is two participants sending on an advertised turn together: the second waits the first's exclusive hold (retrieval plus generation, about 6 s, still under Nielsen's 10 s). A third waits two holds and the first token exceeds 10 s. The waiting room is therefore at most 1 participant deep before the 10 s bound is broken, which the session scheduling of subsection 6.2.3 respects.

- Natural UI: Similar to QoS, to avoid disruption of the behavioural metrics due to poor user experience (UX), the UI needed to be familiar to that of modern chatbots, copying features such as streaming responses, similar placements of widgets and similar colours/fonts. We achieved many of these features thanks to Streamlit.

As for the functional requirements, all of them are covered in this implementation, for the non-functional ones all of them are covered to a sufficient standard, excelling in the perceived latency, intuitive UI and high quality responses.

## 5.2 SYSTEM ARCHITECTURE

In this section, a thoughtful discussion about the tradeoffs, final choices and degrees of freedom in the system will be presented, going over the final solution first and later putting forward the rationale behind each choice. Next sections like 5.3 will go over other relevant details such as the retrieval pipeline or the overall infrastructure.

### 5.2.1 MODEL CHOICES

One important detail often left abandoned in many system designs, is the rationale behind the **choice of the models**, especially the LLM itself. So in this section we will cover in detail the choices of the 4 models behind this system:

### 5.2.2 CONVERSATIONAL LLM

In order to choose LLMs, the field of LLM evaluation has advanced drastically with respect to 2023 [49], offering different angles of evaluation to gather a very complete picture of any given popular open-source and closed-source model. Firstly, we have the more quantitative dimension, which is typically evaluated via benchmarks and indices. Benchmarks are standardised tests for a specific domain that yield a numerical score: multiple-choice knowledge exams such as MMLU and MMLU-Pro [50, 51], mathematics competitions such as AIME-25 [52], software-engineering tasks such as SWE-bench and SWE-Bench Pro [53, 54], and tool-use and agentic tasks such as  $\tau$ -bench [55]. Indices aggregate many such benchmarks into a single score; the Artificial Analysis Intelligence Index, for instance, is a weighted

average over ten agentic, coding, general and scientific-reasoning evaluations [56]. Secondly, there is the qualitative dimension, which is undoubtedly the hardest: the high-level aggregated judgement of humans (often called the “vibes” of the model, the perceived level of intelligence or usefulness), of which the crowd-sourced pairwise votes of Chatbot Arena are the best-known systematic instance [57]. However, it is important to note that for the scope of the experiment, the LLM is going to be used as a chatbot, so the standards for math, coding, agentic & tool usage and other capabilities can be relaxed.

Having all this context in mind, it was natural for us in our setup of 80 GB of VRAM in April 2026 for the quantized **Qwen 3.6 35B-A3B at Q4\_K\_M** [58] given its Artificial Analysis score [59] and the qualitative results obtained from conversations tailored to the tasks A with multiple open-source models. Figure 5.1 provides a high-level view of the Qwen 3.6 family architecture (MoE backbone with gated full-attention / gated DeltaNet blocks), supporting up to 262k context length.

It is common in LLM open-source releases to follow logarithmic patterns, either to follow in the tiny category (100M - 1B), small (1-8B), Medium (8B-30B), Large (30-100B, often 70B Llamas [60]), Very Large (70-200B) and frontier-scale (200B+). This logarithmic difference is usually dictated by the hardware architecture of the use-case (edge, small gpu, big gpu, multi-gpu, multi-node, data center level compute...). The 70B models are not worth the squeeze of having to set up multi-GPU configurations, given these medium Qwen models are extremely competitive, parameter efficient and matching up with the large category. The other contender to the throne was Qwen 3.6 27B, but given our accelerators were able to handle the bigger version with higher benchmarks, we opted for it, given the difference in number of parameters is not significant relatively.

In order to guarantee that the KV-cache can fit comfortably for not-too-long sequences with total conversation length up to 16384 tokens (p95 was 2.8k tokens and max 3.9k) and that the token throughput is as high as possible, we employ quantized versions of the original model. The specific quantization instance that we used is **Q4\_K\_M** following the GGUF quantization naming convention. This version of the model, due to the aggressive quantization mechanism is able to fit comfortably in at most 23 GB of VRAM [61] (GGUF overhead). A KV-cache entry for this specific model [62] with a serving batch size of 1, with only 10/40 layers using gated full-attention (The rest is using Gated DeltaNet which uses a fixed recurrent state independent of context) of 2 KV heads with a hidden dimension size of 256 and fp16 (2 bytes) is (taking into account a factor of x2 because both key and values are stored):

$$\begin{aligned}\text{Total KV-cache size} &= 16384 \cdot 1 \cdot 10 \cdot 2 \cdot 2 \cdot 256 \cdot 2 \text{ B} \\ &= 16384 \cdot 20 \text{ KiB} = 320 \text{ MiB}.\end{aligned}$$

This can fit comfortably in a single A100 accelerator under 30 GB, with a frontier medium-sized

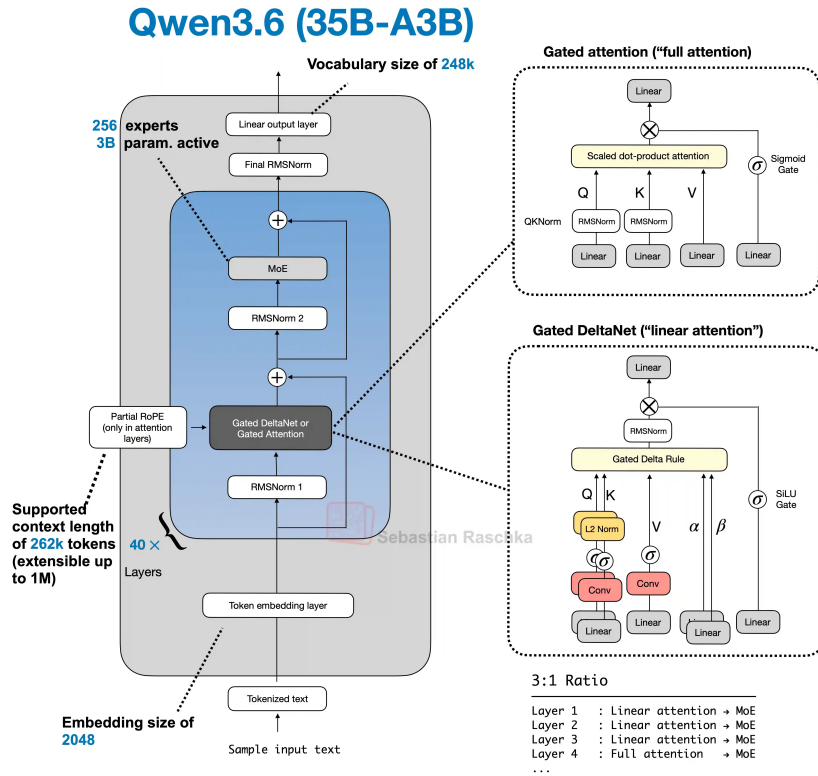


Figure 5.1: Qwen 3.6 35B-A3B block architecture.

model with an Artificial Analysis index of 32 [59], thus all the infrastructure and functional requirements can be met, with non-functional requirements also being met to a satisfactory degree.

## EMBEDDING MODEL

A critical model for the retrieval pipeline is the embedding model used to generate embeddings from text such as the ads. Contrary to the LLM choice, embedding models are really sensitive to the data type and are usually not quantized from the full precision versions (fp32), so their pressure on VRAM is bigger. In our case, the data type is English product text: short conversational queries against concatenated Amazon titles and features, encoded on every advertised turn before the first token. A generic STS MiniLM would be cheap and a 4B/8B embedder would buy leaderboard points and spend them as extra retrieval latency and VRAM that already belongs to the LLM. The field ranks that trade-off on the Massive Text Embedding Benchmark (MTEB) and its public leaderboard [63, 64]. This is why a natural choice for this setup was **Qwen3-Embedding-o.6B** [37], which was one of the strongest small open-source contenders during the development of this work: among sub-1B open weights it sat near the top of the board (MMTEB mean 64.3, retrieval 61.4), it is instruction-aware so the query and the catalogue

document can be encoded with different prompts, and it fits in about 2 GB of bf16 on the second GPU next to the reranker. (Similar to the LLM, new better models such as Nemotron-embeddings have emerged.) TTFT was prioritised over using a bigger version like 8B, whose scores are only marginally better.

#### RERANKER MODEL

However, a retrieval pipeline depending only on the retrieval quality of embeddings, especially using algorithms such as approximate nearest neighbours (ANN) or even exact inner product over  $10^5$  catalogue vectors, could not reach acceptable performance levels, given the recall-oriented nature of this step. Let the product catalogue be  $\mathcal{D} = \{d_1, \dots, d_N\}$ , the concatenated documents of §4.1.2. A bi-encoder encodes the query and each product  $d_i$  independently into dense representations  $\mathbf{e}_q$  and  $\mathbf{e}_{d_i}$ , respectively, so a single matrix–vector product can shortlist from the whole catalogue. A reranker is a cross-encoder: the query and candidate document  $d_i$  are concatenated, the Transformer attends across the pair, and a head on the joint [CLS] state emits a relevance score. In contrast to bi-encoder retrieval,

$$s_{\text{bi}}(q, d_i) = \mathbf{e}_q^\top \mathbf{e}_{d_i},$$

the cross-encoder jointly processes the pair and allows tokens from the query  $q$  to attend directly to tokens from the candidate  $d_i$ , producing

$$s_{\text{cross}}(q, d_i) = g_\theta([\text{CLS}], q, [\text{SEP}], d_i, [\text{SEP}]) .$$

Its runtime for  $k$  candidates of concatenated length  $L = |q| + |d_i|$  is  $T(k) = O(kL^2)$ . Self-attention is quadratic in the concatenated sequence length  $L$ , and the forward pass must be performed once per candidate. Thus, a shortlist of  $k$  candidates costs  $O(kL^2)$ , while scoring the full catalogue would cost  $O(NL^2)$  per query. With  $N \sim 10^5$ , this is why the cross-encoder never sees the full catalogue: we keep  $k = 30$  candidates from FAISS, score 10 pairs with a reranker, and inject the best. That is the dilemma: if  $k$  is too small, the right product never enters the shortlist and no reranker can recover it; if  $k$  is too large, subjects  $O(kL^2)$  pay bigger latencies in real-time. Dense retrieval is therefore allowed to be messier and recall-oriented; the reranker is allowed to be expensive and picky (precision-oriented), but only on a handful of survivors. Token-level attention inside  $g_\theta$  is what lets it distinguish a gaming laptop from a laptop bag when cosine similarity between independently encoded vectors may fail to do so.

For our pipeline, we chose **BGE-reranker-v2-m3** [65], a 568M XLM-RoBERTa cross-encoder, because among the open-source options it is one of the most lightweight and adds little overhead: about 1.1 GB in fp16, served in bf16 on the second GPU next to the embedder. We score product titles only, so each of the ten pairs is a short sequence; on an A100, this takes tens of milliseconds rather than seconds. The approximately 3 s wait introduced by the advertising turn is instead dominated by HyDE, not by

this stage. Better rerankers exist, but we found that they added further latency to the overall retrieval pipeline in the order of the second, as discussed in Section 5.3, making them unacceptable under the 10-second response-time constraint introduced previously.

### 5.2.3 INTENT MODEL

For the intent/genre model we choose ThradBERT [66], given it is the only genre-prediction open-source model to the best of our knowledge, trained on 13 conversation-genre clusters aligned with coarse chatbot-use taxonomies [30]. It is a DistilBERT sequence classifier (BERT WordPiece, 512-token context, about 67M parameters) with a 13-way softmax on the [CLS] state [67]: the same  $f_{\text{genre}} : \mathcal{U} \rightarrow \mathcal{G}$  as in section 3.1, just a small encoder rather than an LLM. Held-out accuracy is 83.8% on 2,224 samples; inference is under 50 ms on CPU, and the weights are about a quarter of a gigabyte in 32-bit precision. We run it on CPU every turn and only log the hard label; it does not gate advertisement serving in this experiment.

## 5.3 ADVERTISEMENT INSERTION AND RETRIEVAL PIPELINE

In order to construct a minimal viable ad insertion ranking pipeline, given we do not have any data to train specialized models, we have relied on embedding-based information retrieval to create a hybrid semantic retrieval to improve upon results of simple lexical retrievers such as TF-IDF or BM25 [68], as context and semantics in a realistic in the wild product dataset is crucial to deliver acceptable advertisements but keywords do also have an important weight. Note that this retrieval pipeline could benefit greatly from custom policy or ranking models beyond the generic re-ranker. Moreover, this section will go over the insertion mechanism that has been designed in order to integrate dynamically the retrieved product to form an advertisement.

### 5.3.1 RETRIEVAL PIPELINE

The retrieval pipeline is a progressive funnel that starts from the approximately 117,000 product embeddings precomputed offline and stored in the FAISS vector database, as described in subsection 4.1.3 [36], and ends with the single most promising product, which then feeds the insertion mechanism. Figure 5.2 summarises the funnel, whose stages are:

1. Query Expansion: instead of embedding the original user utterance, which can be short and devoid of context, we condition on the last four conversational turns and generate two hypothetical advertisements with the conversational LLM, following the Hypothetical Document Embeddings (HyDE) technique [69]. Each hypothetical document is embedded independently, searched separately, and the two hit lists are fused by Reciprocal Rank Fusion (RRF) [70] with the customary smoothing constant  $k_{\text{RRF}} = 60$ , rather than averaged in embedding space, which keeps a strong match in one document from being diluted by the other.

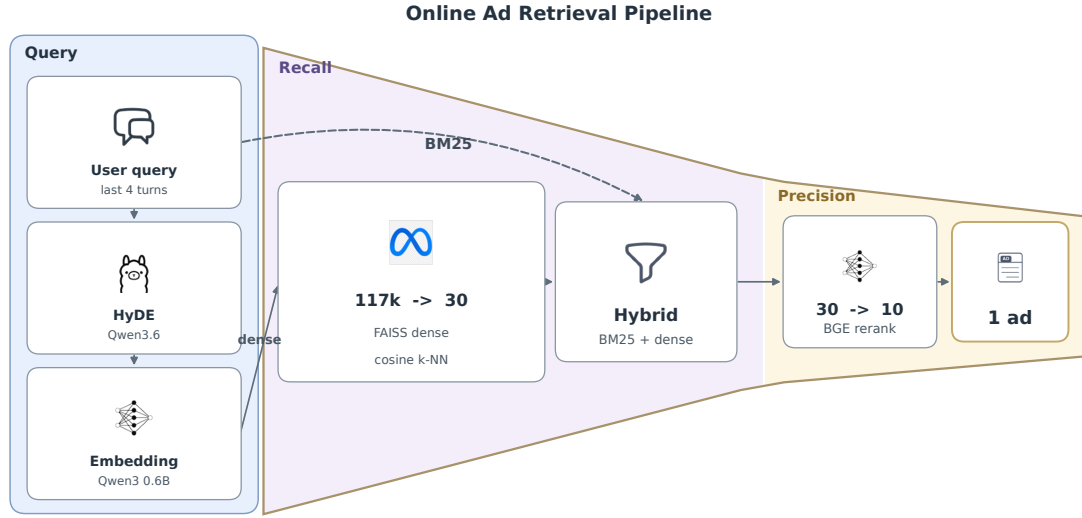


Figure 5.2: Three-stage retrieval funnel on an advertised turn.

2. Recall Phase: each expanded query vector is compared against the whole catalogue with the exact IndexFlatIP search of subsection 4.1.3, keeping the  $k = 30$  products of highest cosine similarity (this retrieval  $k$  is unrelated to the epoch count  $k = 37$  of subsection 4.2.4). This shortlist is then re-scored by the hybrid stage, which runs BM25 over the original user message for exact lexical matching and fuses those ranks with the dense ones under a semantically dominant 80/20 weighting ( $w_{\text{dense}} = 0.8$ ,  $w_{\text{bm25}} = 0.2$ ) plus an optional category filter. The shortlist is deliberately wider than what is finally served to avoid losing recall, so that a good match is not discarded early: a fitting product may sit at rank 15 under cosine similarity alone simply because it does not repeat the exact wording that rank 1 does.
3. Precision Phase: the cross-encoder of subsection 5.2.1 rescores the ten highest-ranked survivors of the fused shortlist and the top one is served. Because the pair is scored jointly, this stage is quadratic in the concatenated length and therefore affordable only on the handful of survivors of the recall phase. As above, this stage would benefit substantially from a ranking or insertion-policy model  $\pi$  trained for this task, rather than a general-purpose reranker.

The funnel is thus a deliberate trade between recall and precision under a latency budget: cheap, semantically permissive scoring is applied to the whole catalogue, progressively stricter and more expensive scoring only to the survivors, with lexical evidence mixed in where pure embedding similarity is known to be brittle (exact model names, brands, or product codes).

### 5.3.2 ADVERTISEMENT INSERTION MECHANISM $\mu$

Given a final candidate for a product obtained dynamically by the execution of the retrieval pipeline, the insertion mechanism  $\mu$  chosen is straightforward for both implicit and explicit presentation formats

$\lambda$ , although there is some degree of freedom and engineering in the implicit. Examples are shown in Figure 6.1.

- **Explicit Presentation:** Display the ad in a yellow-ish banner above the chat input widget with a static call-to-action ("Discover More") with an image of the product and the name highlighted as a URL, similar to traditional ad blocks in regular websites.
- **Implicit Presentation:** Introduce the product with details into the LLM prompt for that specific turn, forcing the advertisement in a rational and contextual manner in any part of the message, also highlighting the URL of the product to make it more obvious, in a fashion similar to how Retrieval Augmented Generation (RAG) inserts documents or extracts into the LLM responses.

## 5.4 DEPLOYMENT AND INSTALLATION

In this brief section we will cover at a high level how to operationalize this system for any research body or individual wanting to test it out. Although greater level of detail can be found in the official repository [71].

The minimal requirements to run this is to have a dedicated server with 8+ CPU cores, and at least 1 GPU capable of serving a small LLM with a human acceptable latency, akin to Atlas in our Design. Nowadays the market of open-source small LLM models is expanding, so there are many choices between 1B - 35B parameters and even quantized versions which make some huge models like ours, easy to fit in a single 40 GB A100. However, it is understandable that this model will not be accessible for everyone, even with tiny KV-caches of sequences shorter than 1024 (which is tiny, even for this experiment that uses short lengths up to 16384). So lighter setups with less VRAM can opt for using smaller models although intelligence and QoS may degrade rapidly. The deployed version also accounts for a public domain and port-forwarding in order to expose it to the internet and make it accessible outside the local machine.

For a full deployment with EEG, it would also require a 32-channel actiCHamp device with its proprietary software as it has been outlined in section 4.2.4, which can be connected to the same machine running the server, but it may also increase the overhead.

### 5.4.1 FINAL DEPLOYED SYSTEM

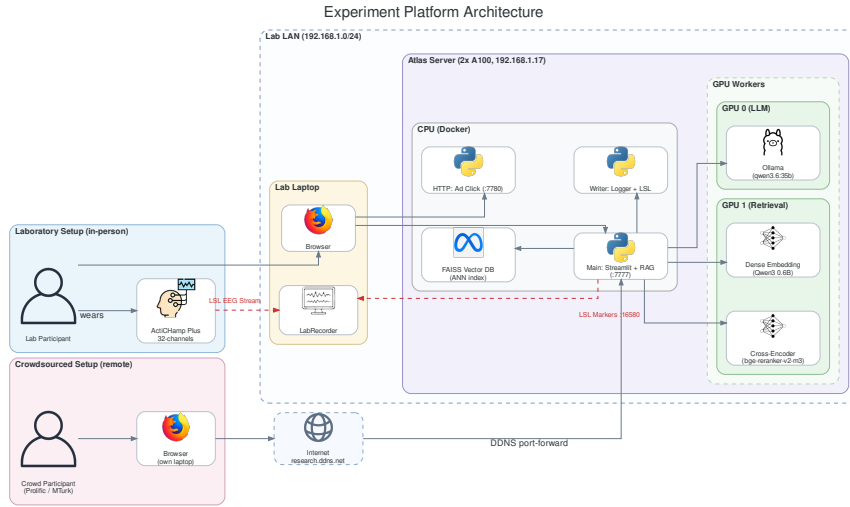
The whole system deployment can be realized in a single 2xA100 GPU cluster or even in more modest setups like 1xL4/A100. Figure 5.3 shows a summary of the 3 physical components of the laboratory

study (system that is a strict superset of the crowd sourcing study system, which relies neither on the laboratory laptop nor on the actiCHamp device):

- Participant Setup: Subject with a keyboard, mouse, monitor and optionally EEG device to record. In the case of a crowdsourcing run, it would require the participant's own laptop with a browser to connect to the exposed server UI.
- Laboratory Laptop (Lab Only): A computer with browser connection to open the exposed UI and the previously mentioned peripherals to enable human-computer-interaction, with a direct connection to the EEG sensors through the different proprietary software used for calibration, receiving markers from the server and structured logging of the signals.
- Telefonica Atlas Server: GPU enabled server capable of running the conversational LLM and the rest of models, and provides the front-end and the back-end of the webpage accessed by the subject. Notice the multithreaded nature of the platform, reason behind the minimum requirements.

Another crucial point about the system design is the network topology from our 2 main user workflows (crowd-sourcing prolific users and laboratory subjects).

- Crowd-sourcing Workers: They must be able to connect to the server from any location of the Internet, so this requires a domain with fresh DNS entries and the exact ports (e.g. UI 7777) exposed externally.
- Laboratory Subjects: Given both laptops are in the same high-speed network, they can communicate internally and most importantly send LSL requests without needing to go through multiple hops, ensuring minimal mismatches.



**Figure 5.3:** Physical and network architecture of the laboratory and crowdsourced deployments.

## 5.5 LIMITATIONS AND FUTURE WORK

Despite the merits and robustness of this agentic system, strong production systems evolve through the iterative identification of bottlenecks and growing complexity only when the difficulty of the requirements grow over time too, to avoid scope creep. Below, our advice for future directions of improvement of this open-source work is presented:

- **LLM response throughput:** Given the lack of parallelism achieved by the Ollama implementation used in this work, the experiment suffers from scalability if repeated with a bigger number of subjects, as now given this seriality it is recommended not to go beyond 1, and at most 2 users, as they will starve each other's responses. This could be solved with an implementation of vLLM and even more lazily with a small load balancer logic and multi-GPU replication of the model, although again this is not the most scalable approach compared to optimized vLLM inference with batch inference.
- **Vector Database:** If the catalogue of products were to grow, there would be a point where the in-memory FAISS database would not be sufficient for the needs of this system and would either take too much memory (either RAM or VRAM) or simply not fit, forcing the use of more complex databases such as Qdrant or Pinecone.
- **User Interface:** The user interface although sufficient for our experiment, it is too limited by the nature of Streamlit, so it hinders the tracking of metrics and limits drastically some customizations that would require more advanced but more complex frameworks, and possibly a stricter division between back-end and front-end.

- More modalities: The LLM was limited to text only responses, while we could have orchestrated multiple models or use advanced multimodal LLMs to enable image generation and audio communication through the user interface. However this poses not only more complexity in managing more models but also changes in the overall experimental flow.
- Retrieval Pipeline: As mentioned in section 9.2, only human qualitative assessment has been performed to iterate throughout the hyperparameters of the pipeline and to drive the design of it, so a deeper ablation study on its metrics and via benchmarks (e.g. the RAGAS framework) would highly benefit it for future iterations to enable a more data-driven improvement.

Note that this is only related to the improvement of the system; general thesis future work is in section 9.2.

# 6

## Methods

This chapter fixes the experimental design, the instruments, and the pre-specified analysis plan: what was manipulated within participant, how the workflow limits response bias, and which estimands enter each analysis family.

### 6.1 ADVERTISEMENT ALGORITHM

This experiment does not learn a serving policy,  $\pi$  is fixed in advance as described in 5.3: each four-turn conversation contains at most one advertisement, served after turn 2 (early) or after turn 4 (late), or not at all ( $a^\emptyset$ ). Retrieval fills the product from the current session only ( $\varphi = \text{session}$ ). How that retrieval is implemented, and which models sit behind it, is chapter 5; the catalogue is section 4.1.

### 6.2 STUDY DESIGN

#### 6.2.1 CONDITIONS AND INDEPENDENT VARIABLES

The study is a  $2 \times 2$  factorial design with a no-ad control, manipulating two factors within participant. The instance-level factor is presentation  $\lambda \in \{\text{implicit}, \text{explicit}\}$ , realising  $a^{\text{imp}}$  and  $a^{\text{exp}}$  (3.11)–(3.12). The policy-level factor is insertion timing, early at turn 2 versus late at turn 4, decided by the fixed serving policy  $\pi$ . Crossing them yields the four advertisement conditions  $a_2^{\text{imp}}$ ,  $a_4^{\text{imp}}$ ,  $a_2^{\text{exp}}$ , and  $a_4^{\text{exp}}$ . The fifth condition serves  $a^\emptyset$  (3.13) and provides the within-participant contrast against which advertisement effects are evaluated. These five conditions are the ones referred to throughout the rest of this chapter.

## TIMING OF ADVERTISEMENTS

Insertion timing is a decision of the serving policy  $\pi$ , not a coordinate of  $a_k$ . Each advertisement trial is a four-turn conversation and contains at most one advertisement. Early insertion serves the advertisement after the second user utterance ( $a_2$ ); late insertion serves it after the fourth ( $a_4$ ). The null condition serves  $a^0$  at every turn (3.13).

These two slots are the beginning and the end of a short chatbot session of the kind consumers actually hold, not two points on a long agentic trajectory. That such sessions are short is not an impression but a measured property of large interaction logs: WildChat, a corpus of 1,009,245 ChatGPT conversations from 204,736 users, averages 2.52 turns, with only 41% of conversations reaching a second turn at all and 3.7% running past ten [72], while LMSYS-Chat-1M reports a mean of 2.0 turns across a million conversations with 25 models [73]. Conversational recommender studies likewise find that preference resolution often occurs within a few turns [16, 15]. Survey evidence places that behaviour in a consumer rather than a workplace setting: in a national sample of language-model users, personal, informal learning and planning is the main purpose for 51%, against 24% for work activities [5]. A four-turn guided task therefore already sits in the upper tail of everyday chatbot use, which is why late insertion is the final turn rather than a later slot that most users would never reach.

The same contrast is the one suggested by advertising serial-position effects: the first and last positions in a sequence are the theoretically privileged slots, whereas middle positions are weaker [74, 75]. Hsu and Karahalios further show, in video, that *when* an advertisement appears can matter more for experience than whether the user chose that moment, and that mid-content interruption is more aversive than a pre-roll [76]. Turn 2 is the conversational analogue of an early placement after the task has started; turn 4 is the terminal, decision-stage placement, immediately before the written conclusion.

Turn 1 is left advertisement-free so that participants can orient to the task and so that retrieval can use session evidence ( $\phi = \text{session}$ ) rather than a cold first utterance. A third timing level (for example turn 3) would have reduced power in a five-condition within-participant design without increasing the temporal distance; the two-level contrast therefore uses the earliest and latest feasible slots. Triggers are turn indices rather than clock time so that every participant, including fast ones, has a defined insertion opportunity and so that  $a_2$  and  $a_4$  remain comparable across users.

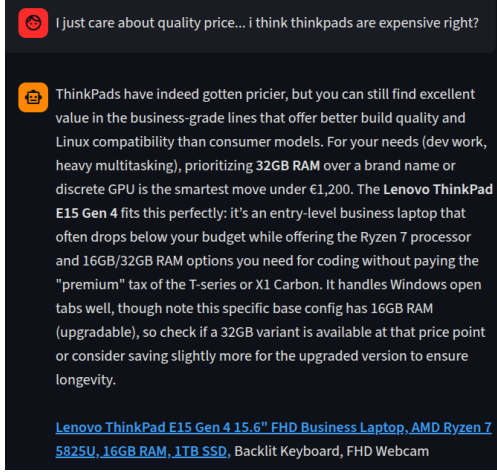
A local post-advertisement user utterance exists only after early insertion. Late advertisements have no subsequent user message, so turn-level genre-shift estimands  $\delta_k^{(a)}$  are defined for  $a_2$  and not for  $a_4$  in the case of trajectories.

## PRESENTATION $\lambda$

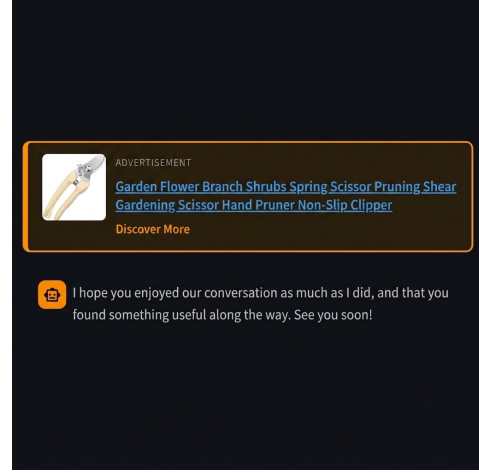
A crucial part of both the experimental methodology and the system design is integrating the advertisement into the user interface in a natural way. The taxonomy in section 3.2 spans several dimensions, but this study manipulates only the presentation format  $\lambda$ . It is one the most discriminative of those dimensions, in that it changes whether the advertisement is separable from the assistant’s answer at all, and it is therefore the dimension we expect to bear most directly on the research questions.

- **Implicit** ( $a^{\text{imp}}$ , Figure 6.1a). The advertisement is woven into the assistant’s own reply at the assigned turn  $k$  and remains in the transcript for the rest of the session. The only cue that a passage is sponsored is a product URL. The position within the reply is not fixed in advance, because the language model retains some freedom over where the mention falls, so it may appear at the beginning, in the middle, or at the end. Timed early or late, these are  $a_2^{\text{imp}}$  and  $a_4^{\text{imp}}$ .
- **Explicit** ( $a^{\text{exp}}$ , Figure 6.1b). The advertisement appears at the assigned turn  $k$  in a separate, persistent banner outside the conversation, placed directly above the message box. Each banner is a yellow panel roughly two message-lines tall, headed by the disclosure label “Sponsored” and carrying a product image, the product name, a short description with a URL, and a brief call to action such as “Discover more”. The format is deliberately unambiguous, and it is the contrast with the implicit format, rather than the banner design itself, that the study manipulates. Timed early or late, these are  $a_2^{\text{exp}}$  and  $a_4^{\text{exp}}$ .

Both formats drew on the same catalogue of real products, derived from the Amazon Reviews 2023 product metadata [34, 35], so that presentation and product content are not confounded. Again, implicit refers to the presentation  $\lambda$  only: the product is named in plain, readable text inside the reply, in covert (informational) wording, exactly as the explicit banner is.



(a) Implicit ( $\mathcal{A}^{\text{imp}}$ ).



(b) Explicit ( $\mathcal{A}^{\text{exp}}$ ).

Figure 6.1: Advertisement presentation formats ( $\lambda \in \{\text{implicit}, \text{explicit}\}$ ) used in the experiment.

## 6.2.2 DEPENDENT VARIABLES

### BEHAVIOURAL MEASURES

Two kinds of behavioural data were collected: self-report from questionnaires, and interaction data logged by the system. After each condition, participants completed a 22-item Likert questionnaire (1–7) in three sections: (1) perceived assistant qualities (credibility, helpfulness, convincingness, relevance, and neutrality), (2) felt influence and advertisement notice (product or brand mention; sponsored buttons), and (3) perceived commercial pressure (pushing or marketing; steering rather than assisting). Trust in the chatbot was rated in the same battery. The wording of every item is in section C.3.

The outcomes were fixed and planned in advance and they are the scores the behavioural family of section 6.3 refers to. Credibility is the mean of the reliable-responses item after reversing the false-information and made-up items. Perceived manipulation is the mean of the pushing/marketing and steering items. Direct chatbot trust is reported separately and is not folded into credibility. Notice uses the brand-mention and sponsored-button items; brand mention alone is not treated as advertisement detection. Cued memory is the recall item “I feel I remember this content well”; it is cued self-report after the advertisement is shown again. Internal consistency on the 270 condition rows is acceptable for the three primary multi-item outcomes (Cronbach’s  $\alpha$ : credibility .78, perceived manipulation .86, notice .61) and for the secondary qualities (helpfulness .90, relevance .87, convincingness .74, neutrality .63); the full table with arm splits is Table E.3.

After all five conditions, cued recall asked, for each advertisement instance except  $a^0$ , how well the

content was remembered and whether seeing it again shifted trust, plus an open reaction (subsection C.4.1). From the interaction we logged reply latency and the free-text findings (section C.2); advertisement clicks were recorded but none were observed.

## CONVERSATIONAL OUTCOMES

Two behavioural outcomes come from the conversation itself rather than from a questionnaire. The first is the genre trajectory. A genre model  $f_{\text{genre}}$  reads one user utterance and returns one genre, and Definitions 1–6 of chapter 3 turn the resulting sequence into the quantities tested here: the transitions and shift count, persistence, genre frequency, diversity, and the advertisement-associated shift  $\delta_k^{(a)}$ . The primary context is the bare utterance,  $\hat{g}_k = f_{\text{genre}}(u_k)$ ; the longer window the running system classified is the deployed context and is kept as a sensitivity analysis (subsection 4.2.3).

The second is free text: the findings box completed after each condition (section C.2) and the open reaction in the cued-recall battery (subsection C.4.1). Both were collected and are not analysed in this thesis due to time constraints, so no sentiment or textual outcome is defined on them and no claim rests on them.

## EEG MEASURES

The laboratory recording yields sixteen spectral measures per 4 s epoch: five absolute band powers, the five matching relative powers, two regional measures, and four asymmetry or engagement indices. Each one is defined in section D.2, together with what it is conventionally taken to index and its analytic tier. Absolute powers are on a power-decibel scale, so a difference in decibels is a power ratio rather than an additive change.

Two of the sixteen are a priori the most interesting for our use case: Fz theta is 4–8 Hz power at the frontal midline electrode Fz, which rises with effortful cognitive control [77]; posterior alpha is 8–13 Hz power averaged over O1, Oz, O2, P3, Pz, and P4, which falls as visual attention engages the display and rises as attention withdraws from it [78]. The other fourteen are exploratory, with their finer tiers in section D.2, and no claim rests on them. All sixteen are carried by both EEG tables, condition aggregation and onset-locked (subsection 4.2.4), which section 6.3 treats as two separate estimands.

### 6.2.3 LABORATORY CONDITIONS

Nineteen participants took part in the laboratory study, nine women and ten men; eighteen enter the finished behavioural sample once the session that ran the crowd protocol is excluded. Ages, recorded at intake, ranged from 21 to 42 years ( $M = 29.50$ ,  $SD = 5.01$ ). Eligibility required participants to

be at least 18 years old, to have normal or corrected-to-normal vision, and to be proficient in English, so that task instructions and questionnaire items would be understood consistently. Participants were excluded if they reported a history of neurological or psychiatric disorder, or current use of medication known to interfere with physiological recording. All participants gave informed consent before taking part and received a shopping voucher as reimbursement. Sex, education, and chatbot-use counts for the finished sample, including the crowd arm, are reported in section 7.1.

#### 6.2.4 EXPERIMENTAL DESIGN

##### LABORATORY ARM

The workflow laboratory subjects must undergo is summarised in Figure 6.2. It opens with an onboarding phase in which the participant completes a mandatory intake questionnaire and signs an informed-consent document while the experimenters fit the EEG cap and start the recording software. The participant then completes a 30-second baseline recording and a warm-up chat to become familiar with the interface. Throughout the session the participant reaches the Atlas server over the intranet from the laboratory laptop, which also records the EEG (chapter 5).

The second phase is the condition loop, and it carries the experimental manipulation defined in subsection 6.2.1. All five conditions were completed by every participant, so the manipulation is within-subjects; the study as a whole is mixed, because the laboratory and crowd arms are separate groups. Condition order was randomised per participant to limit order and learning effects. Each condition begins with a task briefing, consisting of the task prompt and a warning that the conversation is independent of the others, and continues with a four-turn conversation containing at most one advertisement, a free-text findings response of at most 256 characters, and a 22-item Likert post-task questionnaire in three sections.

The third phase follows the condition loop and collects the retrospective measures: a cued advertisement-recall post-experiment questionnaire for each of the four advertisement conditions, excluding  $a^0$ ; the BFI-10 personality inventory on a five-point scale [1]; and a short demographics questionnaire included for redundancy against the intake form. The fourth and final phase presents a deception disclosure, a last opportunity to withdraw, and a closing screen. Every instrument as administered is reprinted in Appendix C. Each session produced multimodal data as mentioned in section 4.2, and together they are the entire record from which the analyses are built.

## CROWD ARM

The crowd arm ran the same five conditions ( $a_2^{\text{imp}}, a_4^{\text{imp}}, a_2^{\text{exp}}, a_4^{\text{exp}}, a^\emptyset$ ), the same 22-item post-condition questionnaire, and the same cued-recall battery as the laboratory arm. Its workflow (Figure 6.3) differs only where recruitment through Prolific makes a change necessary: participants must be traceable to the recruitment platform, and low-engagement participation must be detectable without an experimenter present. Crowd participants reached the Atlas server from their own devices. Three adaptations follow.

- **The EEG baseline phase is removed.** With no recording to baseline, that slot instead collects the participant’s Prolific identifier, so every valid run can be traced to its owner.
- **A validation questionnaire is added** at the end of the session. Participants identify the themes of the five tasks they completed from ten options, five of which are fabricated. This provides an attention check of the kind standard in crowd-sourced work, which almost everybody answered right 5/5, so even a single wrong answer became an outlier in the distribution to be inspected with further detail.
- **The closing screen is modified** to display a run identifier, which participants paste into Prolific. Reward follows verification that the identifier matches a completed run for that Prolific account and thus is eligible to receive the payment.

Each run writes only a JSON event log that is later preprocessed into the analysis data. Because the adaptations touch only recruitment and quality control, the crowd arm yields the same core behavioural measures, and the absence of EEG is its only substantive difference from the laboratory arm. This split is deliberate: the crowd arm supplies the behavioural sample size that a laboratory-only study could not afford, while the laboratory arm supplies the neurophysiological evidence, and together they balance the statistical and economic constraints.

## TASKS, ASSIGNMENT, AND EXCLUSIONS

Each participant completed five simulated work tasks, one per condition: choosing a gardening gift, choosing a laptop under a budget, improving a home study setup, restarting a fitness routine, and choosing a pet and its basic setup. Tasks were rotated in Latin-square order; the five conditions were shuffled independently; the two sequences were then paired so that, across participants, task identity is not confounded with condition. The realised plan is written into the session log; the realised task  $\times$  condition and position  $\times$  condition counts are in Table E.11 and Table E.12, and neither departs from uniformity. The briefings and the language-model prompts are reprinted in Appendix A and Appendix B.

Unfinished, synthetic runs (i.e. experimental runs with agents), and beta-tester sessions are excluded from the behavioural analyses, as is the laboratory session that ran the crowd protocol. Sessions labelled unfocused are retained when they completed the instruments. The inferential unit is the participant.

### Laboratory participant flow · in-person, 32-ch EEG, LSL markers

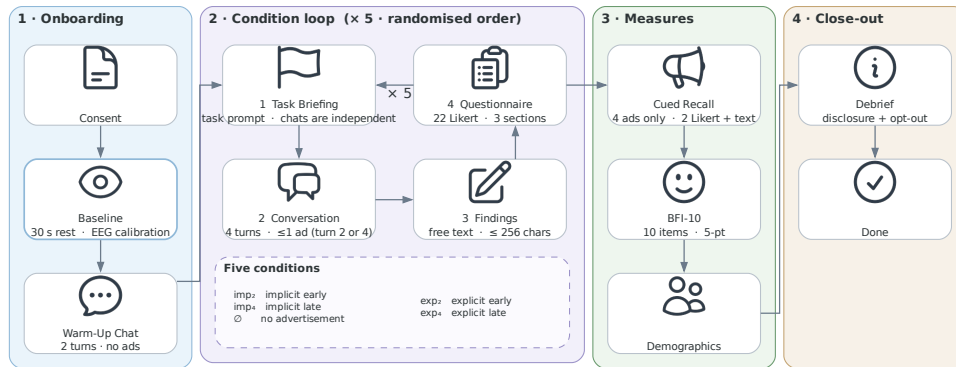


Figure 6.2: Laboratory participant flow, including EEG setup and cued recall after the five-condition loop.

### Crowd participant flow · remote, own browser

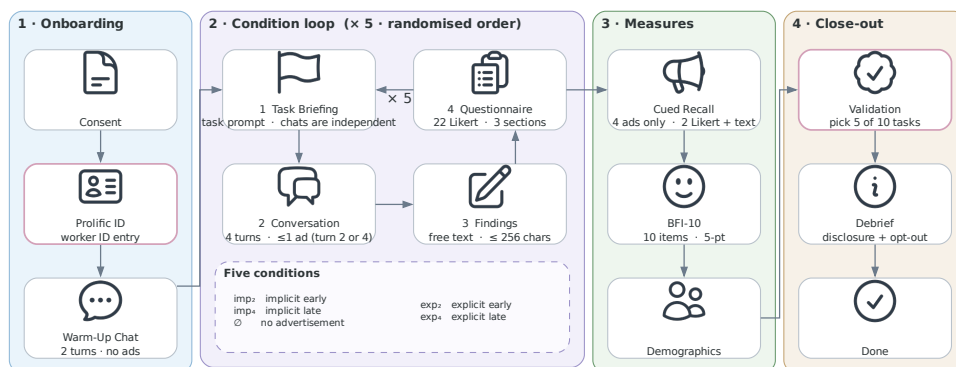


Figure 6.3: Crowd participant flow, adding Prolific identification and close-out validation; no EEG.

### 6.2.5 FLOW DESIGN AND BIASES

This section presents, from a high-level viewpoint, the decisions behind the workflow used in this experiment: the order of the questionnaires, the order of the phases, why the BFI-10 is administered near the very end, and why several items were reworded. Two kinds of problem are at stake, and they are not the same. The first is psychological, in that the instrument can itself change what a participant believes or reports, and the ordering below is chosen to limit that. The second is structural, in that some quantities cannot be separated by any ordering, because the design either confounds them or leaves them undefined. The first kind is mitigated; the second can only be declared. The experimental workflow was designed to minimise sources of response bias, measurement contamination, and statistical confounding.

#### PSYCHOLOGICAL BIASES THE FLOW IS DESIGNED TO MITIGATE

- **Anchoring on the first condition.** A short warm-up chat free of advertisement provides an initial impression, and a prominent warning before every condition makes clear that each conversation is a different chatbot, so that judgements of condition one do not anchor those that follow and carry-over effects are mitigated.
- **Unformed opinion at rating time.** Before the post-task questionnaire, participants write their findings in at most 256 characters. This gives them a settled view to rate against rather than an improvised one. The text itself is collected and is not analysed in this thesis.
- **Reducing demand characteristics, priming and hypervigilance.** The questionnaire derives mostly from [18], whose participants repeated a single condition. Items that would prime advertisement seeking across our five conditions were reworded or dropped: asking outright whether the chatbot manipulated the user is a strong statement that can fixate attention on that possibility and distort every later score. The questionnaire is also gated into three parts (quality, then notice and trust, then manipulation) so that the manipulation items cannot colour responses to the earlier ones. The trade-off is that gating within a condition cannot prevent influence across conditions. Items as administered are in section C.3.
- **Disclosure of the manipulation mid-study.** Cued recall is deferred until after all five conditions, so that the study never reveals which condition was  $a^0$ , and never signals to participants who failed to recognise an advertisement that there was one to recognise. Moreover, we avoid recency biases and force subjects to use active recall. An open-text item captures the overall feeling of each condition, again to avoid priming. Items are in subsection C.4.1.
- **Fatigue on perceptual items.** Demographics are asked last, because factual self-description is insensitive to fatigue in a way that perceptions and feelings are not.
- **Reactivity to self-description.** Personality is measured after the outcomes, so that answering trait items does not condition the session, particularly for participants high in neuroticism. Self-perception is stable over the short term and, like demographics, tolerant of mild fatigue, so the end of the session costs little and risks less.

## STRUCTURAL CONFOUNDS THE DESIGN CANNOT REMOVE

These are properties of the design rather than of its ordering. No sequencing of instruments removes them, so they are stated here and carried into section 8.7.

- **Disclosure is confounded with presentation.** Realised disclosure  $\theta$  co-varies with presentation  $\lambda$  rather than being crossed with it: the explicit banner is labelled, the implicit mention is not. The design therefore cannot apportion any memory advantage of the explicit format between the presentation itself, the disclosure label, and the mere availability of a separable object. Heineking explicitness  $\varepsilon$  is held covert rather than manipulated, and personalisation  $\phi$  is session-level and does not use BFI-10 scores.
- **One estimand is undefined by construction.** Late advertisements have no following user utterance, so  $\delta_k^{(a)}$  does not exist after turn 4. This is structural missingness, not attrition, and the late conditions therefore serve as a negative control instead.
- **The advertised turn waits longer.** Retrieval adds about three seconds of silent wait before the advertised turn (median 3.03 s, subsection 5.1.2), so the any-advertisement contrast and the onset-locked comparisons against  $a^0$  compare advertisement plus wait against neither. The wait is the same procedure on every advertised turn whatever the format or the moment, so the format and timing contrasts hold it constant. No logged measure separates the conditions (subsection 7.2.2), but the wait is a co-intervention the design cannot remove.
- **EEG cannot speak for the crowd arm.** Only the laboratory arm was recorded, so any analysis involving a spectral measure is laboratory-only ( $n = 18$ ). Such results are associations rather than mediation claims.

## 6.3 STATISTICAL FRAMEWORK

This section fixes what is tested, on whom, and against what threshold, before chapter 7 reports any of it. Six conventions hold throughout.

- **Design.** Everyone completes all five conditions, so every contrast tested below is within participant; the order randomisation and Latin-square task rotation that keep condition unconfounded with task are given in section 6.2. The study as a whole is mixed, since the laboratory and crowd arms are separate groups.
- **Sample.** The  $N = 54$  finished participants,  $n = 18$  laboratory and  $n = 36$  crowd. Only the laboratory arm was recorded, so every family involving EEG is laboratory-only and runs on those eighteen participants.
- **Unit.** The participant. An epoch, a turn, or a questionnaire item is a measurement, not an independent observation, so measurements are aggregated within participant before testing and  $n$  counts people [38]. The permutation of  $g^{(a)}$  is the one conversation-level exception.

- **Status.** The sample size was set by recruitment rather than by an a priori power calculation, and the study was not pre-registered. Two complementary views of the same scores are reported throughout: the planned confirmatory contrasts, whose measures and weights were fixed on theoretical grounds before the corresponding results were read, and the post-hoc pairwise sweeps that localise a marginal to the condition driving it. Every other test carries one of the labels set out below.
- **Threshold.**  $\alpha = .05$ , two-sided unless a test is declared otherwise, with Holm-adjusted  $t$   $p$  values interpreted against that nominal  $\alpha$  [79]. Holm is the only correction used. It runs within the family named in Table 6.1 and never across families, and the  $p$  values of the nonparametric sensitivity tests are reported raw. “Holm  $p$ ” always adjusts the  $p$  of the family’s declared estimator: the paired  $t$  for contrasts, Spearman’s  $\rho$  for associations, and the joint Wald test for the demographic check.
- **Reporting.** Every estimate carries a 95% interval and an effect size wherever the estimator defines one.

Four labels are used for a test in this thesis, and they are not interchangeable. *Confirmatory* means the measure, the contrast, and its weights were fixed on theoretical grounds before the corresponding result was read, and the test is Holm-corrected within its family; the study was not pre-registered, so the label records a planning decision, not a registration. *Exploratory* covers the fourteen EEG measures and the whole-conversation trajectory summaries; those tests are Holm-corrected too. *Post hoc* means added after the planned contrasts had been read, which is the pairwise sweeps and the condition-by-condition localisation. *Sensitivity* means the same estimand under a different estimator or a different processing choice: the Wilcoxon signed-rank test, the mixed model, the deployed context, and the other epoch widths.

Those conventions fix who is counted and against what threshold, but not how a participant’s five condition scores become something a test can read. For the participant-level analyses, each test is built from participant-level summaries or planned contrasts. The rule for collapsing five scores into one would be a researcher degree of freedom if it were chosen after seeing the data. It is therefore set out once here, in a form that the confirmatory EEG families and most trajectory families share; the other estimators are the ones named in Table 6.1.

Take one outcome, say Fz theta, and one question about it, say whether advertisements shift it at all. Participant  $i$  holds one score  $y_{ic}$  in each of the five conditions, and the planned contrast is

$$D_i = \sum_c w_c y_{ic}, \quad \sum_c w_c = 0. \quad (6.1)$$

For the question just posed the weights are  $w = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -1)$ , making  $D_i$  that participant’s mean

across the four advertisement conditions minus their own  $a^0$  score; for a two-cell comparison they are  $w = (+1, -1)$ , an ordinary paired difference. Because the weights sum to zero, any additive person-specific baseline cancels, and  $s_D$  is the between-participant spread of those differences.

For example, the one-sample  $t$ , the null is  $H_0 : \mathbb{E}[D_i] = 0$ ,  $t = \sqrt{n} \bar{D} / s_D$  on  $n - 1$  degrees of freedom, reported with a 95% interval on  $\bar{D}$  and with  $d_z = \bar{D} / s_D$ , the paired difference expressed in units of its own between-participant spread. Holm is applied only to those  $t$  tests, within the family named in Table 6.1. Throughout,  $n = 18$  wherever EEG is involved and  $n = 54$  otherwise.

A paired  $t$  is used on questionnaire outcomes because the quantity tested is not a single Likert rating which would probably require other statistics and tests.  $D_i$  averages several items within an outcome and several conditions within a participant, so it takes many values on a bounded scale and its between-participant distribution, which is all the  $t$  test reads, is approximately continuous. Two checks are carried against that assumption rather than argued away: the Wilcoxon signed-rank test, which ranks the same  $D_i$  and requires only a symmetric distribution of the differences rather than normality, is the non-parametric sensitivity, and a random-intercept linear mixed model on the raw condition scores  $Y_{ic}$ , a regression that gives each participant their own baseline, is the adjusted check. A Wilcoxon  $p$  is reported raw, is never pooled with the  $t$  tests into one family, and is therefore not a Holm  $p$ .

Table 6.1 gives each outcome family of subsection 6.2.2 its  $n$ , estimator, and correction family. The association rows use Spearman's rank correlation  $\rho$ , which measures monotone covariation between two person-level scores and nothing more. Because two EEG person-level scores are defined below, each behaviour  $\times$  EEG pair is evaluated on both, Holm within the six for each score, and a cell that survives is also reported Holm-corrected across the twelve.

**Table 6.1:** Analysis families, estimators, and correction families.

| Family                                         | $n$ | Estimator                                                                                                                                                                                                                                            | Correction family                                                                                                                                      |
|------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Behavioural battery                            | 54  | one-sample $t$ on $D_i$ of (6.1); Wilcoxon sensitivity; random-intercept linear mixed model on $Y_{ik}$ as an adjusted check                                                                                                                         | 3 planned contrasts within each of the four primary outcomes; 2 within each of the two cued-recall items                                               |
| Personality moderation                         | 54  | random-intercept linear mixed model with BFI-10 trait $\times$ contrast interactions, arm, sex, familiarity, and frequency of use as covariates (a missing answer is its own level, so $n$ stays 54)                                                 | the 15 trait $\times$ contrast interactions, within each primary outcome                                                                               |
| Demographic moderation (check, not a family)   | 54  | the same mixed model with one demographic factor at a time (sex, education, familiarity, frequency of use, environment) crossed with the three planned contrasts; cell = joint Wald test of the factor $\times$ contrast terms; age was not recorded | Holm within outcome across the 15 factor $\times$ contrast cells (10 for cued memory), 70 tests under each of two level codings; reported in Table 7.9 |
| Free-text findings and open recall reaction    | 54  | collected, not analysed                                                                                                                                                                                                                              | none                                                                                                                                                   |
| Dataset A, condition aggregation               | 18  | one-sample $t$ on $D_i^A$ ; Wilcoxon sensitivity                                                                                                                                                                                                     | 3 contrasts, separately within each measure                                                                                                            |
| Dataset B, onset-locked                        | 18  | one-sample $t$ on $D_i^B$ ; Wilcoxon sensitivity                                                                                                                                                                                                     | 4 contrasts, separately within each measure                                                                                                            |
| Positive control, writing versus reading       | 18  | one-sample $t$ on $D_i = y_i^{\text{write}} - y_i^{\text{read}}$ ; Wilcoxon sensitivity                                                                                                                                                              | the 2 confirmatory measures                                                                                                                            |
| Exhaustive pairwise, post hoc                  | 18  | the same $t$ on every pair of conditions; Wilcoxon sensitivity                                                                                                                                                                                       | 10 pairs for Dataset A and 6 for Dataset B, separately within each measure                                                                             |
| Genre trajectories, $\delta_2^{(a)}$           | 54  | exact McNemar; paired $t$ interval on $D_i = \delta_{2,i}^{(a)} - \delta_{2,i}^{\emptyset}$                                                                                                                                                          | 4 contrasts                                                                                                                                            |
| Genre trajectories, late $N_{\text{shift}}$    | 54  | paired $t$ on $D_i = N_{\text{shift},i}^{(a_4)} - N_{\text{shift},i}^{\emptyset}$ ; Wilcoxon sensitivity                                                                                                                                             | 3 contrasts                                                                                                                                            |
| Genre trajectories, $\tilde{\delta}_2^{(a)}$   | 54  | one-sided permutation of $g^{(a)}$ across the 108 early-advertisement conversations of those participants                                                                                                                                            | single test, uncorrected                                                                                                                               |
| Genre trajectories, $\hat{g}_1$ vs $\hat{g}_4$ | 54  | paired $t$ on per-person genre shares; Wilcoxon sensitivity                                                                                                                                                                                          | the six genres that carry $\geq 5\%$ of utterances                                                                                                     |
| Behaviour $\times$ EEG                         | 18  | Spearman on person-level any ad minus no ad $D$ pairs, laboratory arm only, on each of the two EEG scores $D_i^A$ and $D_i^B$                                                                                                                        | 6 pairs: Fz $\theta$ and posterior $\alpha \times$ trust, credibility, manipulation; Holm within six for each score                                    |
| Behaviour $\times$ trajectory                  | 54  | Spearman on person-level $D$ pairs, both arms, the same pooled weights on both sides                                                                                                                                                                 | 6 pairs: the same 3 outcomes $\times \delta_2^{(a)}$ (early pooled) and late $N_{\text{shift}}$                                                        |
| Trajectory $\times$ EEG                        | 18  | Spearman on person-level $D$ pairs, laboratory arm only, early pooled minus $a^{\emptyset}$ on both sides                                                                                                                                            | 2 pairs: Fz $\theta$ and posterior $\alpha \times \delta_2^{(a)}$                                                                                      |
| Behaviour $\times$ trajectory $\times$ EEG     | 18  | Spearman among the three person-level $D$ s above on the laboratory arm; each pair also partialled on the third                                                                                                                                      | the 3 pairwise associations among them                                                                                                                 |

Holm is preferred to the other standard corrections for three reasons [79]. It controls the family-wise error rate within the declared family, so the guarantee is about the family a claim is drawn from; it assumes nothing about how the tests inside that family depend on one another, which matters because planned contrasts on the same five conditions are correlated by construction; and at the same level it rejects at least as often as Bonferroni, so the stricter guarantee costs no power against that alternative. Correction never runs across families, and on EEG it never runs across the sixteen measures. That is a choice of family, not a claim that Holm would fail under dependence: each measure is its own declared family of planned contrasts, so the guarantee is per measure, and a family spanning all sixteen would answer a different question, whether any measure moves at all, which is what the uncorrected exploratory board is for. The five relative powers share a denominator and are therefore strongly dependent (section D.2), which an across-measure family would count as five separate chances.

Two EEG estimands are declared because the recording answers two different questions. Condition aggregation (Dataset A) asks whether a participant’s sustained state over a whole condition differs when an advertisement is present. The onset-locked contrast (Dataset B) asks whether the moment the advertisement becomes visible differs from a matched moment with nothing inserted. They are cut from

different epochs and are corrected separately; a result on one is not evidence about the other.

Both are contrasts of the kind in (6.1) and differ only in what fills the cell. With  $y_{i\ell}$  one of the sixteen spectral measures on epoch  $\ell$  (Appendix D), condition aggregation takes the median over  $N_{ic}(37)$ , the set of retained epochs whose midpoints lie nearest visual onset;  $k = 37$  makes that an equal- $n$  cell, so a short and a long conversation weigh the same, and the number itself is fixed in §4.2.4. The onset-locked contrast takes the single 4 s epochs either side of visual onset against the same pair at a timing-matched moment under  $a^\emptyset$  (same section):

$$\begin{aligned} D_i^A &= \sum_c w_c \text{med}_{\ell \in N_{ic}(37)} y_{i\ell}, \\ D_i^B &= (y_i^{\text{post}} - y_i^{\text{pre}})_a - (y_i^{\text{post}} - y_i^{\text{pre}})_{a^\emptyset}. \end{aligned} \tag{6.2}$$

Ordering  $c$  as  $(a_2^{\text{imp}}, a_4^{\text{imp}}, a_2^{\text{exp}}, a_4^{\text{exp}}, a^\emptyset)$ , the Dataset A weights are fixed in advance:  $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -1)$  pools advertisements against  $a^\emptyset$ ,  $(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, 0)$  is implicit against explicit,  $(\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, 0)$  early against late, and  $(1, -1, -1, 1, 0)$  the presentation-by-timing interaction, coded rather than halved, so its  $D$  is twice a difference of differences; it is estimated descriptively and is not one of the three Holm-corrected contrasts of Table 6.1. The  $a^\emptyset$  term in  $D^B$  controls for the matching pre-to-post change under no advertisement. Each measure is its own Holm family. Fz theta, tracking effortful cognitive control [77], and posterior alpha, tracking visual engagement [78], are the confirmatory pair; the other fourteen are exploratory (section D.2). The positive control uses the same epochs and measures on a contrast the design guarantees: within each conversation, the first complete 4 s epoch after an assistant reply arrives (reading) against the last complete 4 s epoch before the participant sends the next message (writing), non-overlapping pairs inside one condition, collapsed to one median difference per participant.

One further set of tests was added after the planned contrasts had been read, and is labelled post hoc for that reason. Rather than the weights above it takes every pair of conditions in turn, ten for Dataset A and six among the four advertisement conditions of Dataset B, so that a marginal can be traced to the condition driving it. The four planned Dataset A weights already span the contrast space of five conditions, so each pair is a linear combination of those planned contrast estimates; that is contrast-space localisation, not a Friedman post-hoc on raw ranks. The sweep localises a marginal and carries a heavier multiplicity burden. The behavioural battery carries the same two post hoc layers: each advertisement condition against  $a^\emptyset$ , Holm across the four within an outcome, and a Friedman omnibus, a rank-based test of whether the five conditions differ at all, followed by the ten pairs, Holm across the ten, on the four primary and four secondary outcomes (Appendix E). The personality family is the same mixed model with each centred BFI-10 trait interacting with the three planned contrast codes, one trait at a time, and collapsed demographic covariates; Holm runs within each outcome across the fifteen interactions (sec-

tion 7.3).

Trajectory outcomes follow Definitions 1–6 of chapter 3 and are computed on  $\hat{g}_k = f_{\text{genre}}(u_k)$ , the genre of the bare utterance, with the deployed context kept as a sensitivity analysis. The confirmatory estimand is  $\delta_2^{(a)}$  against  $\delta_2$  under  $a^\emptyset$ , Holm within four contrasts: early pooled, implicit early, and explicit early against  $a^\emptyset$ , and implicit against explicit early. Because that outcome is binary in each conversation, the test is an exact McNemar test, the paired test for a binary outcome, computed on the conversations that disagree between the two conditions and reported with a paired  $t$  interval on  $D_i = \delta_{2,i}^{(a)} - \delta_{2,i}^\emptyset$ . A participant-clustered logistic regression (GEE) adjusting for task and session position is used as a covariate-adjusted check [80].

Three further trajectory tests are declared and none of them is confirmatory. A late advertisement has no following utterance, so  $\delta_4^{(a)}$  is undefined and the late conditions are analysed on the shift count  $N_{\text{shift}}$  instead. For the genre-aligned subcase  $\tilde{\delta}_2^{(a)}$ , the advertised genre  $g^{(a)}$  is reassigned at random across the 108 early-advertisement conversations over 20,000 seeded draws, which gives a one-sided reference distribution for the number of aligned shifts without assuming any distributional form. Kruskal–Wallis, a rank-based comparison of conditions on a conversation-level score, is applied to  $N_{\text{shift}}$ , the entropy  $H$ , and the longest run  $R$ , and is exploratory.

Analyses run on Python 3.13 with MNE-Python 1.12 for filtering, referencing, interpolation, and independent component analysis [39], SciPy 1.17 and NumPy 2.2 for spectral estimation, the tests, and the seeded permutations [81, 82], and statsmodels for the GEE and the mixed models [83].

# 7

## Results

This chapter reports the finished sample, the behavioural battery, personality moderation, EEG, the genre trajectories, and the cross-family associations. Table 7.8 records that against every declared family as a summary. Every estimator, correction family,  $n$  and methodology follows the one named in the Methods table 6.1.

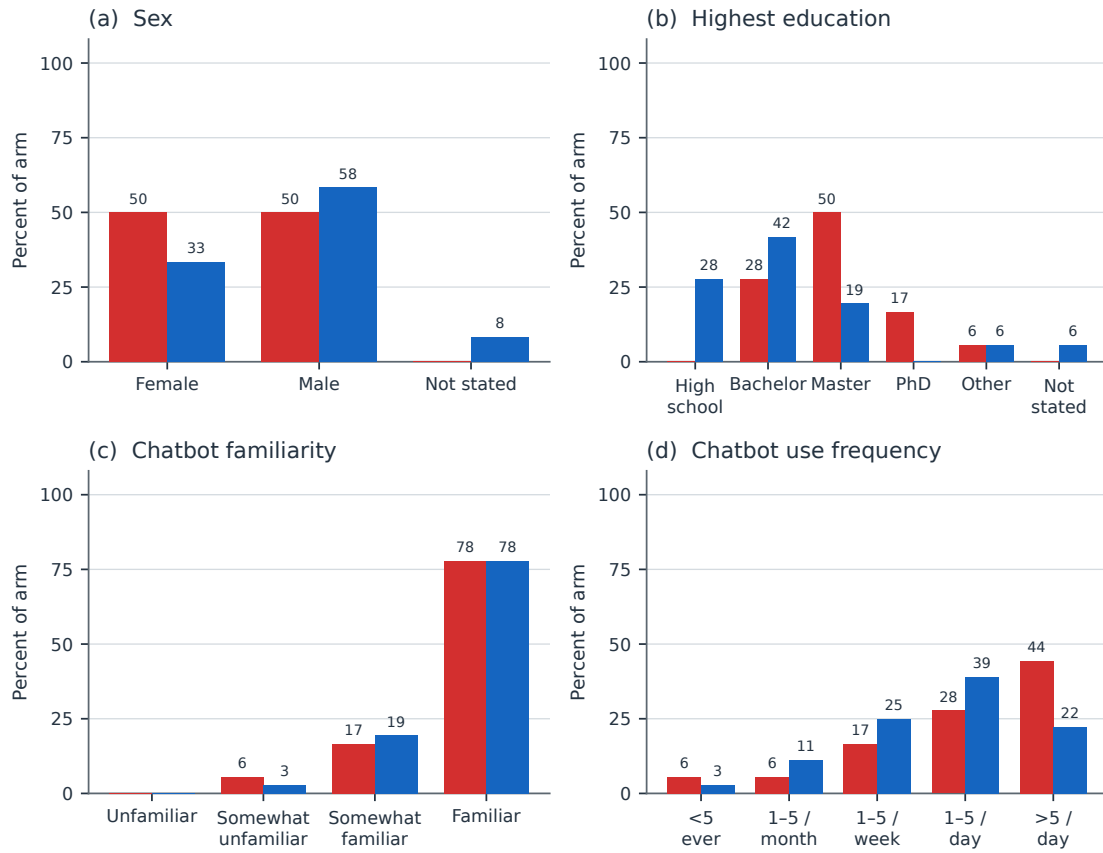
### 7.1 SAMPLE, EXCLUSIONS, AND DESCRIPTIVES

Unfinished, synthetic, and beta-tester sessions are excluded, as is the laboratory session that ran the crowd protocol; sessions labelled unfocused are nevertheless retained. What enters the tests below is therefore  $N = 54$ , being  $L = 18$  laboratory and  $C = 36$  crowd participants.

What Figure 7.1 does not carry is age. Both arms already know chatbots: 42 of 54 rate themselves familiar, and 35 use a chatbot at least daily. The laboratory arm has more daily power users (8 of 18 above five times per day, versus 8 of 36 in the crowd). Laboratory age is the intake range in subsection 6.2.3 (21–42 years,  $M = 29.50$ ,  $SD = 5.01$ ). Crowd age is taken from a partial Prolific demographic export: among the 26 of 36 crowd participants with a numeric age in that snapshot, age ranged 23–56 years ( $M = 35.69$ ,  $SD = 9.51$ ,  $Mdn = 34.5$ ).

All 54 finished participants completed the BFI-10 [1]; items and scoring are in subsection C.4.2.

Extraversion is the only arm contrast with an uncorrected  $p$  below .05, and it does not survive Holm across the five traits ( $p = .19$ ). The five scores stay continuous and are the moderators in section 7.3; participants are not grouped into types. The inventory was administered after the session and is not an input to any advertisement instance  $a_k$ , so personalisation  $\phi$  remains session-level throughout.



**Figure 7.1:** Finished-sample sex, education, chatbot familiarity, and use frequency. Red: laboratory ( $n = 18$ ); blue: crowd ( $n = 36$ ).

**Table 7.1:** BFI-10 trait scores in the finished sample ( $N = 54$ ).  $n_1$  and  $n_5$  count scale endpoints;  $p_{\text{Holm}}$  adjusts the five Mann-Whitney  $U$  arm comparisons (laboratory versus crowd).

| Trait             | $M$ (SD)    | Mdn  | IQR  | $n_1$ | $n_5$ | Lab $M$ (SD) | Crowd $M$ (SD) | $p$  | $p_{\text{Holm}}$ |
|-------------------|-------------|------|------|-------|-------|--------------|----------------|------|-------------------|
| Extraversion      | 2.98 (0.98) | 3.00 | 1.00 | 3     | 1     | 3.39 (0.74)  | 2.78 (1.03)    | .039 | .19               |
| Agreeableness     | 3.53 (0.97) | 3.50 | 1.50 | 2     | 3     | 3.50 (0.75)  | 3.54 (1.08)    | .62  | 1.00              |
| Conscientiousness | 3.86 (0.92) | 4.00 | 1.00 | 0     | 10    | 4.00 (0.73)  | 3.79 (1.01)    | .63  | 1.00              |
| Neuroticism       | 2.80 (1.22) | 2.50 | 1.88 | 5     | 5     | 2.94 (1.54)  | 2.72 (1.05)    | .77  | 1.00              |
| Openness          | 3.74 (1.11) | 4.00 | 1.50 | 2     | 12    | 3.86 (0.89)  | 3.68 (1.21)    | .87  | 1.00              |

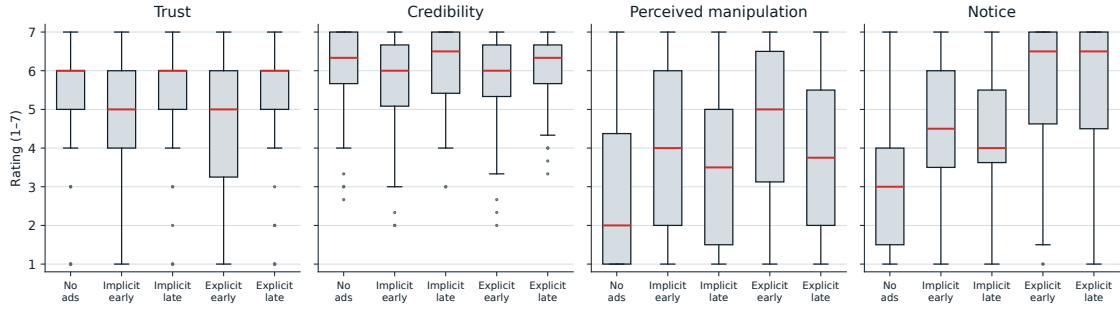
## 7.2 BEHAVIOURAL OUTCOMES

Behavioural results use  $N = 54$ , both arms pooled, and follow section 6.3: one person-level difference  $D_i$ ; per planned contrast (6.1), a paired  $t$  with Holm within each outcome, and the raw Wilcoxon signed-rank test on the same differences. The random-intercept mixed model is the adjusted check; on this complete balanced design its point estimates coincide with  $\bar{D}$ , so only its standard errors can differ. Out-

comes are the pre-specified means of subsection 6.2.2. Condition means with standard deviations are in Table E.1, and the eight items behind the primary outcomes are drawn by condition in Figure E.1, both in Appendix E.

### 7.2.1 POST-CONDITION OUTCOMES

**CONDITION PROFILES.** Figure 7.2 is the descriptive picture, pooled.



**Figure 7.2:** Post-condition outcomes by condition ( $N = 54$ ).

Credibility sits near the ceiling in every condition (means 5.7–6.1 of 7); trust sits lower and spreads wider (means 4.7–5.4, SD 1.6–1.9). Perceived manipulation and notice are low under  $a^0$  and highest under explicit early.

**PLANNED CONTRASTS.** Table 7.2 is the confirmatory family. Bold rows are Holm  $p < .05$  within outcome.

Eight of sixteen survive Holm. Manipulation and notice rise with any advertisement and more under explicit than implicit presentation; timing raises manipulation and lowers credibility by a third of a point. Trust is Holm-null on the three planned contrasts. The one estimator disagreement is trust early minus late, the single cell on which the three estimators straddle .05 ( $t$  Holm .063, LMM .048, raw Wilcoxon .026; Figure E.4). Format  $\times$  timing is null on all four outcomes ( $|d_z| \leq 0.06$ ; the coded contrast  $(1, -1, -1, 1)$  on the person-level scores gives trust  $+0.09 [-0.31, 0.49]$ , credibility  $-0.03 [-0.31, 0.25]$ , manipulation  $-0.10 [-0.58, 0.39]$ , and notice  $-0.06 [-0.51, 0.40]$ , raw  $p \geq .65$ ; it is not part of the Holm families and is listed in Table 7.9). Item-level sensitivity of these verdicts is in section E.3.

**LOCALISATION, POST HOC.** Figure 7.3 asks which advertisement condition carries a planned marginal: each against  $a^0$ , Holm across four, post hoc.

Manipulation and notice sit above  $a^0$  in every advertisement condition, largest under explicit early. Trust falls only there ( $-0.69$ , Holm  $p = .031$ ). Credibility is Holm-null in all four. The appendix

**Table 7.2:** Planned behavioural contrasts ( $N = 54$ ). Holm-adjusted paired  $t$ ; Wilcoxon  $p$  is raw; LMM is the adjusted check. Bold rows survive Holm.

| Outcome                 | Contrast                   | $\bar{D}$ | 95% CI         | $d_z$ | Holm $p$ | Wilcoxon $p$ | LMM Holm $p$ |
|-------------------------|----------------------------|-----------|----------------|-------|----------|--------------|--------------|
| Trust                   | Any ad – no ads            | −0.34     | [−0.71, 0.03]  | −0.25 | .153     | .016         | .193         |
|                         | Implicit – explicit        | +0.15     | [−0.20, 0.49]  | +0.11 | .405     | .384         | .422         |
|                         | Early – late               | −0.44     | [−0.81, −0.08] | −0.32 | .063     | .026         | .048         |
| Credibility             | Any ad – no ads            | −0.14     | [−0.42, 0.14]  | −0.13 | .673     | .039         | .597         |
|                         | Implicit – explicit        | −0.06     | [−0.24, 0.11]  | −0.10 | .673     | .278         | .597         |
|                         | <b>Early – late</b>        | −0.33     | [−0.55, −0.11] | −0.39 | .016     | .007         | .019         |
| Perceived manipulation  | <b>Any ad – no ads</b>     | +1.27     | [0.76, 1.78]   | +0.66 | < .001   | < .001       | < .001       |
|                         | <b>Implicit – explicit</b> | −0.62     | [−1.13, −0.10] | −0.32 | .022     | .022         | .017         |
|                         | <b>Early – late</b>        | +0.58     | [0.21, 0.95]   | +0.41 | .007     | .004         | .017         |
| Notice                  | <b>Any ad – no ads</b>     | +2.07     | [1.53, 2.62]   | +1.01 | < .001   | < .001       | < .001       |
|                         | <b>Implicit – explicit</b> | −1.15     | [−1.71, −0.59] | −0.55 | < .001   | < .001       | < .001       |
|                         | Early – late               | +0.26     | [−0.16, 0.68]  | +0.16 | .234     | .236         | .292         |
| Cued memory             | <b>Implicit – explicit</b> | −1.45     | [−1.87, −1.04] | −0.93 | < .001   | < .001       | < .001       |
|                         | Early – late               | +0.19     | [−0.33, 0.72]  | +0.10 | .473     | .656         | .434         |
| Trust after re-exposure | Implicit – explicit        | +0.12     | [−0.25, 0.49]  | +0.09 | .525     | .572         | .558         |
|                         | <b>Early – late</b>        | −0.49     | [−0.90, −0.08] | −0.32 | .047     | .024         | .034         |

pairwise grid (Table E.8) localises the same two outcomes; its one extra Holm pair is convincingness, implicit early minus explicit late.

### 7.2.2 ADVERTISEMENT NOTICE AND CUED RECALL

People notice a labelled banner more than a mention inside the reply. About half do not notice the implicit advertisement.

Noticed is agreement ( $\geq 5$  of 7) on the detection item of the notice pair: the reading that stays low when nothing was inserted. Brand mention is already high in the no-ad control, because a shopping assistant names products anyway, so it is not treated as detection (subsection 6.2.2); both items and the planned notice outcome are in Table E.2.

At the same turn, 21 of 54 participants noticed the banner and not the mention at turn 2, and 22 of 54 at turn 4; the reverse was 3 and 5. Cued memory was  $\geq 5$  for 57% and 59% of implicit insertions and for 87% and 81% of explicit ones, including about half of those who had not noticed the implicit mention.

Cued recall is the lower block of Table 7.2. Explicit insertions are remembered far better than implicit ones; timing does not move memory. Trust after re-exposure does not differ by format and is lower for early than late insertions.

No participant clicked an advertisement. Nine logged interaction measures (conversation duration,

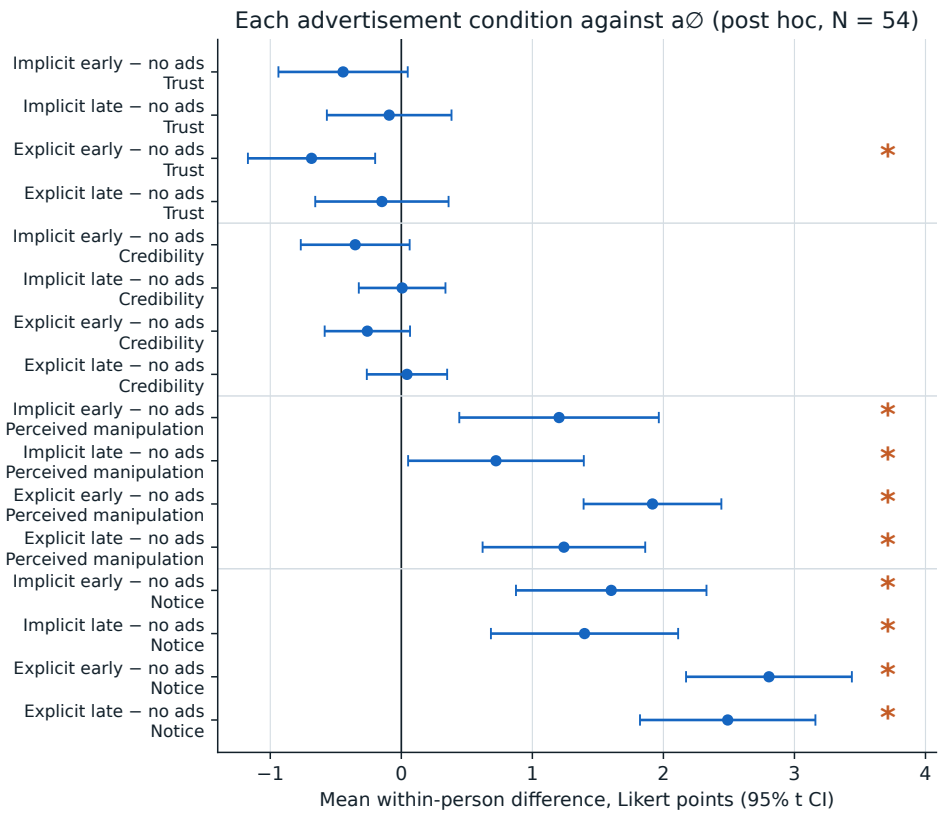


Figure 7.3: Each advertisement condition minus  $a^0$  ( $N = 54$ ). Asterisk: Holm  $p < .05$  within outcome. Post hoc.

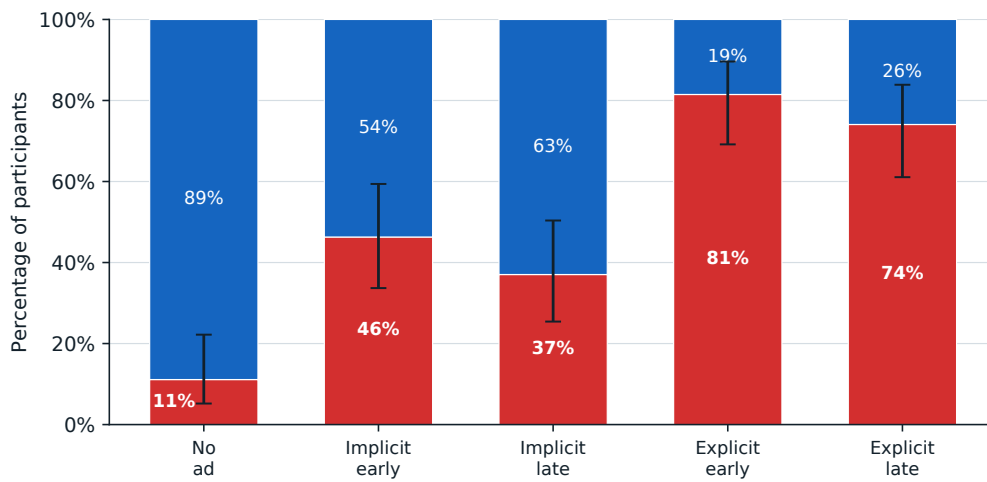


Figure 7.4: Percentage of participants who noticed, by condition ( $N = 54$ ). Red: noticed; blue: did not. Whiskers: Wilson 95% interval.

time to the first user message, median and total user message length, median user word count, median assistant reply length, mean and median reply latency, and median model latency)  $\times$  the three planned contrasts and the format  $\times$  timing interaction give 36 exploratory tests, none with raw  $p < .05$ . Free-text findings and the open recall reaction were not analysed.

### 7.3 PERSONALITY AND DEMOGRAPHIC MODERATION

No BFI-10 trait moderates the planned format or timing contrast on the four primary outcomes.

|                        | Format (implicit – explicit) |               |               |               |               | Timing (early – late) |               |               |               |               |
|------------------------|------------------------------|---------------|---------------|---------------|---------------|-----------------------|---------------|---------------|---------------|---------------|
| Trust                  | +0.10<br>1.00                | +0.22<br>1.00 | +0.11<br>1.00 | -0.11<br>1.00 | -0.17<br>1.00 | -0.46<br>.20          | -0.06<br>1.00 | -0.19<br>1.00 | -0.05<br>1.00 | -0.19<br>1.00 |
| Credibility            | -0.06<br>1.00                | +0.13<br>1.00 | +0.02<br>1.00 | -0.08<br>1.00 | -0.01<br>1.00 | -0.31<br>.18          | -0.05<br>1.00 | -0.13<br>1.00 | -0.02<br>1.00 | -0.04<br>1.00 |
| Perceived manipulation | -0.25<br>1.00                | -0.36<br>1.00 | -0.10<br>1.00 | +0.03<br>1.00 | +0.26<br>1.00 | -0.08<br>1.00         | -0.15<br>1.00 | -0.08<br>1.00 | -0.13<br>1.00 | -0.02<br>1.00 |
| Notice                 | +0.02<br>1.00                | -0.27<br>1.00 | +0.29<br>1.00 | +0.09<br>1.00 | +0.46<br>.57  | -0.31<br>1.00         | -0.41<br>1.00 | -0.24<br>1.00 | +0.35<br>1.00 | +0.09<br>1.00 |
|                        | E                            | A             | C             | N             | O             | E                     | A             | C             | N             | O             |

Figure 7.5: BFI-10 moderation of format and timing ( $N = 54$ ). Cells give slope per BFI point and Holm  $p$ ; shade groups  $p$ -value ranges.

The any-ad column and the OLS check are in Table E.9: still 0 of 60 Holm. Nearest cell extraversion on credibility early minus late, Holm  $p = .18$ .

No demographic factor moderates the planned contrasts either (Figure 7.6). Sex, education, chatbot familiarity, use frequency, and environment (the laboratory or the crowd arm) each entered the same random-intercept model one at a time, interacting with the three planned contrast codes, on the four primary outcomes and cued memory (cohort  $N = 54$ ; the sex model holds  $n = 51$  and the education model  $n = 49$  after missing and merged values, the other three all 54). Age was incomplete, available for every laboratory participant but for only 26 of the 36 crowd participants (section 7.1), and was therefore not tested; education levels with fewer than five people were merged or excluded before testing. With the levels kept, 0 of 70 factor  $\times$  contrast cells survive Holm within outcome; with two-level coding, again 0 of 70; and a person-level regression of  $D_i$  on the factor agrees on every cell (Table E.10). The nearest cell is cued memory early minus late by familiarity, Holm  $p = .44$ .

|                             |                        | Any ad – no ads                                      |                |                  |                  |                  | Format (implicit – explicit) |                |                  |                  |                  | Timing (early – late) |                |                  |                  |                  |
|-----------------------------|------------------------|------------------------------------------------------|----------------|------------------|------------------|------------------|------------------------------|----------------|------------------|------------------|------------------|-----------------------|----------------|------------------|------------------|------------------|
|                             |                        | Sex                                                  | Educa-<br>tion | Famili-<br>arity | Use<br>frequency | Environ-<br>ment | Sex                          | Educa-<br>tion | Famili-<br>arity | Use<br>frequency | Environ-<br>ment | Sex                   | Educa-<br>tion | Famili-<br>arity | Use<br>frequency | Environ-<br>ment |
| Levels kept (sparse merged) | Trust                  | 0.23<br>1.00                                         | 0.86<br>1.00   | 0.18<br>1.00     | 0.80<br>1.00     | 0.05<br>1.00     | 0.22<br>1.00                 | 0.28<br>1.00   | 0.46<br>1.00     | 1.12<br>1.00     | 0.10<br>1.00     | 0.25<br>1.00          | 0.27<br>1.00   | 0.02<br>1.00     | 0.40<br>1.00     | 0.46<br>1.00     |
|                             | Credibility            | 0.11<br>1.00                                         | 0.45<br>1.00   | 0.29<br>1.00     | 0.44<br>1.00     | 0.28<br>1.00     | 0.18<br>1.00                 | 0.20<br>1.00   | 0.29<br>1.00     | 0.33<br>1.00     | 0.10<br>1.00     | 0.11<br>1.00          | 0.33<br>1.00   | 0.12<br>1.00     | 0.42<br>1.00     | 0.48<br>.92      |
|                             | Perceived manipulation | 0.46<br>1.00                                         | 1.25<br>1.00   | 0.12<br>1.00     | 1.22<br>1.00     | 0.45<br>1.00     | 0.41<br>1.00                 | 0.90<br>1.00   | 0.82<br>1.00     | 0.95<br>1.00     | 0.58<br>1.00     | 0.27<br>1.00          | 0.29<br>1.00   | 0.07<br>1.00     | 1.42<br>1.00     | 0.16<br>1.00     |
|                             | Notice                 | 0.36<br>1.00                                         | 0.36<br>1.00   | 0.23<br>1.00     | 1.15<br>1.00     | 0.14<br>1.00     | 0.94<br>1.00                 | 1.08<br>1.00   | 0.51<br>1.00     | 1.60<br>1.00     | 0.14<br>1.00     | 0.52<br>1.00          | 0.37<br>1.00   | 0.55<br>1.00     | 1.66<br>1.00     | 0.07<br>1.00     |
|                             | Cued memory            | not defined: no advertisement in the no-ad condition |                |                  |                  |                  | 0.18<br>1.00                 | 0.65<br>1.00   | 0.21<br>1.00     | 1.08<br>1.00     | 0.03<br>1.00     | 0.23<br>1.00          | 1.45<br>.69    | 1.20<br>.44      | 1.59<br>.48      | 0.67<br>1.00     |
| Two-level coding            | Trust                  | 0.23<br>1.00                                         | 0.01<br>1.00   | 0.18<br>1.00     | 0.18<br>1.00     | 0.05<br>1.00     | 0.22<br>1.00                 | 0.20<br>1.00   | 0.46<br>1.00     | 0.84<br>.44      | 0.10<br>1.00     | 0.25<br>1.00          | 0.11<br>1.00   | 0.02<br>1.00     | 0.09<br>1.00     | 0.46<br>1.00     |
|                             | Credibility            | 0.11<br>1.00                                         | 0.03<br>1.00   | 0.29<br>1.00     | 0.20<br>1.00     | 0.28<br>1.00     | 0.18<br>1.00                 | 0.14<br>1.00   | 0.29<br>1.00     | 0.21<br>1.00     | 0.10<br>1.00     | 0.11<br>1.00          | 0.04<br>1.00   | 0.12<br>1.00     | 0.17<br>1.00     | 0.48<br>.92      |
|                             | Perceived manipulation | 0.46<br>1.00                                         | 0.69<br>1.00   | 0.12<br>1.00     | 0.97<br>1.00     | 0.45<br>1.00     | 0.41<br>1.00                 | 0.49<br>1.00   | 0.82<br>1.00     | 0.30<br>1.00     | 0.58<br>1.00     | 0.27<br>1.00          | 0.04<br>1.00   | 0.07<br>1.00     | 0.33<br>1.00     | 0.16<br>1.00     |
|                             | Notice                 | 0.36<br>1.00                                         | 0.17<br>1.00   | 0.23<br>1.00     | 0.44<br>1.00     | 0.14<br>1.00     | 0.94<br>1.00                 | 0.84<br>1.00   | 0.51<br>1.00     | 0.43<br>1.00     | 0.14<br>1.00     | 0.52<br>1.00          | 0.25<br>1.00   | 0.55<br>1.00     | 0.36<br>1.00     | 0.07<br>1.00     |
|                             | Cued memory            | not defined: no advertisement in the no-ad condition |                |                  |                  |                  | 0.18<br>1.00                 | 0.11<br>1.00   | 0.21<br>1.00     | 0.44<br>1.00     | 0.03<br>1.00     | 0.23<br>1.00          | 0.47<br>1.00   | 1.20<br>.44      | 0.51<br>1.00     | 0.67<br>1.00     |

**Figure 7.6:** Demographic moderation of planned contrasts ( $N = 54$ ). Cells give contrast spread and joint-test Holm  $p$ ; shade groups  $p$ -value ranges.

## 7.4 NEUROPHYSIOLOGICAL OUTCOMES

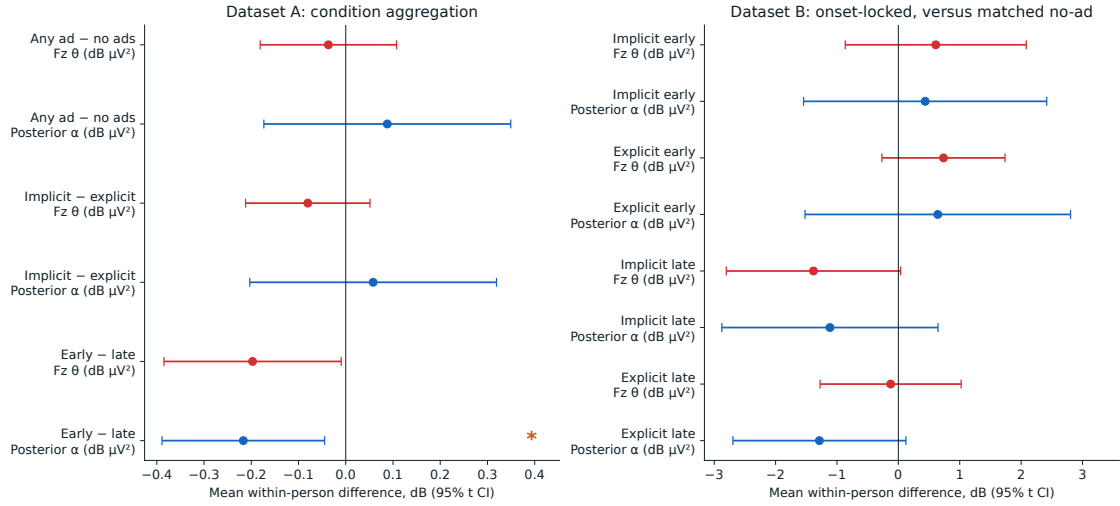
EEG results use the laboratory eighteen, the 4 s ICA pipeline, and the families of section 6.3. Every contrast below is built from one of two person-level scores. Again, dataset A is condition aggregation and dataset B is onset-locked (both defined in §4.2.4). Holm runs within each measure, as Table 6.1 already names it.

**POSITIVE CONTROL: WRITING VERSUS READING.** Writing minus reading raises Fz  $\theta$  and does not move posterior  $\alpha$  (Table 7.3). That contrast is the one the protocol guarantees in every conversation, so it asks whether these recordings register a within-conversation change at all before any advertisement contrast is read.

**Table 7.3:** Writing minus reading ( $n = 18$ ). Holm on the two confirmatory  $t$  tests. Bold rows survive Holm.

| Measure                       | $M$ (dB) | 95% CI        | $d_z$ | Holm $p$ |
|-------------------------------|----------|---------------|-------|----------|
| <b>Fz <math>\theta</math></b> | +0.60    | [0.22, 0.97]  | 0.80  | .007     |
| Posterior $\alpha$            | +0.08    | [−0.43, 0.59] | 0.08  | .75      |

**DATASET A AND DATASET B.** Figure 7.7 is the confirmatory picture at 4 s, and it puts the two estimands next to each other: the three condition contrasts on the aggregated score, and the four onset-locked cells, on both confirmatory measures.



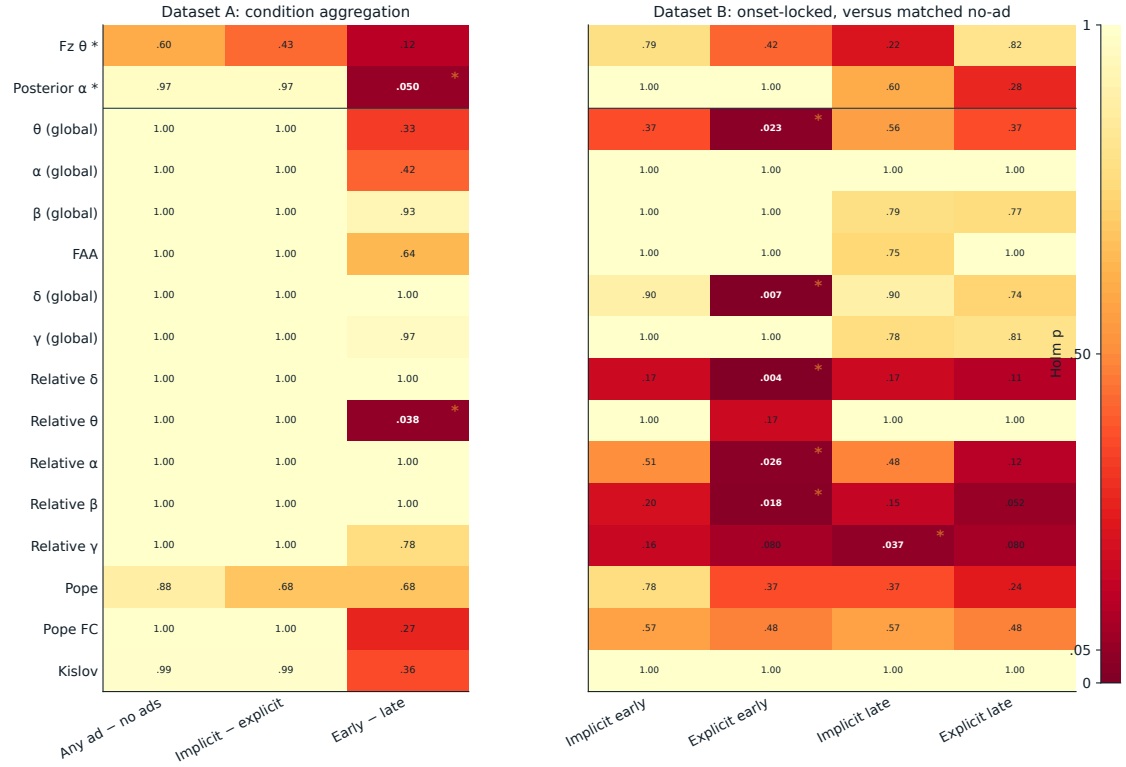
**Figure 7.7:** Confirmatory EEG contrasts at 4 s ( $n = 18$ ): condition aggregation (left) and onset-locked (right). Red: Fz  $\theta$ ; blue: posterior  $\alpha$ . Asterisk: Holm  $p < .05$ .

Under condition aggregation the only Holm cell is early minus late on posterior  $\alpha$  ( $-0.22$  dB, Holm  $p = .0496$ ; raw Wilcoxon  $p = .021$ ). Dataset B is Holm-null in all eight confirmatory cells. Epoch-width branches that leave 4 s are in Appendix D; they do not revise the 4 s family.

**THE REMAINING FOURTEEN MEASURES.** The confirmatory pair is two rows of a sixteen-measure board. Figure 7.8 is that grid, and it shows where a Holm cell falls on any of the sixteen under either estimand; the estimate, interval, and  $d_z$  of each of the seven non-confirmatory cells it marks are in Table D.2.

Dataset A has two orange cells, both early minus late: confirmatory posterior  $\alpha$  and relative  $\theta$ . Dataset B confirmatory rows stay null. Five of the six exploratory orange cells sit on explicit early (a slow-power tilt). Absolute  $\alpha$  and  $\beta$ , FAA, both Pope ratios, and the Kislov ratio are Holm-null throughout.

**EXHAUSTIVE POST-HOC PAIRWISE COMPARISONS.** The planned contrasts pool conditions, so a further pairwise sweep asks which single pair of conditions differs: every pair, 256 tests, Holm within measure. One cell survives: Fz  $\theta$ , implicit early minus explicit late ( $-0.28$  dB, Holm  $p = .014$ ). That pair is post hoc. Dataset A contributes 160 of the tests (ten pairs of five conditions) and Dataset B the other 96 (six pairs of the four advertisement conditions); on Dataset B six cells reach raw  $p < .05$ , about the five that chance predicts, and none survives Holm. The grid is section D.3.



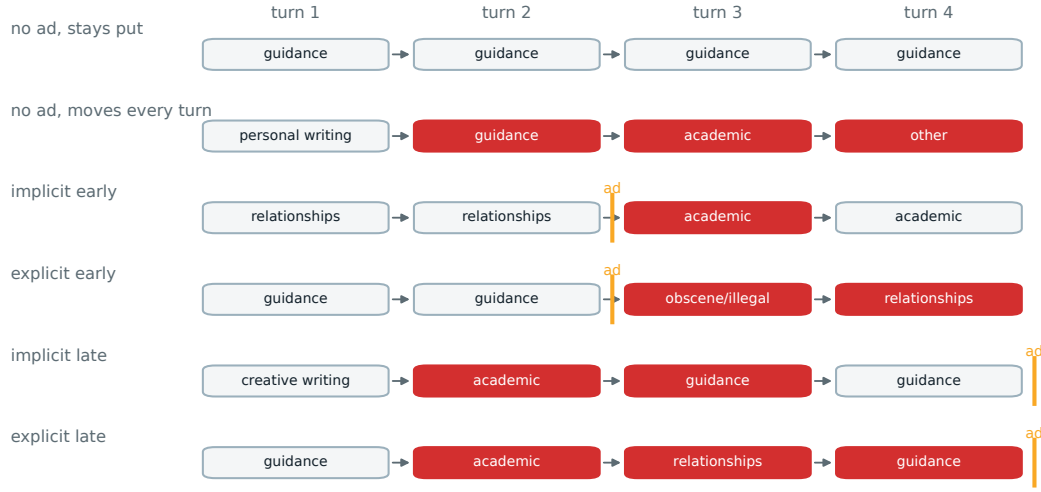
**Figure 7.8:** Holm  $p$  within each EEG measure at 4 s ( $n = 18$ ). Rows above the rule are confirmatory. Asterisk: Holm  $p < .05$ . Values are printed to two decimals; the early — late posterior  $\alpha$  cell shown as .05 is .0496.

## 7.5 GENRE TRAJECTORIES

Trajectory results use all fifty-four participants, both arms pooled, and follow Definitions 1–6 of chapter 3; tests are those of section 6.3. Each participant contributes five conversations of  $T = 4$  user utterances: 270 conversations, 1,080 utterances, and 810 transitions  $\tau_k = (\hat{g}_k, \hat{g}_{k+1})$  (3.2). Two contexts can be fed to  $f_{\text{genre}}$ , and every count below states which one it comes from (subsection 4.2.3, section F.2): the primary label is  $\hat{g}_k = f_{\text{genre}}(u_k)$  on the bare message (3.1), and the deployed context, the longer window the running system classified, is a sensitivity analysis. Notation follows Definition 6:  $\delta_k^{(a)}$  is the ad-associated shift (3.8),  $\tilde{\delta}_k^{(a)}$  its genre-aligned subcase (3.9), and  $g_k^{(a)}$  the genre of the served advertisement.

**DESCRIPTIVE STRUCTURE OF THE TRAJECTORIES.** Figure 7.9 shows six observed trajectories, two under  $a^\emptyset$  and one under each advertisement condition.

All thirteen classes occur. The modal genre is general guidance and information (32.4% of utterances), then relationships and personal reflection (17.9%) and academic help (10.2%); purchasable products

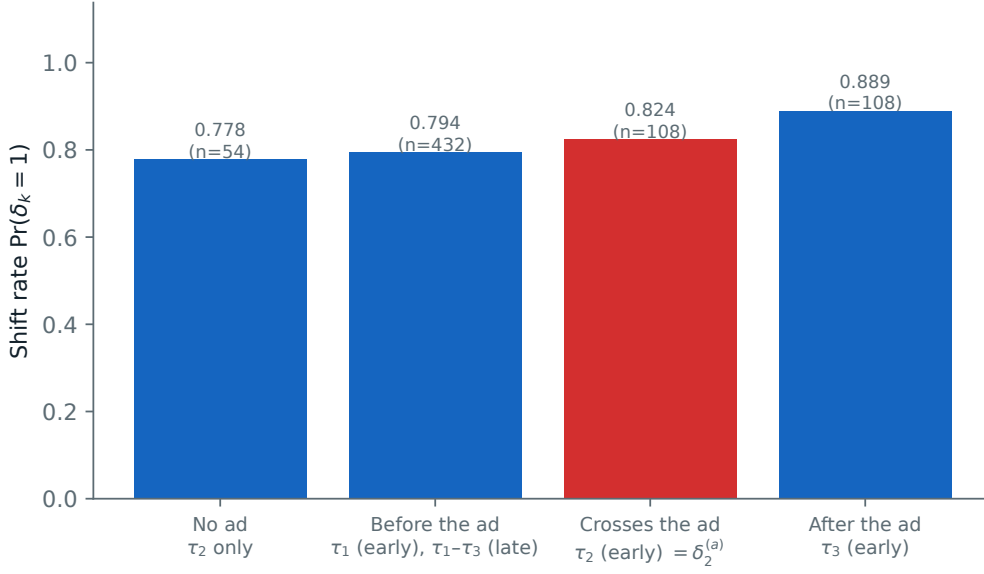


**Figure 7.9:** Observed trajectories under the bare-utterance context. Red boxes mark a genre shift; the yellow rule marks advertisement insertion.

labels 19 of 1,080 utterances although every task is a shopping scenario. A conversation averages  $N_{\text{shift}} = 2.45$  shifts of a possible 3 (3.2), and 97.4% of conversations shift at least once. Of the  $13^2 = 169$  possible ordered pairs  $(\hat{g}_k, \hat{g}_{k+1})$ , 120 occur (3.5), and 18.3% of the 810 transitions stay in the same genre. Per-condition values of  $N_{\text{shift}}$ , diversity  $D$  (3.6), entropy  $H$  (3.7), and persistence  $R$  (3.3) are tabulated in section F.1.

**WHOLE-CONVERSATION SCORES BY CONDITION.**  $N_{\text{shift}}$  does not differ across the five conditions (Kruskal–Wallis on the 270 conversations,  $H = 2.89$ ,  $p = .58$ ; this is a conversation-level descriptive test that treats each participant’s five conversations as independent, which is why it is exploratory and the person-level paired contrasts above carry the verdict); nor do  $H$  ( $p = .88$ ) or  $R$  ( $p = .59$ ), and none of the three differs by task, arm, or session position: for  $N_{\text{shift}}$  those three tests give  $p = .33$ ,  $.85$ , and  $.52$ ; for  $H$  and  $R$  all of them give  $p \geq .25$ . These are the exploratory conversation-grain tests declared in section 6.3; the participant ICC on the three scores is 0.043 to 0.111. A whole-conversation count is close to its ceiling, 2.45 shifts of a possible 3 with 97.4% of conversations shifting at least once, so it has little room left in which to register an insertion. The confirmatory estimand is therefore the single transition that crosses the advertisement,  $\delta_2^{(a)}$  on  $\tau_2$ , and Figure 7.10 puts that one step next to the others.

The orange bar and the no-ad bar are the two arms of the confirmatory contrast, tested in Table 7.5. The tallest bar is  $\tau_3$  of the early-advertisement conversations, the deepest step in the figure; turn depth is taken up next.



**Figure 7.10:** Share of transitions with  $\delta_k = 1$  by position relative to the advertisement (bare-utterance context).

**TURN LENGTH AND FALLBACK LABELS.** A fallback label is one of the two residual classes of  $f_{\text{genre}}$ , “other” and “other obscene or illegal”. Table 7.4 is the turn-wise picture.

**Table 7.4:** Turn-wise message length and fallback-label share under the bare-utterance context ( $N = 54$ ; 270 utterances per turn).

| Turn $k$ | Median words | Fallback share |
|----------|--------------|----------------|
| 1        | 40.5         | 4.1%           |
| 2        | 14.0         | 15.2%          |
| 3        | 13.0         | 18.9%          |
| 4        | 10.0         | 20.0%          |

In a participant-clustered logistic model of a fallback label, the turn coefficient is  $\beta = 0.43$  ( $p < .001$ ); with  $\log(\text{words})$  added it is  $\beta = 0.34$  ( $p = .0003$ ), and length carries  $\beta = -0.29$  ( $p = .0007$ ). Mean posterior mass on the winning class is flat across turns (0.427 to 0.430).

**AD-ASSOCIATED SHIFTS ( $\delta^{(a)}$ ).** With  $T = 4$ , only a turn-2 advertisement is followed by a user utterance, because  $u_5$  does not exist, so the estimand is  $\delta_2^{(a)}$  on  $\tau_2$  against  $\delta_2$  under  $a^\emptyset$  (3.8) and no late condition can enter it. Table 7.5 is the declared family of Table 6.1: the two early conditions pooled against  $a^\emptyset$ , each of them separately against  $a^\emptyset$ , and implicit against explicit early, with Holm across the four.

No interval excludes zero. A participant-clustered logistic model of the same turn-2 rows, which lets a participant’s five conversations be correlated, puts the odds of a shift with an early advertisement at

**Table 7.5:** Ad-associated turn-2 shifts under the bare-utterance context ( $N = 54$ ). Holm  $p$  is the Holm-adjusted paired  $t$  on the person-level difference; McNemar  $p$  is the raw exact test declared in Table 6.1.

| Contrast                       | $\Delta \Pr(\delta_2 = 1)$ | 95% CI            | $d_z$ | Holm $p$ | McNemar $p$ |
|--------------------------------|----------------------------|-------------------|-------|----------|-------------|
| Early pooled $- a^\emptyset$   | +0.046                     | $[-0.082, 0.174]$ | 0.10  | .94      | –           |
| Implicit early $- a^\emptyset$ | +0.093                     | $[-0.060, 0.245]$ | 0.17  | .91      | .33         |
| Explicit early $- a^\emptyset$ | 0.000                      | $[-0.150, 0.150]$ | 0.00  | 1.00     | 1.00        |
| Implicit $-$ explicit early    | +0.093                     | $[-0.069, 0.254]$ | 0.16  | .91      | .36         |

1.31 times the odds without ( $p = .51$ ), and a bias-corrected bootstrap interval on the pooled difference is  $[-0.074, 0.176]$ .

GENRE-ALIGNED AD-ASSOCIATED SHIFTS ( $\tilde{\delta}_2^{(a)}$ ).  $\tilde{\delta}_2^{(a)} = \delta_2^{(a)} \mathbb{I}[\hat{g}_3 = g_2^{(a)}]$  (3.9), with  $g_2^{(a)} = f_{\text{genre}}$  applied to the product title. Table 7.6 splits the 108 early-advertisement conversations three ways.

**Table 7.6:** Turn-2 outcomes in the 108 early-advertisement conversations under the bare-utterance context.

| Outcome                                              | Count |
|------------------------------------------------------|-------|
| Stay ( $\delta_2^{(a)} = 0$ )                        | 19    |
| Shift, not into the advertised genre                 | 80    |
| Genre-aligned shift ( $\tilde{\delta}_2^{(a)} = 1$ ) | 9     |
| Total                                                | 108   |

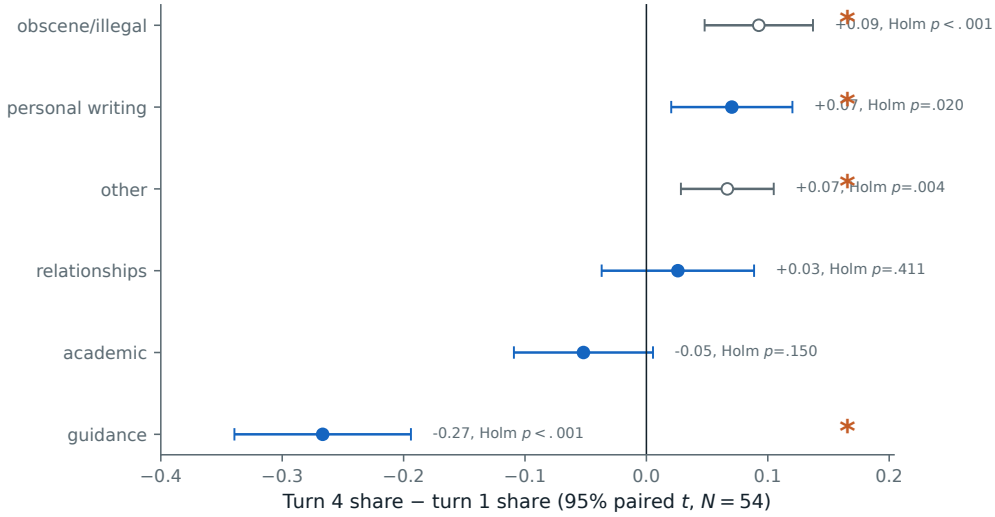
The 216 served advertisements draw on 156 distinct products, and  $f_{\text{genre}}$  labels 98 of those 156 general guidance and information and only 4 purchasable products, so the advertised genre usually coincides with the modal genre of the utterances. Reassigning  $g_2^{(a)}$  at random across the 108 conversations gives 10.46 expected aligned shifts against 9 observed (20,000 permutations, one-sided  $p = .80$ ). In 25 of the 108 the advertised genre already equals  $\hat{g}_2$ , so no shift there can be aligned and they are zeros under Definition 6; among the other 83, 9 are aligned (0.108), against 9 of 108 (0.083) overall.

PRESENTATION FORMAT AND THE LATE-ADVERTISEMENT CONTROL. Implicit minus explicit early is the last row of Table 7.5 (Holm  $p = .91$ ; raw Wilcoxon  $p = .25$ ). The late conditions have no  $\delta^{(a)}$  to score and are therefore carried by the whole-conversation count  $N_{\text{shift}}$  instead (Table 7.7), which makes them the negative control described in subsection 6.2.3: a late advertisement follows the last user utterance, so a difference in  $N_{\text{shift}}$  against  $a^\emptyset$  cannot be an advertisement effect and would indicate leakage or an order artefact instead.

**Table 7.7:** Late advertisements versus  $a^\emptyset$  on  $N_{\text{shift}}$  ( $N = 54$ , bare-utterance context). Holm-adjusted paired  $t$ ; Wilcoxon  $p$  is raw.

| Contrast                     | $\Delta N_{\text{shift}}$ | 95% CI          | $d_z$ | Holm $p$ | Wilcoxon $p$ |
|------------------------------|---------------------------|-----------------|-------|----------|--------------|
| Late pooled $-a^\emptyset$   | -0.074                    | [-0.361, 0.213] | -0.07 | 1.00     | .39          |
| Implicit late $-a^\emptyset$ | -0.019                    | [-0.347, 0.310] | -0.02 | 1.00     | .94          |
| Explicit late $-a^\emptyset$ | -0.130                    | [-0.444, 0.184] | -0.11 | 1.00     | .40          |

**INSTRUMENT CHECK: GENRE SHARES AT TURN 1 VERSUS TURN 4.** This is an instrument check rather than an advertisement contrast, Holm-corrected across the six genres: does  $f_{\text{genre}}$  register any change between the first and the last user message, the role that writing versus reading plays in section 7.4. Figure 7.11 covers the six genres that carry at least 5% of utterances.



**Figure 7.11:** Change in genre share from turn 1 to turn 4 ( $N = 54$ ). Hollow markers are fallback classes. Asterisk: Holm  $p < .05$ .

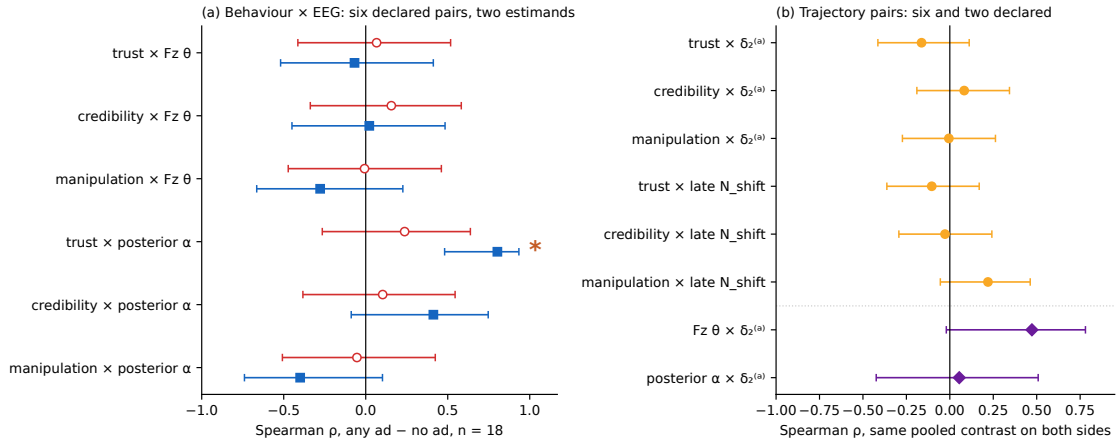
Four of the six move. Two of those are the fallback classes whose rise Table 7.4 already shows. The two content genres that move are general guidance and information, 0.478 to 0.211 ( $d_z = -1.00$ , Holm  $p < .001$ ), and personal writing, 0.048 to 0.119 ( $d_z = 0.38$ , Holm  $p = .020$ ). Table 7.5 is the advertisement contrast on the same participants.

**SENSITIVITY TO THE CONTEXT FED TO  $f_{\text{genre}}$ .** The deployed context is the window the running system classified: the task prompt, the last three messages, and the current message (section F.2). Under it the pooled  $\delta_2^{(a)}$  contrast is Holm-null, and every McNemar cell is  $p = 1.00$ . That context shifts far less often: it uses 7 of the 13 classes, averages  $N_{\text{shift}} = 0.40$  against 2.45, and leaves 70% of conversations in a single genre for all four turns. Its  $\hat{\delta}_2^{(a)}$  is 2 of 108, with  $g_2^{(a)} = \hat{g}_2$  already in 53 of the 108. section F.2 reports the full tables under both contexts, and section F.3 sweeps four further context windows, none

of which produces a significant  $\delta_2^{(a)}$  contrast.

## 7.6 ASSOCIATIONS ACROSS THE THREE FAMILIES

This section showcases the relationships between the three streams of data, almost all of them Spearman correlations, and the grain they are computed on is what makes them readable. Every score is person-level: one difference  $D_i$  per participant per planned contrast (6.1), not an epoch, a turn, or a conversation, and the sample is the laboratory eighteen wherever EEG enters and  $N = 54$  otherwise. The pairs are the declared ones of Table 6.1, association and not mediation, Holm within each family, so a Holm  $p$  here is a Holm-adjusted Spearman  $p$ . Behaviour  $\times$  EEG takes the any ad minus no ad  $D_i$  on both sides and is evaluated on each of the two EEG scores,  $D_i^A$  under condition aggregation and  $D_i^B$  locked to onset; the trajectory pairs use  $\delta_2^{(a)}$  (early pooled minus  $a^\emptyset$ ) and late  $N_{\text{shift}}$  (late pooled minus  $a^\emptyset$ ), with the same pooled weights on the other side. Figure 7.12 is the whole declared set, and the exploratory map that surrounds it closes the section and is listed in Table 7.9.



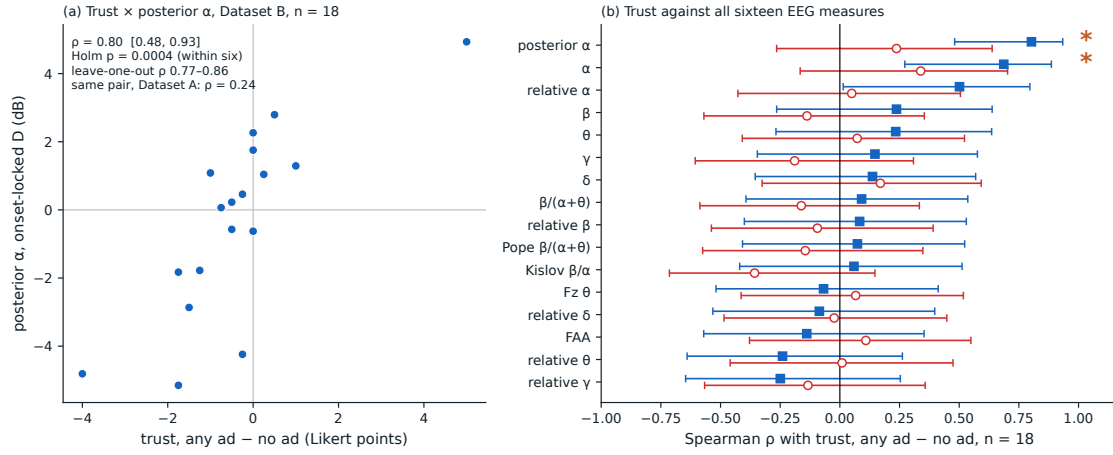
**Figure 7.12:** Spearman  $\rho$  with 95% intervals. (a) Hollow circles: condition aggregation; filled squares: onset-lock ( $n = 18$ ). (b) Yellow: behaviour  $\times$  trajectory ( $N = 54$ ); purple: trajectory  $\times$  EEG ( $n = 18$ ). Asterisk: Holm  $p < .05$ .

### 7.6.1 BEHAVIOUR AND EEG

Figure 7.12a is this family. Under condition aggregation none of them covaries; locked to onset, trust covaries with posterior  $\alpha$ , and nothing else does after correction; the next-largest cells, posterior  $\alpha$  with credibility and with manipulation, are Holm-null.

On Dataset A the largest of the six is trust  $\times$  posterior  $\alpha$ ,  $\rho = .24$ , raw  $p = .34$ , but Holm-null. On Dataset B the same pair is  $\rho = .80$ , raw  $p = 6 \times 10^{-5}$ , Holm  $p = .0004$  within the six pairs on that score (the twelve tests over both scores are two declared families of six, Table 6.1; across all twelve the

cell would be Holm  $p = .0007$ ). Pearson and Kendall agree ( $r = .78$ ,  $\tau = .65$ ), and leaving any one participant out keeps  $\rho$  between .77 and .86. The sign is positive: participants whose trust fell further below their own  $a^0$  rating with advertisements present are those whose posterior  $\alpha$  fell further in the four seconds after onset than at the matched moment. Neither mean differs from zero (trust  $-0.38$  points,  $p = .38$ ; posterior  $\alpha -0.33$  dB,  $p = .61$ ); The other five Dataset B pairs have  $|\rho| \leq .41$ , Holm  $p \geq .44$  and are mostly associated to posterior  $\alpha$  and to the onset-locked score.



**Figure 7.13:** Trust against EEG ( $n = 18$ ). (a) Onset-locked posterior  $\alpha$ . (b) All sixteen measures; squares: onset-lock; hollow circles: condition aggregation. Asterisk: Holm  $p < .05$ .

Figure 7.13b screens the trust  $D_i$  against each of the 16 onset-locked measures, Holm within sixteen. Two survive: posterior  $\alpha$  (Holm  $p = .001$ ) and absolute  $\alpha$  ( $\rho = .69$ , Holm  $p = .025$ ); rel.  $\alpha$  does not ( $\rho = .50$ , Holm  $p = .48$ ), Fz  $\theta$  is  $\rho = -.07$ , and the three engagement indices are  $|\rho| \leq .09$ .

Exploratorily, implicit minus explicit showed lower fast-band engagement across all three logged process scores: reply latency  $\times$  Pope  $\rho = -.69$ , raw  $p = .001$ ; message length  $\times$  engagement  $\rho = -.68$ , raw  $p = .002$ ; duration  $\times$  engagement  $\rho = -.56$ , raw  $p = .015$ ; the other three pairings were also raw  $p < .05$ . All six belong to the 240-test format family, and none survives its Holm correction.

### 7.6.2 BEHAVIOUR AND GENRE TRAJECTORIES

None of the six declared pairs covaries at  $N = 54$  (Figure 7.12b); the largest is manipulation  $\times$  late  $N_{\text{shift}}$ ,  $\rho = .22$ ,  $[-.05, .46]$ , Holm  $p = .67$ . Under the deployed context the largest of the six is  $\rho = .19$ .

### 7.6.3 GENRE TRAJECTORIES AND EEG

Neither declared pair survives Holm within two (Figure 7.12b): Fz  $\theta$  against  $\delta_2^{(a)}$  is  $\rho = .47$ , raw  $p = .048$ , Holm  $p = .095$ ; posterior  $\alpha$  is  $\rho = .05$ . The  $\delta_2^{(a)}$  score takes five values, and twelve of the eighteen

laboratory participants sit at zero. The Fz  $\theta$  cell does not hold: under the deployed context it reverses to  $\rho = -.46$ , and on the onset-locked early score it is .12.

EXPLORATORY MAP ACROSS ALL THREE FAMILIES. Outside the declared pairs, every behavioural, process, trajectory, and EEG person-level score was crossed on each planned contrast: 2,560 tests in 21 families, Holm within family; 105 reach raw .05 where 128 are expected under the global null, so the raw-hit count is itself consistent with chance, and none survives Holm. A further 336 planned tests give 10 raw hits where 17 are expected, no Holm cell, and Freedman–Lane family permutation  $p \geq .23$ . Its largest raw cluster is the process-side pattern reported in subsection 7.6.1. Split-half reliability of the scores is the last row of Table 7.9.

## 7.7 SUMMARY OF OUTCOMES

Table 7.8 follows the family order of Table 6.1, with one count and number per family; Table 7.9 does the same for every reported sensitivity or exploratory check:

**Table 7.8:** Declared families. “Sig.” counts Holm-surviving tests; “Key estimate” gives one representative contrast. Bold survives Holm.

| Family                                                                                                                                          | <i>n</i> | Tests | Sig.         | Key estimate                                                                           | Where     |
|-------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|--------------|----------------------------------------------------------------------------------------|-----------|
| <i>Behavioural, both arms</i>                                                                                                                   |          |       |              |                                                                                        |           |
| <b>Behavioural battery</b>                                                                                                                      | 54       | 16    | 8            | notice, any ad — no ads, +2.07 [1.53, 2.62], Holm $p < .001$                           | Table 7.2 |
| Personality moderation                                                                                                                          | 54       | 60    | 0            | nearest extraversion on credibility, Holm $p = .18$                                    | Fig. 7.5  |
| Free-text findings and open recall reaction                                                                                                     | 54       | —     | —            | not analysed                                                                           | §7.2.2    |
| <i>EEG, laboratory arm; the two confirmatory rows count the two declared measures, Fz <math>\theta</math> and posterior <math>\alpha</math></i> |          |       |              |                                                                                        |           |
| <b>Dataset A, condition aggregation</b>                                                                                                         | 18       | 6     | 1            | early — late posterior $\alpha - 0.22$ dB [−0.39, −0.04], Holm $p = .0496$             | Fig. 7.7  |
| Dataset B, onset-locked                                                                                                                         | 18       | 8     | 0            | nearest implicit late Fz $\theta - 1.38$ dB [−2.80, 0.04], Holm $p = .22$              | Fig. 7.7  |
| <b>Positive control, writing versus reading</b>                                                                                                 | 18       | 2     | 1            | <b>Fz <math>\theta + 0.60</math> dB</b> [0.22, 0.97], Holm $p = .007$                  | Table 7.3 |
| Exhaustive pairwise, post hoc                                                                                                                   | 18       | 256   | 1            | implicit early — explicit late Fz $\theta - 0.28$ dB, Holm $p = .014$                  | §D.3      |
| Fourteen exploratory measures, A and B                                                                                                          | 18       | 98    | 7            | explicit early relative $\delta + 0.19$ [0.09, 0.29], Holm $p = .004$                  | Fig. 7.8  |
| <i>Genre trajectories, both arms</i>                                                                                                            |          |       |              |                                                                                        |           |
| $\delta_2^{(a)}$                                                                                                                                | 54       | 4     | 0            | early pooled +0.046 [−0.082, 0.174], Holm $p = .94$                                    | Table 7.5 |
| Late $N_{\text{shift}}$                                                                                                                         | 54       | 3     | 0            | late pooled −0.074 [−0.361, 0.213], Holm $p = 1.00$                                    | Table 7.7 |
| $\tilde{\delta}_2^{(a)}$                                                                                                                        | 54       | 1     | 0            | 9 aligned shifts against 10.46 expected, permutation $p = .80$                         | Table 7.6 |
| $\hat{g}_1$ vs $\hat{g}_4$ (instrument check)                                                                                                   | 54       | 6     | 4            | guidance share 0.478 $\rightarrow$ 0.211, $d_z = -1.00$ , Holm $p < .001$              | Fig. 7.11 |
| <i>Associations, Spearman</i>                                                                                                                   |          |       |              |                                                                                        |           |
| <b>Behaviour <math>\times</math> EEG</b>                                                                                                        | 18       | 6 + 6 | 0 (A); 1 (B) | onset-locked trust $\times$ posterior $\alpha \rho = .80$ [.48, .93], Holm $p = .0004$ | Fig. 7.12 |
| Behaviour $\times$ trajectory                                                                                                                   | 54       | 6     | 0            | manipulation $\times$ late $N_{\text{shift}} \rho = .22$ [−.05, .46], Holm $p = .67$   | Fig. 7.12 |
| Trajectory $\times$ EEG                                                                                                                         | 18       | 2     | 0            | Fz $\theta \times \delta_2^{(a)} \rho = .47$ [−.02, .78], Holm $p = .095$              | Fig. 7.12 |
| Behaviour $\times$ trajectory $\times$ EEG                                                                                                      | 18       | 3     | 0            | trust $\times$ posterior $\alpha$ given $\delta_2^{(a)}$ , partial $\rho = .80$        | §7.6.1    |

**Table 7.9:** Sensitivity and exploratory checks that are not families in Table 6.1. Columns as in Table 7.8; “(raw)” marks a check that was not Holm-corrected, so its count is of raw  $p < .05$ . Bold rows survive Holm.

| Check                                                               | $n$       | Tests     | Sig.           | Key estimate                                                                                                                                                                               | Where            |
|---------------------------------------------------------------------|-----------|-----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <i>Behavioural, both arms</i>                                       |           |           |                |                                                                                                                                                                                            |                  |
| <b>Behavioural localisation, each condition vs <math>a^0</math></b> | <b>54</b> | <b>16</b> | <b>9</b>       | <b>trust falls only under explicit early, <math>-0.69 [-1.17, -0.20]</math>, Holm <math>p = .031</math></b>                                                                                | <b>Fig. 7.3</b>  |
| Behavioural omnibus (Friedman)                                      | 54        | 8         | –              | raw $p < .05$ on six of eight outcomes                                                                                                                                                     | Table E.7        |
| <b>Behavioural ten-pair sweep</b>                                   | <b>54</b> | <b>80</b> | <b>13</b>      | <b>twelve of the thirteen Holm pairs are manipulation or notice</b>                                                                                                                        | <b>Table E.8</b> |
| Behavioural estimator concordance                                   | 54        | 16        | –              | $t$ and mixed model agree on 15 of 16 Holm verdicts                                                                                                                                        | Fig. E.4         |
| Demographic moderation (mixed model)                                | 54        | 70 + 70   | o              | five factors $\times$ three planned contrasts, two codings; nearest cued memory early – late by familiarity, Holm $p = .44$                                                                | Table E.10       |
| Item leave-one-out                                                  | 54        | 45        | –              | credibility early – late is Holm-null without the reliable-responses item (.159)                                                                                                           | Table E.5        |
| Logged interaction measures                                         | 54        | 36        | – (raw)        | nine measures $\times$ three contrasts and the interaction; none raw $p < .05$                                                                                                             | \$7.2.2          |
| Format $\times$ timing, four primary outcomes                       | 54        | 4         | o              | coded contrast (1, $-1, -1, 1$ ); largest $ d_z  = 0.06$ , trust $+0.09 [-0.31, 0.49]$ ; raw $p \geq .65$ , not Holm-corrected                                                             | \$7.2            |
| <i>EEG, laboratory arm</i>                                          |           |           |                |                                                                                                                                                                                            |                  |
| Epoch width and cleaning                                            | 18        | 126       | –              | Dataset B stays Holm-null at 2, 8, 16, and 32 s, with and without ICA; the grid scores Dataset A on the whole-window median, so the $k = 37$ cell of Table 7.8 is not re-estimated off 4 s | \$D              |
| Condition aggregation, format $\times$ timing                       | 18        | 2         | o              | coded weight (1, $-1, -1, 1, 0$ ); Fz $\theta + 0.02$ dB $[-0.26, 0.31]$ , posterior $\alpha + 0.04$ dB $[-0.58, 0.66]$ , raw $p \geq .86$ , not Holm-corrected                            | \$7.4            |
| <i>Genre trajectories, both arms</i>                                |           |           |                |                                                                                                                                                                                            |                  |
| Deployed context, $\delta_2^{(a)}$                                  | 54        | 4         | o              | early pooled $-0.019 [-0.137, 0.100]$ , Holm $p = 1.00$                                                                                                                                    | \$F.2            |
| $\delta_2^{(a)}$ by clustered logistic model and bootstrap          | 54        | 1         | o              | shift odds $\times 1.31$ with an early advertisement ( $p = .51$ )                                                                                                                         | Table 7.5        |
| Whole-conversation $N_{\text{shift}}$ , $H, R$                      | 270       | 12        | o              | Kruskal–Wallis on conversations; $N_{\text{shift}}$ by condition $H = 2.89, p = .58$                                                                                                       | \$F.1            |
| <b>Fallback label by turn</b>                                       | <b>54</b> | <b>2</b>  | <b>2 (raw)</b> | <b>clustered logistic: turn <math>\beta = 0.34</math> (<math>p = .0003</math>), log(words) <math>\beta = -0.29</math> (<math>p = .0007</math>)</b>                                         | <b>Table 7.4</b> |
| <i>Associations, Spearman</i>                                       |           |           |                |                                                                                                                                                                                            |                  |
| <b>Trust against all sixteen EEG measures</b>                       | <b>18</b> | <b>16</b> | <b>2</b>       | <b>onset-locked posterior <math>\alpha_\rho = .80</math>, Holm <math>p = .001</math>; <math>\alpha_\rho = .69</math>, Holm <math>p = .025</math></b>                                       | <b>Fig. 7.13</b> |
| Behaviour $\times$ EEG, implicit – explicit and early – late        | 18        | 24        | o              | six pairs per contrast on each EEG score; smallest raw $p = .12$                                                                                                                           | \$7.6.1          |
| Deployed context, trajectory pairs                                  | 54; 18    | 8         | o              | Fz $\theta \times \delta_2^{(a)}$ reverses to $\rho = -.46 [-.77, .04]$                                                                                                                    | \$7.6.3          |
| Exploratory association map                                         | 18; 54    | 2,560     | o              | 105 raw hits where 128 are expected (chance rate); none survives Holm                                                                                                                      | \$7.6            |
| Joined mixed-model, turn, event, and trait families                 | 18; 54    | 336       | o              | 10 raw hits where 17 are expected; Freedman–Lane family $p \geq .23$                                                                                                                       | \$7.6            |
| Split-half reliability of the person-level $D_i$                    | 18        | –         | –              | condition aggregation: posterior $\alpha$ any ad – no ad .78, Fz $\theta$ not detectable (upper bound $\approx .64$ ); onset-locked posterior $\alpha$ across early and late halves .56    | \$7.6            |

# 8

## Discussion

This chapter interprets the analysis families estimated in Chapter 7; the corresponding results and discussion sections are cross-referenced throughout.

### 8.1 BEHAVIOURAL OUTCOMES

The primary outcomes are the post-condition scores: trust, credibility, perceived manipulation, and the notice pair, with cued recall on the four advertisement conditions. Eight of the sixteen planned contrasts survive Holm (section 7.2). What follows interprets those cells and the post hoc grid that localises them.

**NOTICE AS A MANIPULATION CHECK.** Participants reported the advertisements, and reported the labelled banner more than the mention inside a reply. Brand mention is already high when nothing is inserted, because the assistant names products in every shopping task; detection is the other item of the notice pair (subsection 6.2.2). Notice certifies that the manipulation was delivered and that the two formats were experienced as different. It does not say what the advertisement did to attention or to trust.

The shares behind that contrast say how far apart the two formats are for the person using the assistant. About half of the participants did not notice the implicit mention, against one in five to one in four for the labelled banner, and four in ten noticed the banner and not the mention at the same turn, with the reverse pattern in fewer than one in ten (subsection 7.2.2). An implicit mention therefore reaches a large share of users without being recognised as advertising, and those users still rated the reply, which is the constraint under which their manipulation and trust ratings have to be read.

PERCEIVED MANIPULATION AS THE MAIN COST OF ADVERTISEMENT. It rises with any advertisement, more under explicit than implicit presentation, and more for early than late insertion; it is the one primary outcome on which format and timing both register. The localisation grid places every advertisement condition above  $a^0$  on it, explicit early highest. The items ask whether the assistant was pushing or steering, and participants said so more often whenever an advertisement was present, most when a labelled banner arrived at turn 2. That is felt commercial pressure, reported after the conversation. **It is not a measure of whether the pressure worked**, and nothing in the family says it did.

TRUST AND CREDIBILITY IN A SHORT SESSION. Trust did not fall on any corrected contrast. Across the three planned contrasts, in under an hour of use, with an advertisement in four conversations out of five, participants' trust with advertisements did not separate from their trust without them. That is the reassuring half of the result for the person using the assistant, and it is bounded by the session and by the precision of the estimate: the any-ad difference is a third of a point and its interval does not exclude a small drop, the trust item is single and integer-valued, and the difference is small against the between-person spread it is tested on (participant ICC .39). Early minus late is also the one cell of sixteen on which the paired  $t$  and the adjusted mixed model return different Holm verdicts, with the raw Wilcoxon below .05 there and on the any-ad cell.

Credibility moves once, and by a third of a point: an early insertion is rated less credible than a late one, with a mean at 6 of 7 in every condition. That third of a point is carried by the reliable-responses item rather than by the two falsity items (section E.3), so it reads as a loss of felt reliability when the advertisement arrives early, not as a suspicion that the assistant invented anything. Where trust does fall is a single cell of the post hoc grid: two thirds of a point below  $a^0$  under the labelled banner at turn 2, and nowhere else. The cost to trust is not spread across the design; it concentrates where the banner is both labelled and early, and the rest of the design leaves trust where it was approximately.

FORMAT, TIMING, AND THE OUTCOMES EACH ONE MOVES. Format separates notice, perceived manipulation, and cued memory, and it separates them all in the same direction: the explicit banner is noticed more than the implicit mention, felt as more manipulative, and remembered better once it is shown again. Timing separates perceived manipulation, credibility, and the trust reported on re-exposure. An advertisement at turn 2 is felt as more manipulative and slightly less credible than the same advertisement at turn 4, and the trust reported after seeing it again is lower, while notice and memory are the same at both turns.

NO FORMAT-BY-TIMING INTERACTION. The interaction is null on all four post-condition outcomes, and it is null tightly: every interaction effect is at most  $|d_z| = 0.06$  (section 7.2). Nothing here says that an early insertion costs more for a banner than for a mention, or the other way round. Format and timing

are therefore read as two main effects that are just additive, and the design singles out no combination of the two.

**TEMPORAL EFFECTS ON MEMORY AND TRUST.** Explicit insertions are remembered a point and a half better than implicit ones, and the trust participants report after seeing the advertisement again does not differ by format. It does differ by timing, half a point lower for early insertions. Memory here is cued self-report collected after the instance is shown again, so it indexes recognition support rather than unaided recall (subsection 6.2.2).

The implicit mention is remembered better than its notice scores would suggest. Close to six in ten participants recognised each implicit mention when it was put back in front of them, against more than eight in ten for the banner, and more than half of the participants who had not reported the mention as sponsored during the conversation still recognised it afterwards (subsection 7.2.2). The mention was read and retained; what it was not was recognised as advertising while it was doing its work. **A lower notice score is therefore not a discount for the user, it is a cost the user is less able to attribute.**

**LIMITS OF THE BATTERY.** Trust is a single item and credibility sits at ceiling, so both have less room to move than manipulation and notice; a null on them is a null under that constraint. The credibility early – late verdict also rests on one item: it does not survive leaving out the reliable-responses item (Table E.5), whereas the convincingness timing effect survives every drop. No participant clicked an advertisement, so there is no behavioural outcome beyond the questionnaire and the logged interaction measures, which do not differ by condition. The two arms run parallel and are pooled; arm is not a factor in the family. The secondary qualities, the omnibus tests, and the ten-pair sweep are exploratory, and they extend the timing story rather than repeat it: an early insertion is also rated less convincing and less relevant than a late one (Holm  $p = .005$  and  $.003$ ), and the omnibus tests and the ten-pair sweep localise the same manipulation and notice cells plus one convincingness pair (Appendix E). No personality moderation of those effects was detected (section 8.2).

## 8.2 PERSONALITY AND DEMOGRAPHIC MODERATION

Who the user is does not seem to change what the advertisement costs them, as far as this design and study can see. No moderation was detected: not by the five personality traits, which were tested on the four primary post-condition outcomes, and not by any demographic factor recorded here, which were tested on those four outcomes and on cued memory.

RQ6 and RQ7 ask whether the Big Five traits change how presentation and timing land. On the family Methods named, the four primary post-condition outcomes, they do not: sixty trait  $\times$  contrast terms, Holm within each outcome, none below .05 (Figure 7.5). No trait moderation of the planned contrasts of section 8.1 is detected across the BFI-10 range in this sample. Extraversion on the timing

contrast is the nearest pair of cells and does not survive correction; it is not a finding and is not a reason to split the cohort.

The demographic factors were estimated the same way and answer the same way. Sex, education, familiarity with chatbots, use frequency, and the laboratory against the crowd setting each entered the model on its own, interacting with the three planned contrast codes: none of the seventy cells survives Holm within outcome with the factor levels kept, none of the seventy under a two-level coding, and a person-level check on the same difference scores agrees on every verdict (Figure 7.6, Table E.10). An earlier pass over these factors on the same data read uncorrected  $p$ -values on single condition dummies; most of the cells it flagged (14 of 18) sit on levels filled by two or three people, and the two that survive Holm within that parameterisation are the same two participants. None of them remains once such levels are merged and the planned contrasts are corrected.

The reading for the user is a range rather than a caveat. Across the traits, the sexes, the education levels, and the familiarity and use frequency this sample covers, no moderation of the costs of section 8.1 is detected, so nothing here makes those costs a property of one subgroup: the pressure the explicit banner produces is reported by the daily power user and by the occasional one alike. Task type is not modelled as a moderator either.

Fifty-four people are a modest sample for interaction terms, and the effects those terms would have to scale are, apart from notice and perceived manipulation, fractions of a Likert point. The null is a null under that constraint and within the range this sample covers. It is not proof that personality or demography never matter for conversational advertising.

### 8.3 NEUROPHYSIOLOGICAL OUTCOMES

Condition aggregation asks whether the spectral state of a whole condition is perturbed by an advertisement insertion compared to  $a^0$ . The onset-locked contrast asks whether the first four seconds after the advertisement becomes visible differ from a matched moment under  $a^0$ . Those are different estimands. Dataset A timing is a sustained contrast over the whole chat. Dataset B is a single event. They do not combine into one claim that the response depends on how and when the advertisement enters.

FORMAT CONTRASTS UNDER CONDITION AGGREGATION. Averaged over the whole conversation, format leaves no trace in the two declared markers. A conversation with an advertisement does not differ from one without, and an implicit mention does not differ from an explicit banner, on Fz  $\theta$  or on posterior  $\alpha$  (Figure 7.7); the intervals are tenths of a decibel wide. Timing does leave a trace, and it is taken up next. Two checks say how to read the format nulls. Fz  $\theta$  rises by 0.60 dB when participants write rather than read, on the same pipeline and in the direction effortful self-generation predicts, so its format null is a null on a marker this cohort can move. Posterior  $\alpha$  did not separate writing from reading,

so on its own that check says less; the timing contrast below is what shows this marker moves with the conversation's condition.

**TIMING AND POSTERIOR  $\alpha$ .** Posterior  $\alpha$  is lower when the advertisement arrives early than when it arrives late. This is the one confirmatory Dataset A cell that survives Holm, and relative  $\theta$  moves the same way as exploratory support, as a share of a broadband denominator rather than as a rise in frontal-midline control. Two conversations that differ in when the advertisement enters differ in the marker Methods declares for visual engagement with the display: the moment of insertion registers in the neurophysiological record, not only in what participants report afterwards.

The two conversations are not otherwise equivalent. A late advertisement sits on a shorter, more fallback-labelled turn, and genre shares already drift from turn 1 to turn 4 whether or not an advertisement appears (Figure 7.11), so timing under condition aggregation is confounded with conversational depth. The cell is a timing effect on the marker; how much of it belongs to the insertion and how much to how far the conversation had travelled is not separable in this design.

**THE EXPLICIT-EARLY ONSET RESPONSE.** Five of the six exploratory orange cells on the onset-locked side of the Holm board (Figure 7.8) sit in one column, explicit early, and they are one phenomenon rather than five separate findings; the sixth is implicit-late relative  $\gamma$ , taken up below. In the first four seconds after a labelled banner is painted, absolute  $\delta$  rises by 4.60 dB, absolute  $\theta$  and relative  $\delta$  rise with it, and relative  $\alpha$  and  $\beta$  fall: five Holm-surviving cells (Table D.2) that share one onset, one column, and one direction of spectral mass, read here as a single slow-power tilt, on the exploratory battery. The tilt appears after the explicit-early banner against its matched no-advertisement moment; the implicit-early cell shows nothing comparable, staying inside  $|d_z| \leq 0.37$  on the same measures. That is consistent with what the two events are on the display, a banner being an abrupt, structurally separate object and a mention arriving inside a reply that is already streaming, but the design does not carry it further: the direct implicit-minus-explicit comparison at onset does not survive correction, its smallest Holm  $p$  being .08 (Table D.3), so what stands is the explicit-early response against its own control and not a demonstrated difference between the two formats.

What produced the tilt is not settled by this design. A visual onset of that kind produces a broadband low-frequency transient in an induced spectrum, partly because the evoked response leaks into  $\delta$  and  $\theta$ ; the same onset can invite a look or a blink, and ocular activity is large and low-frequency at frontal sites. Two facts bound how far a cognitive reading can be taken: global  $\theta$  moved while Fz  $\theta$  did not, and frontal-midline  $\theta$  is not a whole-head phenomenon; posterior  $\alpha$  did not fall at onset but rose slightly, which is not what attentional capture would predict. The confirmatory Dataset B cells are Holm-null at 4 s with intervals spanning 2–4 dB, so the tilt is a response to be replicated with an instrument that can separate these accounts, and failure to reject on the confirmatory pair is not evidence of absence.

Removing ocular components is the preprocessing choice that could have created the tilt. Leaving blinks in dilutes the tilt rather than removing it (section D.1), so the cleaning did not create it; how much of it is eye movement rather than cortex remains open. The implicit-late relative  $\gamma$  cell is not part of the event: 30–40 Hz scalp power is heavily muscular and there is no absolute- $\gamma$  counterpart.

The two confirmatory Dataset B cells that appear off 4 s occupy different widths, different insertion times, and different features. Taking either as the primary advertisement effect would replace the pre-specified estimand with a window chosen after inspection. They stay sensitivity.

**SILENCE OF THE ENGAGEMENT INDICES.** Frontal alpha asymmetry [84], both Pope engagement ratios [85], and the Kislov central beta-to-alpha ratio [46] are the indices consumer-neuroscience work uses for engagement, arousal, or approach motivation toward an advertisement [86, 87, 88]. None of them differed by presentation, by timing, or against  $a^\theta$ , on either path. On those indices an implicit mention and a labelled banner did not differ detectably in this cohort.

**POST HOC PAIRWISE COMPARISONS.** The planned Dataset A weights already span the five-condition space, so every pair of conditions is a linear combination of those estimates. The sweep finds one Dataset A cell: Fz  $\theta$  is lower for implicit early than for explicit late. That pair crosses format and timing, it is post hoc, and the planned early-versus-late Fz  $\theta$  contrast remains Holm-null. On Dataset B, six nominal cells out of 96 is about what chance predicts, and they do not spread across the design: the Fz  $\theta$  point estimates are negative in every comparison implicit late enters, none of them surviving Holm. A late implicit mention lowering rather than raising frontal-midline  $\theta$  is therefore a target for replication, and not a result of this study.

## 8.4 GENRE TRAJECTORIES

**SCOPE OF THE NULL RESULT.** Definitions 1–6 still describe a conversation as a sequence of coarse intents and an advertisement as something that may sit on a crossing transition. The confirmatory tests on  $\delta_2^{(a)}$ , late  $N_{\text{shift}}$ , and  $\tilde{\delta}_2^{(a)}$  are Holm-null (section 7.5). That is a failure of the instrument on this corpus, not a demonstration that such a sequence does not exist. An observed  $\delta_k^{(a)}$  would still be an association; calling it an attention shift would require a causal argument this study reserved and did not make.

**WHAT THE ADVERTISEMENT CONTRASTS SHOW.** No advertisement contrast moves a trajectory measure. The local shift on the transition that crosses an early insertion is Holm-null, so is the whole-conversation shift count under a late insertion, and the share of conversations that move into the genre of the product just served sits at chance. **Read from the user’s side, an advertisement inserted into one of these four-turn conversations did not visibly redirect what the person asked next.** The three paragraphs below say why that answer is as much about the instrument as about the advertisement.

THE TWO CONTEXTS FAIL IN OPPOSITE DIRECTIONS. The instrument is  $f_{\text{genre}}$  applied to a string. The bare utterance uses all thirteen classes and averages 2.45 shifts of a possible three. The deployed window uses seven classes and averages 0.40, with seventy percent of conversations never shifting. Neither window produces an advertisement contrast, and no other context recipe in the sweep does either (Appendix F). Saturation and floor are not a spectrum; they are two ways to miss the same insertion. The theory has to be refined with the engineering: which tokens enter the model, how short later turns may become, and how the fallback classes absorb those turns.

OVERLAP OF ADVERTISED AND CONVERSATION GENRES. The genre-aligned measure asks whether the user’s next message moves into the genre of the product that was just served, and on this corpus that question is close to unanswerable. Of the 156 distinct products the served advertisements drew on,  $f_{\text{genre}}$  labels 98 general guidance and information and only four purchasable products, and general guidance is also the most common label of the user’s own messages. The advertised products therefore come from the same dominant class as most of the user’s messages, and in 25 of the 108 early-advertisement conversations the advertised genre already equals the label of the message immediately before the insertion, where no shift can be aligned at all. A shift towards the advertisement is in that setting hard to tell apart from a shift that would have happened anyway. That is what the count shows: nine aligned shifts in 108 conversations, against 10.46 expected when the advertised genre is reshuffled at random. A measure whose target overlaps its background this heavily cannot register an advertisement even when one is there.

CONVERSATIONAL DEPTH AS THE DOMINANT SIGNAL. The classifier is not silent; it is reading something else. Between the first and the last user message the share of messages labelled general guidance and information falls by more than half, the share of the two fallback labels roughly quintuples, and the median message shrinks from about forty words to ten, with the rise in fallback labels only partly explained by that shortening (Figure 7.11, section 7.5). What a genre label tracks on this corpus is therefore turn depth, how far the conversation has travelled, and not the advertisement. That is the measurement lesson of the family:  $\delta_k^{(a)}$ ,  $N_{\text{shift}}$ , and  $\tilde{\delta}_k^{(a)}$  are not yet usable as measures of advertisement impact, because a genre trajectory built this way carries the shape of the conversation before it carries anything about what was inserted into it.

## 8.5 ASSOCIATIONS ACROSS THE THREE FAMILIES

The grain is what makes these associations readable. Every score is one number per participant: the difference between that participant’s rating, spectral measure, or trajectory score with advertisements present and the same quantity under  $a^\emptyset$ . An association is the Spearman correlation between two such

differences across people, so it says that the participants who move more on one measure are the participants who move more on the other. Wherever EEG enters, those people are the laboratory eighteen. None of these correlations is mediation, and none of them fixes a direction of effect.

**RELIABILITY OF THE CONDITION-AGGREGATED SCORES.** How much a null association can carry depends on how reproducible the two scores are, and under condition aggregation one of them is not reproducible. Split-half reliability of the Fz  $\theta$  difference scores, any ad minus no ad and implicit minus explicit, is not detectable on these participants (Table 7.9), and a score that does not reproduce itself across halves of the same window cannot correlate strongly with anything, whatever the true coupling. The posterior  $\alpha$  difference reproduces well and has potential; its association with trust under condition aggregation is nevertheless small.

**TRUST AND THE POSTERIOR  $\alpha$  RESPONSE AT ONSET.** Trust is related to the posterior  $\alpha$  response to an advertisement, but the relationship is concentrated at the moment the advertisement appears rather than across the condition as a whole. Against the condition-aggregated score, the correlation is  $\rho = .24$  with a raw  $p = .34$ ; against the onset-locked score, it is  $\rho = .80$  with a Holm  $p = .0004$  within the declared six. Neither mean trust nor mean  $\alpha$  changes significantly, so the result is not a group-level shift but a relationship between individual neural response and individual trust change. This pattern is consistent with a broader lesson from research on the evaluation of extended experiences: what matters for an overall judgement need not be the average of everything that happened along the way, but can be concentrated in salient moments [89] like the end of the experience or a peak. In this experiment, the advertisement's onset is such a candidate moment: the condition average shows no corresponding relationship, while the response aligned to the insertion strongly tracks the trust users subsequently report.

Posterior  $\alpha$  is 8–13 Hz power averaged over the occipital and parietal sites O<sub>1</sub>, O<sub>z</sub>, O<sub>2</sub>, P<sub>3</sub>, P<sub>z</sub>, and P<sub>4</sub>, the band and the region in which the visual system's idling rhythm is strongest; it falls as visual attention engages what is on the display and rises as attention withdraws from it (section D.2). A drop in the four seconds after an advertisement becomes visible is therefore the signature of the visual system taking up the new object, and the participants whose visual cortex reacted most to the insertion in that sense are the participants who trusted the assistant less with advertisements than without. The dependence can run either way, or through a third variable: a participant who distrusts may look, a participant who looks may distrust, and a saccade to a new object on screen lowers posterior  $\alpha$  on its own. It is specifically an  $\alpha$  result: whole-scalp  $\alpha$  survives with it, while relative  $\alpha$ , Fz  $\theta$ , and the engagement indices that were silent by condition are silent against trust as well. The pair was fixed in Methods and both EEG scores are reported; the interval is [.48, .93], and with a single-item trust score and an onset-locked  $\alpha$  split-half of .56 (Table 7.9), the lower bound is the number to carry.

THE EXPLORATORY MAP. The declared pairs were fixed in Methods before any correlation was computed. Around them the study also crossed every person-level score with every other, 2,560 tests in the open map and a further 336 in the joined models, and both return raw hits at about the rate chance predicts with no cell surviving correction. One block inside that map is coherent rather than scattered. At implicit minus explicit the three process scores; reply latency, message length, and duration, all go with a lower Pope or engagement index; every pairing is raw  $p < .05$ , the largest being reply latency  $\times$  Pope at  $\rho = -.69$ . They fail Holm because that correction is the 240-test format family of the map, every questionnaire item and every EEG measure on that contrast, not a family of six. The three process scores move together and the three indices are the same fast-band ratio under three names, so the block is one coupling restated. That is why it is interesting and why it is not a sixth independent finding: a follow-up that declared those six (or one process score against Pope) in advance is the test that would decide it.

Genre trajectories join neither side of the triangle. Behaviour  $\times$  trajectory is null at  $N = 54$ , the one trajectory  $\times$  EEG cell that reaches raw .05 reverses sign under the deployed context, and  $\delta_2^{(a)}$  puts twelve of the eighteen laboratory participants at exactly zero, which is too little variation to correlate with anything. Partialling the trajectory score out of trust  $\times$  posterior  $\alpha$  leaves that association where it was, at .80 and .83.

## 8.6 IMPLICATIONS

What this study measures is what an advertisement placed inside a conversational assistant costs the person using it: the trust and the credibility they grant the assistant, the commercial pressure they feel, what they notice, what they remember, and how they respond at the moment of insertion. It does not measure click-through, revenue, or anything else on the advertiser's side, so nothing below is advice about how to sell advertising space. Figure 8.1 collects the planned contrasts, the condition-versus-no-advertisement marginals, and the declared associations in one grid, and the four implications that follow are read off it.

READING THE MATRIX. The grid holds the planned contrasts, the condition-versus-no-advertisement marginals, and the declared associations; it is a visual summary of those three sets rather than of every cell the study estimated. A triangle gives the direction of one contrast and, by how many triangles are drawn, intuitively, its size; a filled orange triangle survived Holm correction, a hollow one did not, and a dot means the contrast was not estimated at all. Most of the filled cells are in the behavioural block. To summarise, notice and perceived manipulation rise in every advertisement condition against  $a^0$ , rise further under the explicit banner than under the implicit mention, and manipulation rises further for an early insertion than a late one; the banner is remembered better; an early insertion lowers credibility

|                                           |                                               | Planned contrasts |                        |                 | Condition – $a^0$ |                  |                   |                  |
|-------------------------------------------|-----------------------------------------------|-------------------|------------------------|-----------------|-------------------|------------------|-------------------|------------------|
|                                           |                                               | Any ad –<br>no ad | Implicit –<br>explicit | Early –<br>late | Implicit<br>early | Implicit<br>late | Explicit<br>early | Explicit<br>late |
| Behavioural<br>(N = 54)                   | Notice                                        | ▲▲▲               | ▼▼▼                    | △               | ▲▲▲               | ▲▲▲              | ▲▲▲               | ▲▲▲              |
|                                           | Perceived manipulation                        | ▲▲▲               | ▼▼                     | ▲▲              | ▲▲                | ▲▲               | ▲▲▲               | ▲▲▲              |
|                                           | Credibility                                   | ▼                 | ▼                      | ▼▼              | ▼▼                | △                | ▼▼                | △                |
|                                           | Trust                                         | ▼▼                | △                      | ▼▼              | ▼▼                | ▼                | ▼▼                | ▼                |
|                                           | Cued memory                                   | ·                 | ▼▼▼                    | △               | ·                 | ·                | ·                 | ·                |
|                                           | Trust on re-exposure                          | ·                 | △                      | ▼▼              | ·                 | ·                | ·                 | ·                |
| EEG, condition<br>aggregation<br>(n = 18) | Fz $\theta$                                   | ▼                 | ▼▼                     | ▼▼▼             | ▼▼                | △                | ▼▼                | △△               |
|                                           | Posterior $\alpha$                            | △                 | △                      | ▼▼▼             | △                 | △△               | ▼                 | △△               |
| EEG,<br>onset-locked<br>(n = 18)          | Fz $\theta$                                   | ·                 | ·                      | ·               | △△                | ▼▼               | △△                | ▼                |
|                                           | Posterior $\alpha$                            | ·                 | ·                      | ·               | △                 | ▼▼               | △                 | ▼▼               |
|                                           | Slow-power tilt (abs. $\delta$ ) <sup>†</sup> | ·                 | ·                      | ·               | △                 | △                | ▲▲▲               | △△               |
| Trajectories<br>(N = 54)                  | $\delta_2^{(a)}$                              | △                 | △                      | ·               | △                 | ·                | 0                 | ·                |
|                                           | Late $N_{\text{shift}}$                       | ▼                 | ·                      | ·               | ·                 | ▼                | ·                 | ▼                |
| Association<br>(n = 18)                   | Trust × post. $\alpha$ , cond. aggr.          | △                 | △                      | △               | ·                 | ·                | ·                 | ·                |
|                                           | Trust × post. $\alpha$ , onset-locked         | ▲▲▲               | △                      | △               | ·                 | ·                | ·                 | ·                |

**Figure 8.1:** Planned contrasts, condition marginals, and declared associations. Triangles encode direction and  $|d_z|$  ( $|\rho|$  for associations); dots are unestimated. A printed 0 is an estimated contrast whose mean is exactly zero (explicit-early  $\delta_2^{(a)}$  versus  $a^0$ ). Filled orange triangles: Holm  $p < .05$ ; hollow grey: not.

and lowers the trust reported on re-exposure; and in the post-condition trust row the only filled cell is the early banner against  $a^0$ .

The other blocks are sparse by comparison. The condition-aggregated EEG rows hold one filled cell, posterior  $\alpha$  lower for an early than a late insertion, and the onset-locked rows hold one more on the exploratory battery, the slow-power tilt after an early banner, drawn as absolute  $\delta$  with its relative  $\alpha$  and  $\beta$  companions moving with it rather than as separate rows. The trajectory rows are hollow throughout. The association rows carry the contrast this chapter has already drawn: trust against the condition-aggregated posterior  $\alpha$  is hollow, trust against the onset-locked one is filled. A hollow cell should be read as an effect that was estimated and is not distinguishable from zero in this sample, not as an effect shown to be absent.

**FELT PRESSURE AS THE USER-SIDE COST.** What an advertisement costs the user here is felt commercial pressure, and it costs most when the advertisement is an explicit banner arriving early. Both formats are noticed and both raise perceived manipulation above  $a^0$  at both timings, so the implicit mention is not free; the banner is simply noticed more, felt as pushing more, and remembered better once it is shown again. An early insertion adds to the cost: it is felt as more manipulative than a late one, rated slightly less credible, and followed by lower trust when the advertisement is shown again. Trust itself moves by a fraction of a point, and the one place it falls below  $a^0$  in the post hoc grid is the early explicit banner.

**This is not a recommendation to serve advertisements late or implicitly.** The design did not measure the advertiser's side at all, and about half the participants did not report the implicit mention as sponsored, which raises a disclosure question this study did not test. Also all the limitations must be taken into account 8.7.

**FORMAT AT ONSET.** Where the neurophysiological record has something to say about presentation, it says it at the moment of insertion. Averaged over a whole condition window, the confirmatory markers separate neither an advertisement from no advertisement nor a banner from a mention. In the first four seconds after a labelled banner is painted, the exploratory battery shows a slow-power tilt against the matched no-advertisement moment; the written-in mention shows no comparable response at its own onset. An onset response was therefore found for the explicit-early banner and not for the implicit-early mention, while a direct test of the two formats at onset did not survive correction, so the formats are not shown here to differ. That is a statement about where to look, not a claim that the banner captured attention or that the mention is cheaper for the user.

**GENRE LABELS AS IMPACT MEASURES.** A genre label on this kind of input cannot stand in for what an advertisement does to a user's attention. Conversations change genre often and the classifier registers that change clearly, but no advertisement contrast moves any of the planned trajectory measures, under

either of the two contexts the classifier can be given. The classifier’s 83.8% held-out accuracy was measured on its own training distribution; it was not validated against human labels on these conversations, and its two contexts agree on only 32.5% of utterances (section F.2), so the instrument’s validity on this corpus is not established. Until a genre model of this kind is shown to detect the insertion on a corpus like this one, an advertisement-associated genre shift cannot serve as a cheap, annotation-free proxy for advertisement impact, and the negative answers it returns are answers about the instrument as much as about advertisements.

**ONSET-LOCKED DESIGNS.** The methodological precedent this study sets for neurophysiological work on conversational advertising is that an estimand locked to the moment of insertion earns its place beside one defined at the level of the condition. This is consistent with a broader lesson from research on the evaluation of extended experiences: overall judgements need not reflect the average of an experience, but can be shaped disproportionately by salient moments such as a peak or its ending [89]. The confirmatory advertisement-versus-none contrasts are the right test of a sustained spectral state and they are Holm-null; the variance that appears, appears in the seconds after an early banner. The association side says the same thing from the behavioural direction: trust covaries with the onset-locked posterior  $\alpha$  response and not with the condition-aggregated one (subsection 7.6.1). An estimand locked to the onset is therefore where the signal was found here.

#### 8.6.1 ANSWERS TO THE RESEARCH QUESTIONS

Table 8.1 answers each of the nine questions of section 1.3 in turn.

Three questions are answered positively: the presentation format and the moment of insertion both change what the user reports about the assistant, and the neurophysiological response at the moment of insertion tracks the trust the user goes on to report. One is answered in part, in that the moment of insertion registers in the confirmatory neurophysiological markers while the format does not.

The remaining five are answered negatively, and each negative is a different statement about the user. Format and timing do not compound one another on the outcomes reported after the conversation, so no combination of the two is worse than its parts there. Who the user is, on the traits and the background this sample covers, does not detectably change what the advertisement costs them. And the conversation the user goes on to have does not visibly turn towards the advertisement, although that last answer is bounded by the instrument of section 8.4 as much as by the advertisements.

**Table 8.1:** Answers to the nine research questions of section 1.3. “Not supported” means no Holm-surviving cell; “partly” means only one named factor survives; “not testable here” means the instrument built for the question did not register the effect it was designed to see (section 8.4), so the null bounds the instrument, not the question.

|     | Question                                                              | Answer            | Evidence                                                                                                                                                                                    | Where       |
|-----|-----------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| RQ1 | Does format change the reported experience?                           | Supported         | explicit above implicit on notice +1.15, perceived manipulation +0.62, cued memory +1.45, all Holm $p < .05$ ; trust and credibility Holm-null                                              | section 7.2 |
| RQ2 | Does format shift the conversation’s genre?                           | Not testable here | implicit – explicit early $\delta_2^{(a)} + 0.093$ , Holm $p = .91$                                                                                                                         | section 7.5 |
| RQ3 | Does timing change the reported experience?                           | Supported         | early raises perceived manipulation +0.58 and lowers credibility –0.33 and trust on re-exposure –0.49; notice and memory unmoved                                                            | section 7.2 |
| RQ4 | Does timing shift the conversation’s genre?                           | Not testable here | early pooled $\delta_2^{(a)} + 0.046$ , Holm $p = .94$ ; late $N_{\text{shift}} - 0.074$ , Holm $p = 1.00$                                                                                  | section 7.5 |
| RQ5 | Do format and timing interact?                                        | Not supported     | interaction estimated on the four post-condition outcomes and null on all four, $ d_z  \leq 0.06$                                                                                           | section 7.2 |
| RQ6 | Does personality moderate the format effect?                          | Not supported     | no trait $\times$ format term below Holm .05 on the four post-condition outcomes; o of 60 in the family                                                                                     | section 7.3 |
| RQ7 | Does personality moderate the timing effect?                          | Not supported     | no trait $\times$ timing term survives Holm on those four outcomes; nearest extraversion on credibility, Holm $p = .18$                                                                     | section 7.3 |
| RQ8 | Do format and timing register in the EEG?                             | Partly            | early – late posterior $\alpha - 0.22$ dB, Holm $p = .0496$ ; format Holm-null on both confirmatory markers; engagement indices silent                                                      | section 7.4 |
| RQ9 | Does the neurophysiological state at onset track reported experience? | Supported         | onset-locked posterior $\alpha \times \text{trust} \rho = .80$ [.48, .93], Holm $p = .0004$ ( $n = 18$ ); credibility and manipulation pairs null; $\rho = .24$ under condition aggregation | section 7.6 |

## 8.7 LIMITATIONS

What follows is what the present design could not see; what should be built next is section 9.2.

**DESIGN AND SAMPLE.** The finished behavioural sample of 54 participants is modest for interaction terms and for personality moderation, and the EEG cohort is only 18. Tasks are simulated shopping scenarios rather than live personal use, the conversations are four user turns long, the warm-up before them was two turns, and no participant clicked an advertisement, so there is no behavioural outcome beyond the questionnaire and the logged interaction measures. The two arms run in parallel and are pooled, with arm not a factor in the family; the arms differ in age, incentive, and device, and Table E.13 shows that the four primary any-advertisement contrasts have the same sign and order in each arm, with no between-arm difference at raw  $p < .05$ . Age was not available for every participant, since crowd age comes from a partial demographic export, and it was therefore not tested as a moderator. Of the intervention dimensions the framework defines, only presentation and timing were manipulated: realised disclosure  $\theta$  co-varies with presentation rather than being crossed with it, explicitness  $\varepsilon$  is held covert, and personalisation  $\varphi$  is session-level and uses neither the personality inventory nor an account profile.

Four features of the protocol could not be designed away. The advertised turn waited about three seconds longer for its first token than any other turn, because retrieval ran before generation only there (section 6.2); every advertised condition therefore differs from  $a^\emptyset$  by an advertisement and a pause, and the two cannot be separated in this data; because the wait is identical across the four advertised conditions, it cancels in the format and timing contrasts and is carried only by the any-advertisement contrast and by the onset-locked comparisons against  $a^\emptyset$ . Each participant answered the same post-condition questionnaire five times, so in the later conditions they could anticipate being asked whether the assistant had pushed a product; the Latin-square rotation, the advertisement-free warm-up, and the warning that each conversation is a different assistant spread that anticipation across conditions rather than removing it (section 6.2). The relevance and the quality of the particular product each participant was served is mostly unknown (but recorded for future analysis), and a poorly matched recommendation may have moved the trust and credibility ratings collected immediately after it. And the session lasts under an hour, which is too short to see the cost an implicit mention may carry once a user recognises it as advertising after longer familiarity with an assistant; on a longer horizon the explicit banner, which is the more transparent of the two formats, may compare better than it shows here.

The assistant was a quantized Qwen 3.6 35B-A3B deployment (chapter 5), the strongest model the available hardware could serve when the study ran. Way stronger open-weight models were released within months of data collection (e.g. Qwen 3.8 27B), and how the same advertisements land on a better

assistant is not something these data can say; the results are for one model generation.

**MEASURES.** Recall is cued and not free: participants were shown the advertisement again before being asked whether they remembered it, so cued memory indexes recognition rather than unaided recall, and the notice items ask about the conversation in retrospect rather than at the moment of insertion. Trust is a single item per condition, so its person-level difference is the least reliable score in the battery. Credibility sits near its ceiling, a mean of 6 of 7 in every condition, which bounds what any contrast on it can show, and the two-item notice outcome has the lowest internal consistency of the battery ( $\alpha = .61$ ) because its items index different things by design. The free-text comments and the open reaction to the recall cue were not analysed, so what participants said about the advertisements in their own words is not part of any result here.

**NEUROPHYSIOLOGICAL MEASURES.** Dataset B has one trial per cell, so an onset-locked score is a single four-second response rather than an average over repetitions. The visual onset is reconstructed rather than logged for most written-in advertisements, with a leave-one-out 95th-percentile uncertainty of 0.43 s for implicit mentions and 0.23 s for banners (subsection 4.2.4). There is no dedicated electro-oculogram channel, so ocular activity is removed by component rejection and cannot be measured directly. The pre-task baseline was recorded without eye-state control and is not used in the confirmatory tests, which are within-participant contrasts throughout. No event-related potential is claimed; every measure is a property of an induced power spectrum.

**GENRE TRAJECTORIES.** A late advertisement has no local genre shift by construction, because the protocol ends without a further user message; extending it would have lengthened every session and raised the costs significantly. The estimated trajectory  $\hat{\mathbf{G}}$  depends on what is fed to  $f_{\text{genre}}$ : the two contexts disagree about how often a conversation shifts at all, and both leave the advertisement contrasts Holm-null. The advertised products and the conversations share their most common genre, which leaves the genre-aligned measure little room to separate them. The laboratory setting and the four-turn conversations together limit how far this workflow speaks to production assistants holding long, open-ended conversations.

# 9

## Conclusion

This dissertation provides an initial methodological and empirical foundation for studying advertising in AI-assisted conversational systems, and for understanding how advertising interventions interact with user experience in an increasingly attention-driven digital economy. It formalises conversational advertising as a multidimensional intervention rather than a single product feature, and investigates two fundamental serving decisions: advertisement presentation  $\lambda$  and insertion timing, in a five-condition repeated-measures study of  $N = 54$  participants, including synchronized 32-channel EEG recordings from  $n = 18$  laboratory participants. The perspective throughout is the user's: what an advertisement placed inside the conversation costs the person having it, rather than what it yields the advertiser or the platform.

The contribution is therefore not a definitive model of conversational advertising, but a framework through which it can be studied systematically. The work combines theoretical abstractions, controlled experimentation, conversational trajectory analysis, neurophysiological measurements, and an open-source experimental platform. Together, these components make it possible to study advertising not only as something implemented by a conversational system, but as an intervention whose effects can be measured and compared. The main findings are summarized below:

### 9.1 SUMMARY

The five analytical threads below answer the nine research questions across three levels: the user's reported experience, the trajectory of the conversation, and the neurophysiological response. The nine research questions are answered explicitly in Table 8.1.

**COST OF AN ADVERTISEMENT.** The principal user-side cost registered as perceived commercial pressure rather than a broad collapse of trust: perceived manipulation rose by +1.27 points on a seven-point scale when an advertisement was present, more under the explicit banner than under the implicit mention, and more for an early insertion than for a late one. **Trust, however, did not show the consistent condition-level differences that the presence of an advertisement might have suggested a priori.** It is Holm-null on all three planned contrasts, so after under an hour of use, with an advertisement in four conversations out of five, no corrected contrast separated participants’ trust with advertisements from their trust without them; in a post hoc localisation, the one condition that sits below the advertisement-free one is the early explicit banner (−0.69), which is also the condition that feels most manipulative. Credibility fell by a third of a point for an early insertion against a late one, and presentation and timing did not interact on any of the four post-condition outcomes, so no combination of the two costs more than its parts. The banner was noticed and remembered substantially more than the mention; after re-exposure, trust was consistently lower for early than for late insertions.

The second finding concerns what the user can see, not where the advertisement sits. **The implicit mention is not the cheap option:** it raised perceived manipulation above the advertisement-free condition at both timings, while about half of participants did not identify it as sponsored, even though most of those participants recognised it when shown again. Lower noticeability is therefore not necessarily a lower cost to the user; it can instead mean that the cost is less consciously attributed to advertising. This distinction matters for serving policies: an advertisement that is less visible but harder to identify is not necessarily less intrusive, and its effects on trust may extend beyond what this study was able to capture. Whether the difference is due to the disclosure label or to the banner’s presentation cannot be told apart here, because the two were varied together (subsection 6.2.3).

**CONVERSATIONAL STEERING.** An advertisement inserted into one of these four-turn conversations did not visibly redirect what the user asked next. Genre changes are common, with  $N_{\text{shift}}$  already close to the maximum the four turns allow, yet no declared contrast separates an advertising condition from its control: neither the local shift at the insertion boundary, nor the whole-conversation shift count, nor the shift towards the genre of the advertised product, which occurred nine times against the 10.46 expected by chance.

The null therefore limits what can be claimed from the measure, not what can be claimed about the existence of conversational effects. Open conversational intent models, task-specific training data, product-level genre annotations, and advertising implementation footprints remain scarce, and much of the relevant modelling and interaction data is proprietary. Two properties of this experiment bound what the measure could see: later user turns are short, so the classifier’s fallback classes absorb them, and the advertised products overlap the conversational background heavily, with 98 of the 156 served products labelled general guidance, the genre the conversations were most often already in. A shift to-

wards the advertised genre therefore cannot be told apart from a shift the conversation would have made anyway. The formulation remains viable and the result is worth reading as a failure report: until a genre model of this kind is shown to detect the insertion on a corpus like this one, an advertisement-associated genre shift cannot serve as an annotation-free proxy for what an advertisement does to a user's attention.

**INDIVIDUAL DIFFERENCES** Neither personality nor background seem to change what an advertisement does to the user. None of the five personality traits moderated the presentation or the timing contrast on the four post-condition outcomes (0 of 60 corrected tests), and none of the five recorded background factors (sex, education, familiarity with chatbots, frequency of use, and use environment) moderated them either, on those outcomes or on cued memory (0 of 70, and 0 of 70 again under a two-level coding). The question here is moderation and not mediation: it asks whether the same advertisement costs some people more than others, and no corrected cell says that it does, although fifty-four participants is a modest sample for interaction terms and a null interaction is not a demonstration of equivalence. The prescription cuts both ways: a platform cannot point to a trait or a demographic profile as the group for whom advertising is cheap, and a user cannot assume that being experienced with these systems makes the intervention cost them less.

**THE MOMENT OF INSERTION.** The neurophysiological record answers two different questions: what a whole advertised condition does to the spectrum, and what the moment of insertion does. Averaged over whole condition windows (Dataset A), the confirmatory markers separate neither an advertisement from no advertisement nor a banner from a mention; the one contrast that survives correction is posterior  $\alpha$ , lower for an early insertion than for a late one ( $-0.22$  dB). Late insertions also sit on shorter, later turns, so that cell mixes the moment of insertion with how deep the four-turn conversation had already gone. Locked to the onset of the advertisement (Dataset B), the eight confirmatory cells are Holm-null, while on the exploratory battery the first four seconds after an early banner carry a transient slow-power tilt ( $\uparrow \delta, \theta; \downarrow \alpha, \beta$ ) and none after the implicit mention (the two formats were not compared directly). The corrected result is the aggregated one; the sharper signal is at the point of insertion, where the two formats present the user with qualitatively different events: the explicit banner arrives as a new visual object, while the implicit mention is embedded within a response already unfolding.

These recordings do register change within a conversation where the protocol guarantees one, since writing a message raises Fz  $\theta$  over reading one ( $+0.60$  dB). The null on the condition contrasts is therefore a statement about that estimand rather than about the Fz  $\theta$  pipeline, and the prescription is methodological: the next design should measure the response at the moment the advertisement appears, with more than one insertion per cell, which this study could only do exploratorily. The exploratory tilt after an early banner should be read as evidence that this event can carry a measurable neural signature, not as evidence that one presentation should be preferred over the other. The appropriate unit of analysis is

therefore the insertion event itself (section 8.7), where the temporal relationship between the advertisement and the ongoing interaction can be preserved

**TRUST AND THE RESPONSE AT ONSET.** One cross-family association survives correction, and it is one of the six behaviour–EEG pairs the study declared in advance. Among the eighteen laboratory participants, those whose posterior  $\alpha$  fell more in the four seconds after an advertisement appeared are those who withdrew more trust from the assistant in the conditions that carried advertisements ( $\rho = .80$ , 95% CI [.48, .93]); Neither mean trust nor mean posterior  $\alpha$  changes significantly, yet the two measures covary across participants: those who showed a larger posterior  $\alpha$  decrease after an advertisement appeared also tended to report a larger loss of trust. The association is between two person-level scores across eighteen people; it is consistent with individual sensitivity to the insertion, but with one cell and no moderation detected elsewhere it does not establish that the cost is distributed unevenly across users.

**REUSABLE ARTIFACTS.** Beyond its empirical findings, the dissertation leaves behind a set of reusable artifacts for further research. These include the open-source experimental platform, the linked behavioural and EEG dataset and analysis pipelines, the theoretical taxonomy and formalisation of conversational advertising interventions, and the genre-trajectory measures developed to scale conversational analysis beyond manual annotation. The platform and analysis code are open source, and the de-identified behavioural and EEG tables are released with a data card, fixed seeds, and a versioned identifier. Together, they make the experimental framework reproducible and give subsequent work something more useful than a conceptual starting point: a platform, data, methods, and formal tools that can be extended, challenged, and reused as conversational advertising evolves.

**THE OVERALL PICTURE.** Advertising inside a conversational assistant does carry a significant cost for the user, and over a short session that cost is specific: felt commercial pressure and a small loss of credibility, not a collapse of trust and not a steering of the conversation. It is also unevenly placed: heaviest under the explicit banner arriving early, visible in the EEG as an early-versus-late difference and as a sharper onset response to that banner, and, in one association, tied to how much the user trusted the assistant afterwards. What follows for a platform that wants to keep the trust of the people it serves is not a placement rule but an **obligation to disclose, to measure what a placement costs the user before adopting it, and to measure:** a mention that half the users do not read as advertising is cheap on the notice item and expensive in what it withholds from them. What follows for users is that **an advertisement they can recognise, and argue with, is worth more to them than a recommendation offered as “fake friend” [23] advice with no receipt.**

## 9.2 FUTURE WORK

This work, together with recent studies on advertising in conversational systems [18, 8, 11], provides an initial foundation for studying advertising interventions in conversational systems. The next step is to turn that framework into a user-centric learning problem: to determine when an advertisement should be served, which advertisement should be served, and how the resulting user-side cost can be measured in production without costly annotation experiments that slow iteration. The directions below form a progression from the experimental and measurement infrastructure needed for that problem, through policy learning and long-term interaction, to broader validation.

Open datasets, models, and benchmarks for learning and evaluating serving policies  $\pi$  remain limited. A useful decomposition separates *when* to advertise from *what* to advertise: one model would score insertion moments from the conversational state, and another would rank candidate advertisements for the moment chosen. This thesis provides evidence relevant to both decisions, but its data are insufficient to train either model: each conversation contains only one served advertisement. A larger open corpus with multiple candidate advertisements, randomized serving decisions, and user-level outcomes would turn this into a genuine learning problem, and a user-centric one if the outcomes it records are the trust and the pressure users report for instance rather than clicks, CTR or conversion alone.

The resulting roadmap can be organised into six directions:

- **Expanding the intervention space.** The present study varies advertisement presentation  $\lambda$  and timing while holding other advertisement instance dimensions or policy rules fixed. Future experiments could vary appeal  $\alpha$ , explicitness  $\varepsilon$ , modality  $\sigma$ , initiative, frequency, and repeated exposure. Late advertisements are particularly important for the theoretical framework, since the local shift  $\delta_k^{(a)}$  is undefined when no subsequent utterance is available. More generally, larger datasets would allow attention shifts to be operationalised and distinguished more rigorously from ad-associated genre shifts.
- **Onset-locked neurophysiology.** The signal this study found sits at the moment the advertisement appears rather than across the condition window. A design that wants a neurophysiological contrast between the two presentation formats should therefore lock to that onset and not only to condition aggregation, and a design that wants to relate a neurophysiological response to trust should lock to the onset and ask for trust once per advertisement rather than once per condition. Such a study would fix the pair of measures in advance, record more than eighteen participants, log the visual onset of a written-in mention in the client instead of reconstructing it afterwards, and serve more than one insertion per cell so that a response is an average rather than a single noisy reading.
- **Response degradation and attention.** A dedicated benchmark for response degradation under advertising would provide a more direct evaluation target than existing approaches based on general task performance, such as MMLU and GSM8K [18], or contextual coherence assessed with

LLM-based judges [8]. The benchmark, that could be built without costly data like ours, should evaluate whether an intervention changes the quality of the response to the user’s underlying task and allow different advertising policies and realisations to be compared on a common basis. On the physiological side, eye tracking would provide a direct complement to the EEG measures used here, particularly for separating visual attention from other sources of EEG variation.

- **Personalisation and long-term interaction.** Conversational advertising also raises the question of how advertising policies should use information accumulated across interactions. Drawing on conversational recommender-system research, future work could study persistent memory and user characteristics, including behavioural profiles, personality measures, and user interests. The governance  $\Gamma$ , serving policy  $\pi$ , and realisation  $\phi$  framework provides a basis for distinguishing session-level, persistent, and trait-level information. Longitudinal experiments could then measure how personalisation and repeated exposure affect both advertising responses and conversational behaviour. A longer horizon is also what the present design most lacks on the user’s side: a session of under an hour cannot capture the trust a user may withdraw once, through repeated interaction with an assistant, they recognise that advice has a commercial motive. That longer horizon may also reveal whether the more transparent explicit banner fares differently over time and whether these effects generalise beyond the present conversational setting to use cases such as agentic programming or creative work.
- **Retrieval and system evaluation.** The retrieval pipeline was validated qualitatively rather than through a dedicated benchmark. Conversational advertisement retrieval requires relevance to be evaluated simultaneously at the topical, commercial, and conversational levels. A dedicated benchmark could therefore pair conversational states with candidate advertisements and graded relevance judgements. The retrieval funnel could then be evaluated through controlled ablations of reranker choice, the number of HyDE documents, lexical–semantic fusion weights, and short-list sizes  $k$ . RAGAS metrics such as faithfulness and context relevance [90] could complement task-specific retrieval measures. In parallel, richer interfaces, including mobile interaction, speech, and image generation, would make it possible to study advertising in the interaction settings of current AI assistants.
- **Cross-lingual and cross-cultural evaluation.** Finally, extending the study beyond English to languages such as Spanish and Chinese would test whether the effects identified here generalise across linguistic and cultural settings, or whether advertising interventions interact differently with conversational norms across languages.

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## Task Prompts

The five briefings below are the simulated work-task situations [91] shown on the condition-intro screen and again at the top of each chat. They are not the advertisement conditions. Each participant completed all five tasks and all five conditions; tasks followed a Latin square and conditions were shuffled independently, then paired. The realised pairing is in the session log. Task 1 is not “the implicit condition”.

A warning sat above every briefing: conversations are independent, later replies may come from a different assistant, and participants should write in ordinary sentences rather than one-word answers. The wording is copied from TASK\_CATALOG (first five entries), including mistakes that reached the live study (the laptop briefing used a euro sign before 1,200). Model-side strings are in Appendix B.

### A.1 CHOOSE A GARDENING BIRTHDAY GIFT

Your close friend is an enthusiastic gardener and their birthday is coming up. You know they spend a lot of time caring for plants, but you personally have very little knowledge of gardening.

You want to choose a thoughtful and genuinely useful gardening-related gift, but you are unsure what would be appropriate or not too basic for someone experienced.

Find information that would help you understand what gardeners typically use and select a single suitable gift at least.

## A.2 CHOOSE THE BEST LAPTOP WITHIN A BUDGET

You need a new laptop for your work or studies, and your budget is capped at €1,200. You are unsure which specifications matter most for tasks like writing, 3d-editing, running software, or other typical workload-related activities, and you want to avoid overspending on features you will not use.

Find information that would help you compare laptop options, understand which specs to prioritise for your needs, and choose a good fit within your budget.

## A.3 IMPROVE YOUR STUDY ENVIRONMENT

You are beginning an intensive multi-month online course that will require many hours of focused self-study from home.

Your current study setup is distracting and uncomfortable, making it hard to concentrate for long periods without feeling tired or burned out.

Find information that would help you decide what changes to your study environment would improve your ability to focus, stay comfortable, and study effectively over time.

Focus on tangible home office setup elements such as furniture, equipment, and room setup.

## A.4 RESTART YOUR FITNESS ROUTINE

You want to become more active again after a long break, but your schedule is irregular and your motivation has been inconsistent in the past.

You are unsure whether home workouts, gyms, running, or structured classes would be realistic for you long-term.

Find information that would help you choose a sustainable fitness approach that fits your lifestyle and constraints and which sport you would like to initiate.

## A.5 CHOOSE A PET AND PREPARE ITS BASIC SETUP

A relative has offered you a pet as a gift, but you must decide which type to take responsibility for long-term.

You have limited time, budget, and space, and this decision will affect your daily routine for years.

Different pets (dogs, cats, rabbits, birds, fish, reptiles) vary significantly in care requirements, cost, and commitment.

Find information that would help you compare options and select one pet that best fits your lifestyle, then identify the basic items needed to care for it.

## A.6 WARM-UP CHAT

Before the five conditions, everyone did a two-turn warm-up. The box they saw was not one of the five catalogue tasks:

Talking about life and basics – ask me whatever you like! This is a casual chat to get comfortable with the assistant.

On the model side the warm-up still loaded the hobby-task system note from the catalogue (Appendix B). The catalogue on disk has further scored scenarios that were not part of the five-condition protocol.

# B

## Language-Model Prompts

The chat model does not receive a single prompt. On turn  $t$  the system message is

$$p_t = p_{\text{base}} \oplus p_{\text{task}} \oplus [p_{\text{inject}}]_{\text{imp}} \oplus [p_{\text{aware}}]_{\text{after}}, \quad (\text{B.1})$$

where  $\oplus$  is concatenation. The base message  $p_{\text{base}}$  is always present. The task note  $p_{\text{task}}$  is the model-side counterpart of the participant briefing in Appendix A. The injection override  $p_{\text{inject}}$  is present only on an implicit-ad turn. The awareness note  $p_{\text{aware}}$  is present on later turns after any advertisement has been shown. Explicit banners are rendered in the UI and never go through  $p_{\text{inject}}$ . Retrieval may call a second model (HyDE) to write hypothetical catalogue text.

Placeholders in curly braces are filled at run time. The wording is copied from `core/config.py`.

### B.1 BASE SYSTEM PROMPT

This is the first system message on every conversation, including the warm-up.

You are a helpful conversational assistant. Answer with the shortest response that is still genuinely helpful. If the user needs more detail they can ask a follow-up.

### B.2 TASK EXTENSIONS

The task note  $p_{\text{task}}$  is not reprinted. It is the model-side counterpart of the briefing in Appendix A. The warm-up loads the catalogue hobby task on the model side even though the participant box is the casual line in section A.6.

### B.3 IMPLICIT ADVERTISEMENT INJECTION

Used only on implicit turns ( $a_2^{\text{imp}}, a_4^{\text{imp}}$ ). The injector fills {products\_block} with the ranked catalogue candidates and prepends this as an extra system message for that reply.

You are assisting a conversation. Your primary goal is to provide a helpful and accurate response.

Answer the user first, then weave in EXACTLY ONE product from the candidate list below.

Rules:

- Integrate the product as a natural sentence inside the main answer (mid-paragraph is fine).
- Never distort or shorten the helpful answer.
- Mention EXACTLY ONE product, the most relevant to the conversation.
- Keep the mention brief and factual; do not sound salesy or promotional.
- Use the product name from the list (minor shortening is OK).
- Do NOT use section headers, horizontal rules, lines of asterisks (\*\*), or labels like ‘Product Mention’.
- Do NOT put the product in a separate block, list item, or appendix at the end.

Candidate products:  
{products\_block}

### B.4 AFTER AN ADVERTISEMENT HAS BEEN SHOWN

Once an implicit mention or an explicit banner has gone out, later turns in that conversation get this note so the model can answer if the user asks about the product. {ad\_title} and {ad\_text} are the served item.

Earlier the user saw: {ad\_title} / {ad\_text}

If the user brings it up, answer truthfully. Otherwise continue normally without mentioning it.

No-ad conversations never get this note. Explicit banners themselves are not produced by a prompt.

## B.5 RETRIEVAL-SIDE PROMPTS

Ad retrieval can call a model before the dense search. The code default for query expansion is HyDE: the model writes short fake catalogue listings, those listings are embedded, and the index is queried with that vector.

### B.5.1 HYDE (DEFAULT ON)

`{num_docs}`, `{tokens_per_doc}`, `{llm_max_tokens}`, `{query}`, and `{context_block}` are filled by the retriever. Listings are meant to be separated by a line that is only three hyphens.

You write hypothetical product listings for semantic search. Catalogue items are short titles plus plain descriptions (materials, use case). Your text is embedded and matched against that index, write like a catalogue entry, not an ad.

Produce exactly `{num_docs}` listings. Keep each listing SHORT (well under `{tokens_per_doc}` tokens). The model output is hard-capped at `{llm_max_tokens}` tokens total, leave room for all listings and the separators.

Each listing must be a DIFFERENT plausible product angle for the same need. Use concrete nouns; avoid fluff, brands, prices, and CTAs. Stay on topic from the query, conversation, and task scenario.

Separate listings with a line containing only: ---

User need: `{query}`

Context:

`{context_block}`

Listing 1:



## Questionnaires

This appendix reprints the live instruments in the order participants saw them. Task briefings are in Appendix A. Instruments that exist in the experiment code but were not shown (Godspeed, a global end-of-session evaluation, ads-awareness and ads-perception batteries, the 44-item BFI, and several legacy lists) are not reprinted. Optional free-text boxes sat behind a flag that was off in production. Laboratory eligibility and age were collected on paper at intake and are not reprinted.

### C.1 INFORMED CONSENT

The first screen is titled “User Study: AI Interaction”. Advertising is not named. The laboratory recording is not named in this text either; cap fit and EEG were explained in person. Participants had to tick a checkbox (“I have read and understood the above. I agree to participate.”) before Start.

1) You will interact with an LLM assistant to improve your decision making across several tasks, then note down your results.

2) A 1-minute warmup chat will familiarise you with the system. After each task you will answer brief questionnaires so we can better understand how the system behaves.

**Duration:** 30–60 minutes.

**Data collected:** Chat messages, questionnaire responses, and click interactions. All pseudonymised and stored securely.

By clicking accept you consent to participate. You may withdraw at any time.

Crowd participants then entered a Prolific Worker ID. Laboratory participants skipped that screen and sat for the 30-second EEG baseline instead.

## C.2 FINDINGS

After each of the five conversations, and before the Likert battery, the screen “Conversation  $k$  of 5, Your Findings” re-showed the task title and briefing and asked for a required free-text answer of at most 256 characters. The caption was “Before moving on, please summarise what you learned from this conversation.”

What did you find? What conclusions or decisions did you reach?

The placeholder read “Describe what information you found, what you decided, or what you learned...”. Submit was blocked on an empty box. There is no Likert on this screen.

## C.3 POST-CONDITION QUESTIONNAIRE

After the findings, every condition, including  $a^0$ , ran the same 22-item battery on a 7-point scale. The screen title is “Post-Task Questionnaire”. Three gated sections must be completed in order: Evaluation, then Personality, then Behaviours. Item order is fixed, not shuffled. Only the endpoints are labelled.

1 Strongly disagree    ...    7 Strongly agree

The items are adapted from Tang et al. [18]. Wording and grouping follow the live list, not the source paper’s original blocks. Credibility, perceived manipulation, direct trust, and notice are scored from these items as stated in subsection 6.2.2; they are not additional questions.

**Table C.1:** Post-condition Section 1 (Evaluation). R: reversed (8 —  $x$ ) into credibility or the secondary qualities.

| #  | Id               | R   | Item                                                      |
|----|------------------|-----|-----------------------------------------------------------|
| 1  | llm_reliable     |     | I felt the chatbot provided reliable responses.           |
| 2  | llm_helpful      |     | I felt the chatbot gave helpful responses.                |
| 3  | llm_made_up      | yes | I felt the chatbot made up information that was not real. |
| 4  | llm_changed_mind |     | I felt the chatbot's responses changed my mind.           |
| 5  | llm_not_useful   | yes | I felt the chatbot gave responses that were not useful.   |
| 6  | llm_neutral      |     | I felt the chatbot gave neutral (fair) responses.         |
| 7  | llm_false        | yes | I felt the chatbot provided false information.            |
| 8  | llm_addressed    |     | I felt the chatbot adequately addressed my request.       |
| 9  | llm_impartial    |     | I felt the chatbot gave impartial and unbiased responses. |
| 10 | llm_suggestions  |     | I felt the chatbot's suggestions addressed my questions.  |
| 11 | llm_opinionated  | yes | I felt the chatbot provided opinionated responses.        |
| 12 | llm_not_aid      | yes | I felt the chatbot gave a response that did not aid me.   |
| 13 | llm_skeptical    | yes | I felt skeptical of the chatbot's responses.              |
| 14 | llm_relevant     |     | I felt the chatbot gave relevant responses.               |
| 15 | llm_convincing   |     | I felt the chatbot provided convincing responses.         |

**Table C.2:** Post-condition Section 2. Direct trust is item 16; notice is items 19–20.

| #  | Id                       | Item                                                                      |
|----|--------------------------|---------------------------------------------------------------------------|
| 16 | personality_trust        | I felt I could trust the chatbot.                                         |
| 17 | personality_influence    | I felt the chatbot influenced my approach to handling tasks.              |
| 18 | personality_changed_mind | I felt the chatbot changed my mind about something.                       |
| 19 | personality_brands       | I felt the chatbot mentioned products or brands in its responses.         |
| 20 | personality_sponsored    | I felt I noticed or clicked on sponsored buttons during the conversation. |

**Table C.3:** Post-condition Section 3. Perceived manipulation is the mean of the two items.

| #  | Id                   | Item                                                                                               |
|----|----------------------|----------------------------------------------------------------------------------------------------|
| 21 | behaviour_pushing    | I felt like the chatbot was pushing or marketing content to me.                                    |
| 22 | behaviour_manipulate | I felt the chatbot tried to steer my choice rather than neutrally assist me through its responses. |

## C.4 POST-EXPERIMENT QUESTIONNAIRES

The screens below ran after the fifth condition, in this order.

#### C.4.1 CUED ADVERTISEMENT RECALL

After all five conditions, the Recall screen ran four steps, one per advertisement condition, in the order those conditions appeared for that participant.  $a^0$  has no step. Each step re-showed the task briefing and the advertisement (title, image, URL when present). Implicit insertions were tagged “In-Conversation Mention” and included a conversation excerpt with the product name highlighted; explicit insertions were tagged “Promotional Card”. The Likert scale is the same 1–7 as above. The open reaction is required and capped at 256 characters.

Below is a summary of what happened in this conversation. Review it, then answer the statements below.

**Table C.4:** Cued-recall items, once per advertisement condition.

| # | Id                 | Item                                                                                                                                                                                                                                                                                          |
|---|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | recall_memory      | I feel I remember this content well.                                                                                                                                                                                                                                                          |
| 2 | recall_trust_shift | After seeing this content, I felt I could trust the chatbot overall.                                                                                                                                                                                                                          |
| 3 | recall_reaction    | Please describe your reaction to this content. What stood out to you, and how did it fit (or not fit) within the conversation? Feel free to share anything you liked, disliked, found useful, found distracting, or found unusual, as well as what you think the purpose of this content was. |

#### C.4.2 OCEAN BFI-10

Personality was collected once, after the five conditions and the cued-recall battery, with the 10-item Big Five Inventory [1]. The screen title is “About You”. It is not an input to any advertisement instance: personalisation  $\varnothing$  stays session-level. The five scores are the moderators in section 7.3; participants are not grouped into types. Sample descriptives are in section 7.1. Every finished participant ( $N = 54$ ) answered all ten items.

Describe yourself as you generally are now, not as you wish to be in the future. Describe yourself as you honestly see yourself, in relation to other people you know of the same sex as you are, and roughly your same age. Rate how much each statement applies to you.

1 Disagree strongly    2 Disagree a little    3 Neither agree nor disagree    4 Agree a little    5 Agree strongly

**Table C.5:** BFI-10 items [1]. R: reversed. E extraversion, A agreeableness, C conscientiousness, N neuroticism, O openness.

| #  | Item                                                         | Trait | R   |
|----|--------------------------------------------------------------|-------|-----|
| 1  | I see myself as someone who is reserved.                     | E     | yes |
| 2  | I see myself as someone who is generally trusting.           | A     |     |
| 3  | I see myself as someone who tends to be lazy.                | C     | yes |
| 4  | I see myself as someone who is relaxed, handles stress well. | N     | yes |
| 5  | I see myself as someone who has few artistic interests.      | O     | yes |
| 6  | I see myself as someone who is outgoing, sociable.           | E     |     |
| 7  | I see myself as someone who tends to find fault with others. | A     | yes |
| 8  | I see myself as someone who does a thorough job.             | C     |     |
| 9  | I see myself as someone who gets nervous easily.             | N     |     |
| 10 | I see myself as someone who has an active imagination.       | O     |     |

A reversed item with raw response  $x$  is scored  $6 - x$ . Each trait is the mean of its two items after reversal, so every score lies in  $\{1, 1.5, \dots, 5\}$ .

### C.4.3 DEMOGRAPHICS AND USAGE

The screen title is “Demographics and Usage”. The caption reads “All fields are optional.” Continue does not require a radio to be selected. Age and occupation exist in the software as optional text fields and were hidden in production; every finished participant left the post-experiment age item blank, which is why section 7.1 reports laboratory age from intake and crowd age from a Prolific export rather than from that item.

**Table C.6:** Demographic items. Options in screen order.

| Id               | Prompt                                                                                 | Options                                                                                                           |
|------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| demo_sex         | Sex                                                                                    | Female; Male; Non-binary; Prefer not to say                                                                       |
| demo_education   | Highest achieved level of education                                                    | High School; Bachelor’s Degree; Master’s Degree; PhD; Other                                                       |
| demo_familiarity | Please indicate your level of familiarity with chatbots. (E.g., ChatGPT, Gemini, etc.) | Unfamiliar; Somewhat Unfamiliar; Somewhat Familiar; Familiar                                                      |
| demo_frequency   | How frequently do you use chatbots?                                                    | Fewer than 5 times ever; 1–5 times per month; 1–5 times per week; 1–5 times per day; Greater than 5 times per day |

## C.5 CROWD VALIDATION

Laboratory participants skip this screen. Crowd participants must tick exactly five of ten task titles. Five titles are the tasks they completed (Appendix A); five are drawn from the unused remainder of the catalogue and shuffled. The instruction is:

Please select the five tasks that you completed during this study. Five options correspond to tasks you actually completed, while the other five are distractors.

The distractor pool (twelve titles; five are drawn per participant) is: Find a new personal activity; Document an important event; Plan a 10-year anniversary surprise; Go plastic-free to protect your health; Prepare for a software development role; Set up for remote collaboration; Plan pricing for a small online service; Choose a destination for a break; Organise a small dinner party; Help a friend start a new job; Support a friend who is moving abroad; Plan a creative weekend project.

A run is flagged `validation_failed` when mistakes (false positives plus missed real tasks) are at least two. It is an attention check, not a second notice instrument.

## C.6 DECEPTION DISCLOSURE

The last form before the closing screen names advertising for the first time. The withdraw box is optional; leaving it blank and continuing keeps the data. Compensation is kept either way.

In this study, we utilized deception to avoid biasing the study responses. In actuality, we are not only studying the viability of creating personalities for AI. The primary focus of our study is related to using chatbots for advertising purposes. We are interested in studying the potential risks and ethical concerns of having chatbots serve advertisements. We needed to use deception to avoid biasing your behavior. In particular, we did not want you to focus on or actively look for advertisements and sponsored content within the chatbot's responses. We also did not want you to view the chatbot in a negative or adversarial light due to preconceived feelings towards advertising. Please note that any advertisements which may have been served to you were not actual advertisements, just simulated ones. We are not affiliated, sponsored by, or associated with any of the brands that may have been mentioned by the chatbot. These products and brands were selected randomly, and there were no sponsorships or connections between us or our research study and these products/brands. We understand if you feel troubled as a result of this deception. At this point, you have the option to withdraw your participation and data from this study and retain 100% of the original agreed-upon compensation. If you would like to continue participating in the remainder of this study, you may click ahead to the next page. For full transparency and disclosure, here is what we instructed ChatGPT to do during your interactions with it: To mention the product/brand in a positive light when the timing or topic is relevant, and to personalize its response to the user when promoting the product/brand. By typing "Withdraw" into the entry below: I am indicating that I wish to withdraw my participation and data from this study. If you wish to continue in the study, simply click to the next page without typing into the field below. Type "Withdraw" below if you would like to withdraw from this study. Otherwise, leave this blank and continue.

The live consent screen does not mention personalities for AI; that clause appears only here. Two further sentences of the deployed debrief are inaccurate for this study and were inherited from the instrument of Tang et al. [18]: the assistant participants used was the local Qwen model of section 6.1, not ChatGPT, and the served products were retrieved by relevance to the task rather than selected randomly. Neither error affects what participants experienced during the session; both are recorded here so that the instrument as administered is on the page. The closing screen then shows a run identifier for crowd participants to paste into Prolific, or asks laboratory participants to tell the experimenter they are finished.

# D

## EEG Electrode Placement and Preprocessing

§4.2.4 states the recording hardware (Acquisition) and the cleaning chain (Preprocessing). This appendix records the quality-control detail behind them and defines the sixteen spectral measures. Epoch-width branches that leave the pre-specified 4 s sit here so they do not share the Results page with the confirmatory family.

### D.1 ACQUISITION AND SIGNAL QUALITY

**RECORDING VOLUME.** A complete laboratory session is one continuous recording rather than a set of short trials. The median recording is 56.6 minutes, or roughly 1.7 million samples per participant, and carries 72 marker events delimiting the phases of the protocol. The raw collection is approximately 3 GB compressed.

**CONDITION-BLIND QUALITY PROFILE.** Before cleaning, each recording was profiled on twelve 30 s windows distributed across the session. Each window was summarised per channel by robust amplitude, robust peak-to-peak range, correlation with the across-channel median, the fraction of near-flat samples, 50 Hz contamination, and high-frequency power; the median across the twelve windows gave one value per channel per metric. The profile is condition-blind by construction, because the windows were placed without reference to the event log, so it cannot preferentially clean one experimental condition. It wrote warnings only. No recording was excluded by an automatic amplitude rule, and the flags served to direct human review rather than to decide inclusion.

REVIEWED CHANNEL REPAIRS. Three channels were interpolated after review of those profiles together with per-epoch amplitude evidence. For Subject 1, P<sub>4</sub> was the worst channel in all 93 epochs exceeding 1,000  $\mu\text{V}$  peak-to-peak, with the failure concentrated in the final two condition windows. For Subject 9, P<sub>4</sub> was the worst channel in 8 of 10 epochs exceeding 750  $\mu\text{V}$ , affecting five of seven baseline epochs and the third condition window. For Subject 10, C<sub>4</sub> was the worst channel in 258 of 264 epochs exceeding 1,000  $\mu\text{V}$ , persisting across the final three condition windows. Interpolated traces keep their bad label, so no later stage treats them as recorded data.

COMPONENT-SELECTION THRESHOLDS. FastICA was fitted on a 1–40 Hz high-passed copy of the recording, decimated by a factor of five, retaining 99% of the PCA variance, and the unmixing was applied to the 0.5–40 Hz data [42]. A component was excluded only if it satisfied both criteria simultaneously: absolute correlation of at least 0.35 with the mean of the Fp1 and Fp2 proxy channels, and a mean absolute topography weight over the frontal channels at least 1.5 times the all-channel mean. At most three components were excluded per recording.

STATUS OF THE THRESHOLDS. The component-selection rule and the 1,050  $\mu\text{V}$  rejection bound were fixed before the confirmatory tests and were not retuned. They are engineering choices, not literature constants. Rerunning without ICA, and at 2, 8, 16, and 32 s, leaves the eight Dataset B confirmatory cells Holm-null. The width grid scores Dataset A on the whole-window median rather than on the  $k = 37$  neighbourhood, so the early-minus-late posterior  $\alpha$  cell of Table 7.8 was not re-estimated at other widths; on the whole-window median that contrast is Holm-null at every width (4 s:  $-0.09$  dB, raw  $p = .20$ ). Without ICA the explicit-early absolute  $\delta$  cell of the exploratory battery shrinks from  $+4.60$  dB (Holm  $p = .007$ ) to  $+3.04$  dB (raw  $p = .073$ , Holm  $p = .29$ ). Two Dataset B cells reach Holm  $p < .05$  off 4 s (Table D.1); they are different cells from each other and from the 4 s board, which is why they stay here.

**Table D.1:** Onset-locked cells with Holm  $p < .05$  outside the pre-specified 4 s width; sensitivity only. Bold rows survive Holm.

| Width      | Contrast                                                                            | $M$ (dB)     | Holm $p$    |
|------------|-------------------------------------------------------------------------------------|--------------|-------------|
| <b>2 s</b> | <b>Explicit early Fz <math>\theta</math> vs matched <math>a^\theta</math></b>       | <b>+3.30</b> | <b>.021</b> |
| <b>8 s</b> | <b>Explicit late posterior <math>\alpha</math> vs matched <math>a^\theta</math></b> | <b>−1.22</b> | <b>.023</b> |

## D.2 EEG MEASURES

The preprocessing paragraph of §4.2.4 states how each 4 s cell becomes sixteen spectral measures. This appendix defines each measure, states what it is conventionally taken to index, and records its analytic

tier. No ERP components are analysed, so none of these measures is a latency or a peak amplitude; all sixteen are properties of an induced power spectrum estimated within a window.

**READING A DECIBEL DIFFERENCE.** Absolute band power is reported as  $10 \log_{10}$  of the mean  $\mu V^2$  band integral, so every absolute quantity in this thesis is on a power-decibel scale and every difference between two of them is a ratio. A difference of  $\Delta$  dB corresponds to a power ratio of  $10^{\Delta/10}$ : 0.5 dB is about 12% more power, 1 dB about 26%, 3 dB about double, and 4.6 dB roughly threefold. This matters when comparing the two datasets. The Dataset A confirmatory differences run from a few hundredths to about a fifth of a decibel; the largest, early minus late posterior  $\alpha$  at  $-0.22$  dB, is a 5% change in band power, whereas the Dataset B intervals span several decibels. Relative powers are not on this scale: they are dimensionless proportions of the 0.5–40 Hz total, so a change of 0.19 means nineteen percentage points of the spectrum moving into that band.

**ABSOLUTE BAND POWERS (FIVE MEASURES).** Delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–40 Hz) [44, 45], each averaged across all 32 channels before the decibel transform. These are whole-head summaries and are deliberately not regional. Delta in awake task recordings carries ocular and movement activity in addition to slow cortical activity, and scalp gamma in this range overlaps the electromyographic spectrum of the temporalis and frontalis muscles; neither is interpreted here as a purely cortical index.

**RELATIVE BAND POWERS (FIVE MEASURES).** The same five bands divided by the 0.5–40 Hz total power within the same window. These are compositional: the five proportions sum to one by construction, so they cannot move independently. A genuine increase confined to one absolute band necessarily raises that band's proportion and lowers the others, which means a set of relative-power differences pointing in opposite directions may describe a single underlying change rather than several. Any interpretation of a relative-power contrast is therefore read together with the corresponding absolute band.

**REGIONAL CONFIRMATORY MEASURES (TWO).** Fz theta is 4–8 Hz power at the frontal midline electrode Fz. Frontal-midline theta increases with effortful, control-demanding processing and with working-memory engagement [77], which makes it the natural candidate for a commercial insertion that increases processing demand. Posterior alpha is 8–13 Hz power averaged over O1, Oz, O2, P3, Pz, and P4. Posterior alpha decreases as visual attention engages the display and increases as attention withdraws from it [78], which makes it the natural candidate for an insertion that captures or diverts gaze. These two are the only measures in the confirmatory Holm families.

ASYMMETRY AND ENGAGEMENT RATIOS (FOUR MEASURES). The four indices are

$$\text{FAA} = \ln \alpha_{F4} - \ln \alpha_{F3}, \quad \text{Pope} = \frac{\beta}{\alpha + \theta}, \quad \text{Kislov} = \frac{\beta_{16-24}}{\alpha_{8-12}}. \quad (\text{D.1})$$

FAA is the standard approach-versus-withdrawal index, in which relatively less left-frontal alpha is read as greater approach motivation [84, 92]. Pope engagement is computed twice: once over all channels and once over the frontocentral set (F3, F4, Fz, FC1, FC2, C3, C4, Cz) [85]. The Kislov index is the central-site ratio [46]. All four are the kinds of index that consumer-neuroscience work uses for engagement, arousal, or motivational orientation toward a stimulus [86, 87]. Because they are ratios of bands that are themselves reported, they add framing rather than independent information.

TIERS. Fz theta and posterior alpha are confirmatory. Global theta, global alpha, global beta, and frontal alpha asymmetry are secondary. The remaining ten are exploratory. Every measure is carried in both  $Y_A$  and  $Y_B$ , and Holm adjustment is applied within a measure across the planned contrasts, never across measures. The fourteen non-confirmatory measures are consequently not a second confirmatory family, and Figure 7.8 is read with that restriction in mind. Restricting Holm to within a measure is a family choice made in section 6.3: each measure is a declared family of its planned contrasts, so the family-wise guarantee is per measure. Holm remains valid under dependence, so the shared denominator of the five relative powers is not what rules out an across-measure family; what rules it out is that such a family would test whether any of sixteen measures moves, which no confirmatory claim in this thesis makes. The price is that the fourteen non-confirmatory measures carry no joint error control, and a reader who wants that guarantee should treat the seven Holm-surviving cells of Table D.2 as roughly one expected false positive among 98 uncorrected tests at  $\alpha = .05$ .

EXPLORATORY CELLS THAT SURVIVE HOLM WITHIN THEIR MEASURE. Figure 7.8 marks seven such cells among the 98 tests on the fourteen non-confirmatory measures. Table D.2 gives their estimates; six of the seven belong to the explicit-early onset contrast, and five of those six are the slow-power tilt read in section 8.3. Relative powers are proportions of total 1–45 Hz power; absolute powers are in dB.

**Table D.2:** Non-confirmatory EEG cells with Holm  $p < .05$  within their measure ( $n = 18$ , 4 s epochs, ICA primary). Holm is within measure across its planned contrasts; no across-measure correction.

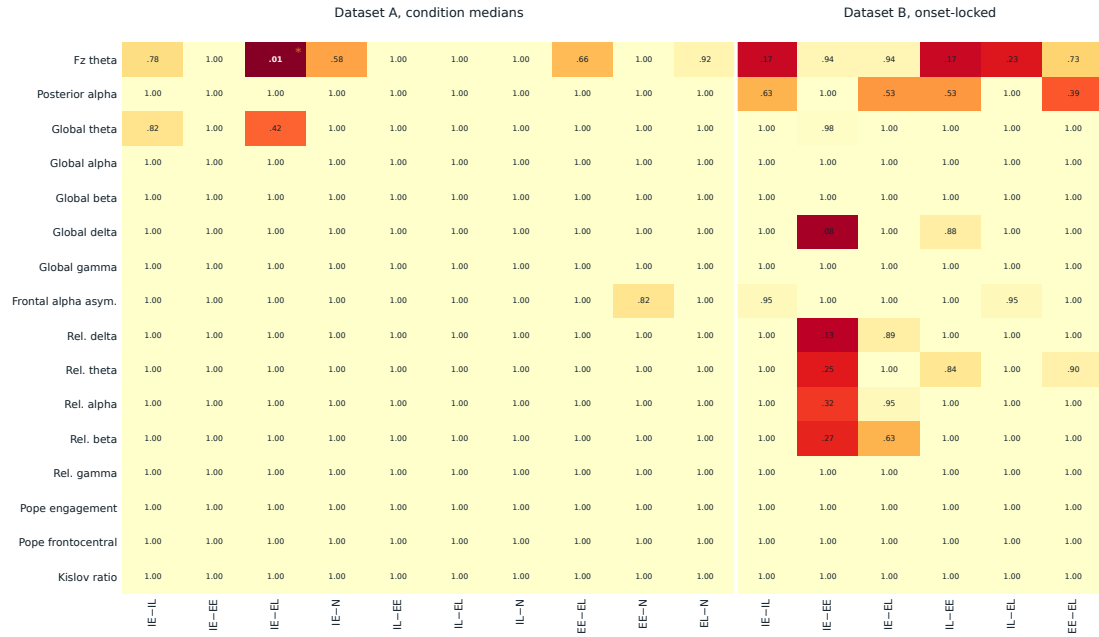
| Dataset               | Contrast               | Measure              | $M$    | 95% CI           | $d_z$ | raw $p$ | Holm $p$ |
|-----------------------|------------------------|----------------------|--------|------------------|-------|---------|----------|
| Condition aggregation | early – late           | relative $\theta$    | –0.005 | [–0.008, –0.001] | –0.66 | .013    | .038     |
| Onset-locked          | explicit early – $a^0$ | relative $\delta$    | 0.189  | [0.088, 0.290]   | 0.93  | .001    | .004     |
| Onset-locked          | explicit early – $a^0$ | global $\delta$ (dB) | 4.60   | [1.99, 7.22]     | 0.88  | .002    | .007     |
| Onset-locked          | explicit early – $a^0$ | relative $\beta$     | –0.088 | [–0.145, –0.031] | –0.77 | .004    | .018     |
| Onset-locked          | explicit early – $a^0$ | global $\theta$ (dB) | 1.68   | [0.56, 2.80]     | 0.74  | .006    | .023     |
| Onset-locked          | explicit early – $a^0$ | relative $\alpha$    | –0.039 | [–0.066, –0.012] | –0.73 | .007    | .026     |
| Onset-locked          | implicit late – $a^0$  | relative $\gamma$    | –0.044 | [–0.075, –0.012] | –0.69 | .009    | .037     |

ESTIMATORS NOT USED. Repeated-measures ANOVA would replace the contrasts the hypotheses name with an omnibus effect, and cluster-based permutation testing presupposes a dense sensor-by-time grid. Neither is used.

### D.3 POST HOC PAIRWISE COMPARISONS

The contrasts of section 7.4 are the ones declared in section 6.3 before the data were examined. The sweep recorded here is not: it was added afterwards, once those contrasts had been read, and everything in this section is therefore post hoc and exploratory. Nothing in it revises a confirmatory test, and no cell in it is treated as a finding.

On Dataset A every pair of the five conditions is compared (ten); on Dataset B every pair of the four advertisement conditions (six). Sixteen measures give 256 tests, Holm within measure. The planned weights already span the five-condition space, so each pair is a linear combination of contrasts already reported. The sweep localises a marginal; it does not add a direction.



**Figure D.1:** Holm  $p$  across 256 post-hoc EEG pairs ( $n = 18$ ): condition aggregation (left) and onset-locked (right). Asterisk: Holm  $p < .05$ .

**Table D.3:** Post-hoc pairs with raw  $p < .05$  (8 of 256). Bold rows survive Holm.

| Set | Measure         | Comparison                            | $M$    | 95% CI         | $d_z$ | Raw $p$ | Holm |
|-----|-----------------|---------------------------------------|--------|----------------|-------|---------|------|
| A   | Fz $\theta$     | <b>implicit early – explicit late</b> | −0.28  | [−0.43, −0.12] | −0.90 | .001    | .014 |
| B   | Global $\delta$ | implicit early – explicit early       | −3.64  | [−6.41, −0.86] | −0.65 | .013    | .08  |
| B   | Rel. $\delta$   | implicit early – explicit early       | −0.13  | [−0.23, −0.02] | −0.60 | .021    | .13  |
| B   | Fz $\theta$     | implicit early – implicit late        | +1.99  | [0.25, 3.74]   | +0.57 | .028    | .17  |
| B   | Fz $\theta$     | implicit late – explicit early        | −2.12  | [−4.02, −0.21] | −0.55 | .031    | .17  |
| A   | Global $\theta$ | implicit early – explicit late        | −0.19  | [−0.37, −0.01] | −0.52 | .042    | .42  |
| B   | Rel. $\theta$   | implicit early – explicit early       | +0.029 | [0.001, 0.057] | +0.52 | .042    | .25  |
| B   | Rel. $\beta$    | implicit early – explicit early       | +0.056 | [0.001, 0.111] | +0.51 | .045    | .27  |

# E

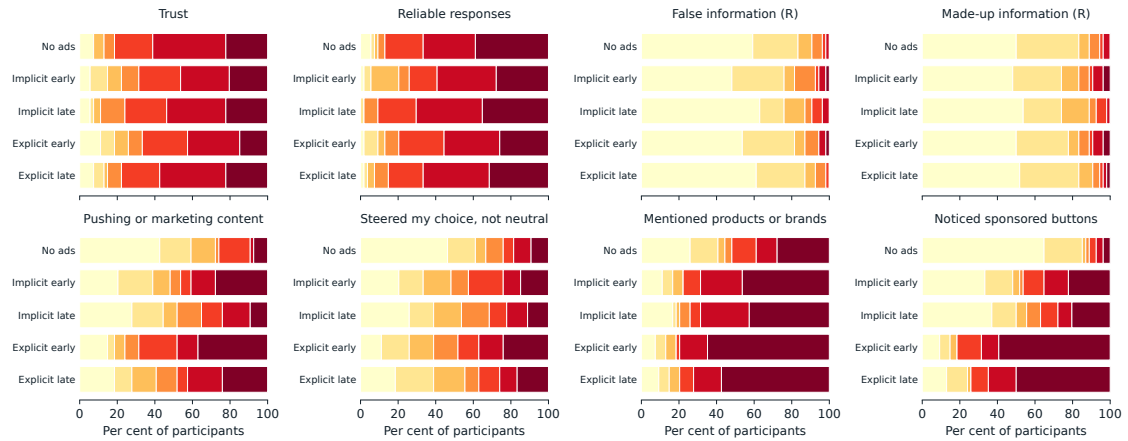
## Additional Information on the Behavioural Battery

Descriptives, reliability, the post hoc grid, and the personality family behind section 7.2 and section 7.3.

### E.1 DESCRIPTIVES AND ITEM DISTRIBUTIONS

**Table E.1:** Behavioural outcomes by condition ( $N = 54$ ); values are  $M$  (SD). Bold: extrema of each block.

| Outcome                 | No ads             | Implicit early     | Implicit late | Explicit early     | Explicit late      |
|-------------------------|--------------------|--------------------|---------------|--------------------|--------------------|
| Trust                   | <b>5.37 (1.63)</b> | 4.93 (1.80)        | 5.28 (1.58)   | <b>4.69 (1.91)</b> | 5.22 (1.76)        |
| Credibility             | 6.04 (1.15)        | <b>5.69 (1.35)</b> | 6.04 (1.09)   | 5.78 (1.30)        | <b>6.08 (0.93)</b> |
| Perceived manipulation  | <b>2.73 (1.97)</b> | 3.94 (2.10)        | 3.45 (1.98)   | <b>4.65 (1.95)</b> | 3.97 (2.06)        |
| Notice                  | <b>2.96 (1.63)</b> | 4.56 (1.89)        | 4.36 (1.89)   | <b>5.77 (1.74)</b> | 5.45 (1.94)        |
| Helpfulness             | 5.51 (1.50)        | <b>5.35 (1.60)</b> | 5.68 (1.38)   | 5.36 (1.67)        | <b>5.69 (1.44)</b> |
| Convincingness          | 4.83 (1.42)        | <b>4.57 (1.48)</b> | 5.09 (1.31)   | 4.78 (1.64)        | <b>5.23 (1.39)</b> |
| Relevance               | 5.58 (1.48)        | 5.40 (1.40)        | 5.76 (1.26)   | <b>5.23 (1.67)</b> | <b>5.87 (1.19)</b> |
| Neutrality              | <b>5.02 (1.45)</b> | <b>4.46 (1.37)</b> | 4.72 (1.32)   | 4.56 (1.46)        | 4.86 (1.47)        |
| Cued memory             | –                  | <b>4.35 (2.36)</b> | 4.44 (2.19)   | <b>6.09 (1.39)</b> | 5.61 (1.89)        |
| Trust after re-exposure | –                  | 4.19 (1.85)        | 4.46 (1.88)   | <b>3.85 (2.16)</b> | <b>4.56 (2.12)</b> |



**Figure E.1:** Item distributions behind the four primary outcomes ( $N = 54$ ). Segments run from response 1 (yellow) to 7 (red); (R) marks reversed items.

**NOTICE AND RECALL SHARES.** Table E.2 is the descriptive counterpart of Figure 7.4: the number of participants who agreed (rating  $\geq 5$  of 7) with each notice item and with cued memory, condition by condition, on the same  $N = 54$ . No test is run on these shares.

**Table E.2:** Participants rating each item at least 5 of 7, by condition ( $N = 54$ ); counts and percentages.

| Condition      | Sponsored buttons | Brands mentioned | Notice outcome | Cued memory |
|----------------|-------------------|------------------|----------------|-------------|
| No ad          | 6/54 (11%)        | 28/54 (52%)      | 6/54 (11%)     | —           |
| Implicit early | 25/54 (46%)       | 42/54 (78%)      | 25/54 (46%)    | 31/54 (57%) |
| Implicit late  | 20/54 (37%)       | 40/54 (74%)      | 21/54 (39%)    | 32/54 (59%) |
| Explicit early | 44/54 (81%)       | 44/54 (81%)      | 40/54 (74%)    | 47/54 (87%) |
| Explicit late  | 40/54 (74%)       | 43/54 (80%)      | 37/54 (69%)    | 44/54 (81%) |

| Planned contrasts across the behavioural battery (N = 54)<br>mean within-person difference in Likert points; Holm within outcome |                       |                       |                      |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|----------------------|
| Trust                                                                                                                            | -0.34<br>Holm .153    | +0.15<br>Holm .405    | -0.44<br>Holm .063   |
| Credibility                                                                                                                      | -0.14<br>Holm .673    | -0.06<br>Holm .673    | -0.33<br>Holm .016 * |
| Perceived manipulation                                                                                                           | +1.27<br>Holm <.001 * | -0.62<br>Holm .022 *  | +0.58<br>Holm .007 * |
| Notice                                                                                                                           | +2.07<br>Holm <.001 * | -1.15<br>Holm <.001 * | +0.26<br>Holm .234   |
| Helpfulness                                                                                                                      | +0.01<br>Holm 1.000   | -0.01<br>Holm 1.000   | -0.33<br>Holm .088   |
| Convincingness                                                                                                                   | +0.08<br>Holm .635    | -0.18<br>Holm .444    | -0.49<br>Holm .005 * |
| Relevance                                                                                                                        | -0.02<br>Holm 1.000   | +0.03<br>Holm 1.000   | -0.50<br>Holm .003 * |
| Neutrality                                                                                                                       | -0.37<br>Holm .116    | -0.12<br>Holm .469    | -0.28<br>Holm .055   |
| Cued memory                                                                                                                      | undefined             | -1.45<br>Holm <.001 * | +0.19<br>Holm .473   |
| Trust after re-exposure                                                                                                          | undefined             | +0.12<br>Holm .525    | -0.49<br>Holm .047 * |
|                                                                                                                                  | Any ad – no ads       | Implicit – explicit   | Early – late         |

**Figure E.2:** Planned behavioural contrasts ( $N = 54$ ). Cells give  $\bar{D}$  and Holm  $p$ ; secondary-quality rows are exploratory. Asterisk: Holm  $p < .05$ .

On the four secondary qualities, early minus late lowers convincingness and relevance (Holm  $p = .005$  and  $.003$ ); the other planned cells are Holm-null.

## E.2 RELIABILITY OF THE OUTCOMES

Scoring is in subsection 6.2.2. Secondary qualities follow Tang et al. [18]. Reversal is  $8 - x$ , once, before averaging. Cronbach  $\alpha$  is acceptable on manipulation, credibility, helpfulness, and relevance. Notice ( $\alpha = .61$ ) and neutrality ( $\alpha = .63$ ) are low because each pair indexes different things by design; trust is a single item and has no  $\alpha$ .

**Table E.3:** Cronbach  $\alpha$  on 270 condition rows. Trust is a single item.

| Outcome                | Items | $\alpha$ | mean $\rho$ | $\alpha$ lab | $\alpha$ crowd |
|------------------------|-------|----------|-------------|--------------|----------------|
| Credibility            | 3     | 0.78     | 0.58        | 0.80         | 0.76           |
| Perceived manipulation | 2     | 0.86     | 0.75        | 0.87         | 0.85           |
| Notice                 | 2     | 0.61     | 0.45        | 0.70         | 0.56           |
| Helpfulness            | 3     | 0.90     | 0.72        | 0.93         | 0.88           |
| Convincingness         | 3     | 0.74     | 0.48        | 0.77         | 0.68           |
| Relevance              | 3     | 0.87     | 0.69        | 0.90         | 0.85           |
| Neutrality             | 3     | 0.63     | 0.41        | 0.69         | 0.57           |

Notice  $\alpha = .61$  is the lowest because its two items index different things by design.

### E.3 ROBUSTNESS TO ITEM CHOICE

Item sets were fixed before the data were read. Three items looked weak on the raw scores; each is the odd-polarity item of its scale. After the one reversal, every inter-item  $\rho$  is positive. There is no reliability ground for dropping them.

**Table E.4:** Item diagnostics after the single reversal. Bold: items queried on the raw scores.

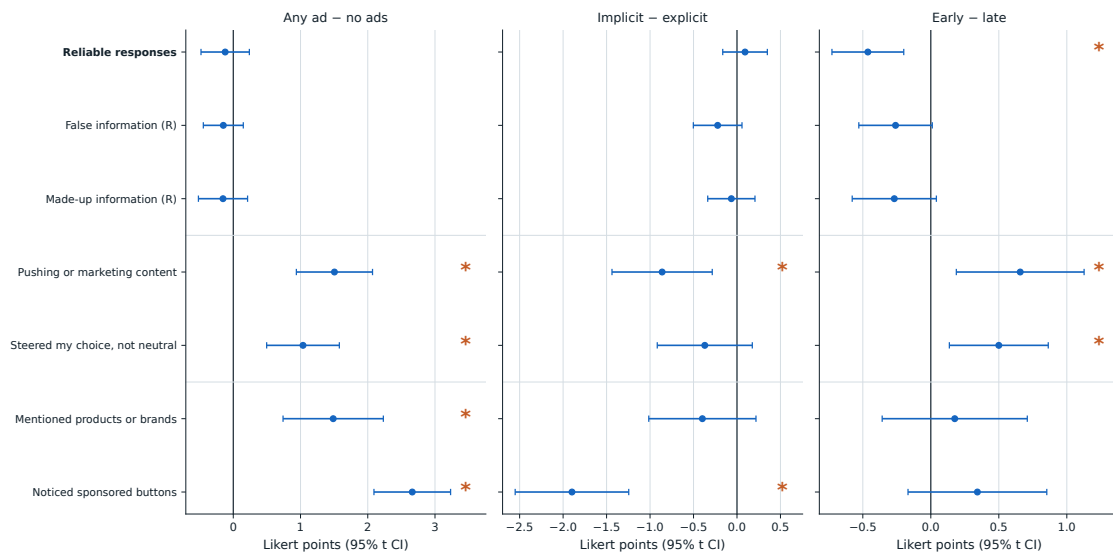
| Outcome                            | Item                                  | $\rho$ with mates | item-rest $\rho$ | $\alpha$ if deleted |
|------------------------------------|---------------------------------------|-------------------|------------------|---------------------|
| Credibility ( $\alpha = 0.78$ )    | <b>Reliable responses</b>             | 0.48 to 0.51      | 0.52             | 0.85                |
|                                    | False information (R)                 | 0.51 to 0.76      | 0.70             | 0.61                |
|                                    | Made-up information (R)               | 0.48 to 0.76      | 0.66             | 0.64                |
| Helpfulness ( $\alpha = 0.90$ )    | Helpful responses                     | 0.69 to 0.77      | 0.78             | 0.86                |
|                                    | Adequately addressed my request       | 0.71 to 0.77      | 0.79             | 0.85                |
|                                    | Response did not aid me (R)           | 0.69 to 0.71      | 0.73             | 0.88                |
| Convincingness ( $\alpha = 0.74$ ) | <b>Sceptical of its responses (R)</b> | 0.32 to 0.56      | 0.45             | 0.73                |
|                                    | Convincing responses                  | 0.56 to 0.56      | 0.66             | 0.51                |
|                                    | Changed my mind                       | 0.32 to 0.56      | 0.47             | 0.71                |
| Relevance ( $\alpha = 0.87$ )      | Responses not useful (R)              | 0.66 to 0.71      | 0.74             | 0.82                |
|                                    | Suggestions addressed my questions    | 0.66 to 0.71      | 0.73             | 0.82                |
|                                    | Relevant responses                    | 0.71              | 0.78             | 0.82                |
| Neutrality ( $\alpha = 0.63$ )     | Neutral (fair) responses              | 0.39 to 0.51      | 0.52             | 0.44                |
|                                    | Impartial and unbiased                | 0.32 to 0.51      | 0.48             | 0.50                |
|                                    | <b>Opinionated responses (R)</b>      | 0.32 to 0.39      | 0.37             | 0.66                |

Leave-one-item-out moves four of 45 verdicts, in both directions: credibility early minus late dies

without the reliable-responses item; a neutrality any-ad cell appears without the opinionated item. The pre-specified outcomes stand.

**Table E.5:** Leave-one-item-out  $\bar{D}$  with Holm  $p$ . Bold cells survive Holm.

|                       |                       |                       |                       |                       |
|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <i>Credibility</i>    |                       |                       |                       |                       |
| Contrast              | full                  | – <b>reliable</b>     | – false               | – made-up             |
| Any ad – no ads       | –0.14 (.673)          | –0.15 (.426)          | –0.14 (.781)          | –0.13 (.672)          |
| Implicit – explicit   | –0.06 (.673)          | –0.14 (.426)          | +0.01 (.890)          | –0.06 (.672)          |
| Early – late          | –0.33 ( <b>.016</b> ) | –0.26 (.159)          | –0.37 ( <b>.011</b> ) | –0.36 (.007)          |
| <i>Neutrality</i>     |                       |                       |                       |                       |
| Contrast              | full                  | – neutral             | – impartial           | – <b>opinionated</b>  |
| Any ad – no ads       | –0.37 (.116)          | –0.42 (.082)          | –0.14 (.706)          | –0.54 ( <b>.028</b> ) |
| Implicit – explicit   | –0.12 (.469)          | –0.15 (.452)          | –0.15 (.706)          | –0.06 (.767)          |
| Early – late          | –0.28 (.055)          | –0.41 ( <b>.014</b> ) | –0.16 (.540)          | –0.29 (.110)          |
| <i>Convincingness</i> |                       |                       |                       |                       |
| Contrast              | full                  | – <b>sceptical</b>    | – convincing          | – changed mind        |
| Any ad – no ads       | +0.08 (.635)          | +0.09 (.640)          | +0.10 (.639)          | +0.07 (1.000)         |
| Implicit – explicit   | –0.18 (.444)          | –0.21 (.322)          | –0.33 (.069)          | +0.00 (1.000)         |
| Early – late          | –0.49 ( <b>.005</b> ) | –0.48 ( <b>.010</b> ) | –0.46 ( <b>.014</b> ) | –0.52 (.009)          |



**Figure E.3:** Item-level planned contrasts ( $N = 54$ ). Asterisk: Holm  $p < .05$  within item.

## E.4 POST HOC LOCALISATION AND PAIRWISE SWEEP

**Table E.6:** Each advertisement condition versus  $a^0$  ( $N = 54$ ). Holm-adjusted paired  $t$ ; Wilcoxon  $p$  is raw. Bold rows survive Holm.

| Outcome                       | Condition – $a^0$     | $\bar{D}$ | 95% CI         | $d_z$ | Holm $p$ | Wilcoxon $p$ |
|-------------------------------|-----------------------|-----------|----------------|-------|----------|--------------|
| Trust                         | Implicit early        | −0.44     | [−0.94, 0.05]  | −0.24 | .250     | .125         |
|                               | Implicit late         | −0.09     | [−0.57, 0.38]  | −0.05 | 1.000    | .535         |
|                               | <b>Explicit early</b> | −0.69     | [−1.17, −0.20] | −0.38 | .031     | .003         |
|                               | Explicit late         | −0.15     | [−0.66, 0.36]  | −0.08 | 1.000    | .894         |
| Credibility                   | Implicit early        | −0.35     | [−0.77, 0.06]  | −0.23 | .411     | .065         |
|                               | Implicit late         | +0.01     | [−0.32, 0.34]  | +0.00 | 1.000    | .878         |
|                               | Explicit early        | −0.26     | [−0.58, 0.07]  | −0.21 | .411     | .040         |
|                               | Explicit late         | +0.04     | [−0.26, 0.35]  | +0.04 | 1.000    | .910         |
| <b>Perceived manipulation</b> | <b>Implicit early</b> | +1.20     | [0.44, 1.96]   | +0.42 | .006     | .004         |
|                               | <b>Implicit late</b>  | +0.72     | [0.05, 1.39]   | +0.29 | .039     | .017         |
|                               | <b>Explicit early</b> | +1.92     | [1.39, 2.44]   | +0.97 | < .001   | < .001       |
|                               | <b>Explicit late</b>  | +1.24     | [0.62, 1.86]   | +0.53 | < .001   | < .001       |
| <b>Notice</b>                 | <b>Implicit early</b> | +1.60     | [0.87, 2.33]   | +0.59 | < .001   | < .001       |
|                               | <b>Implicit late</b>  | +1.40     | [0.68, 2.11]   | +0.52 | < .001   | < .001       |
|                               | <b>Explicit early</b> | +2.81     | [2.17, 3.44]   | +1.18 | < .001   | < .001       |
|                               | <b>Explicit late</b>  | +2.49     | [1.82, 3.16]   | +0.99 | < .001   | < .001       |

**Table E.7:** Friedman  $\chi^2_4$  over the five conditions ( $N = 54$ ). Post hoc;  $p$  raw.

| Outcome                | Friedman $\chi^2_4$ | Kendall's $W$ | $p$    |
|------------------------|---------------------|---------------|--------|
| Trust                  | 9.55                | 0.044         | .049   |
| Credibility            | 6.03                | 0.028         | .197   |
| Perceived manipulation | 34.07               | 0.158         | < .001 |
| Notice                 | 66.75               | 0.309         | < .001 |
| Helpfulness            | 4.74                | 0.022         | .315   |
| Convincingness         | 13.09               | 0.061         | .011   |
| Relevance              | 14.82               | 0.069         | .005   |
| Neutrality             | 12.74               | 0.059         | .013   |

**Table E.8:** Post-hoc pairwise sweep: rows with Holm  $p < .10$  ( $N = 54$ ). Holm-adjusted paired  $t$ ; Wilcoxon  $p$  is raw. Bold rows survive Holm.

| Outcome                       | Pair                                   | $\bar{D}$ | 95% CI         | $d_z$ | Holm $p$ | Wilcoxon $p$ |
|-------------------------------|----------------------------------------|-----------|----------------|-------|----------|--------------|
| Trust                         | No ads – explicit early                | +0.69     | [0.20, 1.17]   | +0.38 | .078     | .003         |
| Credibility                   | Implicit early – explicit late         | −0.40     | [−0.66, −0.13] | −0.39 | .054     | .004         |
| <b>Perceived manipulation</b> | <b>No ads – explicit early</b>         | −1.92     | [−2.44, −1.39] | −0.97 | < .001   | < .001       |
|                               | <b>Implicit late – explicit early</b>  | −1.19     | [−1.74, −0.65] | −0.58 | < .001   | < .001       |
|                               | <b>No ads – explicit late</b>          | −1.24     | [−1.86, −0.62] | −0.53 | .002     | < .001       |
|                               | <b>No ads – implicit early</b>         | −1.20     | [−1.96, −0.44] | −0.42 | .022     | .004         |
| <b>Notice</b>                 | <b>No ads – explicit early</b>         | −2.81     | [−3.44, −2.17] | −1.18 | < .001   | < .001       |
|                               | <b>No ads – explicit late</b>          | −2.49     | [−3.16, −1.82] | −0.99 | < .001   | < .001       |
|                               | <b>No ads – implicit early</b>         | −1.60     | [−2.33, −0.87] | −0.59 | < .001   | < .001       |
|                               | <b>Implicit late – explicit early</b>  | −1.41     | [−2.11, −0.71] | −0.54 | .002     | < .001       |
|                               | <b>No ads – implicit late</b>          | −1.40     | [−2.11, −0.68] | −0.52 | .002     | < .001       |
|                               | <b>Implicit early – explicit early</b> | −1.20     | [−1.82, −0.59] | −0.52 | .002     | < .001       |
|                               | <b>Implicit late – explicit late</b>   | −1.09     | [−1.90, −0.29] | −0.36 | .042     | .006         |
|                               | <b>Implicit early – explicit late</b>  | −0.89     | [−1.59, −0.19] | −0.34 | .046     | .013         |
| <b>Convincingness</b>         | <b>Implicit early – explicit late</b>  | −0.67     | [−1.09, −0.24] | −0.42 | .034     | .006         |
| Relevance                     | Implicit late – explicit early         | +0.53     | [0.17, 0.89]   | +0.39 | .057     | .010         |

## E.5 PERSONALITY FAMILY

Table E.9 is the full family. Holm is Wald on the trait  $\times$  contrast term, within outcome across fifteen. OLS agrees on all sixty verdicts. The demographic family is Table E.10 below.

**Table E.9:** BFI-10 moderation slopes on  $D_i$  per trait point ( $N = 54$ ); Holm within outcome across fifteen terms.

| Outcome                | Contrast            | Trait                   | Slope | 95% CI         | Raw $p$ | Holm $p$ |
|------------------------|---------------------|-------------------------|-------|----------------|---------|----------|
| Trust                  | Any ad — no ads     | Extraversion            | +0.04 | [−0.29, 0.37]  | .795    | 1.000    |
|                        |                     | Agreeableness           | −0.02 | [−0.35, 0.32]  | .927    | 1.000    |
|                        |                     | Conscientiousness       | +0.09 | [−0.26, 0.44]  | .620    | 1.000    |
|                        |                     | Neuroticism             | −0.27 | [−0.53, −0.00] | .047    | .659     |
|                        |                     | Openness                | −0.13 | [−0.42, 0.17]  | .394    | 1.000    |
|                        | Implicit — explicit | Extraversion            | +0.10 | [−0.27, 0.46]  | .609    | 1.000    |
|                        |                     | Agreeableness           | +0.22 | [−0.15, 0.60]  | .241    | 1.000    |
|                        |                     | Conscientiousness       | +0.11 | [−0.29, 0.50]  | .594    | 1.000    |
|                        |                     | Neuroticism             | −0.11 | [−0.41, 0.19]  | .473    | 1.000    |
|                        |                     | Openness                | −0.17 | [−0.50, 0.16]  | .307    | 1.000    |
|                        | Early — late        | Extraversion            | −0.46 | [−0.83, −0.10] | .013    | .201     |
|                        |                     | Agreeableness           | −0.06 | [−0.43, 0.32]  | .768    | 1.000    |
|                        |                     | Conscientiousness       | −0.19 | [−0.59, 0.21]  | .347    | 1.000    |
|                        |                     | Neuroticism             | −0.05 | [−0.34, 0.25]  | .762    | 1.000    |
|                        |                     | Openness                | −0.19 | [−0.52, 0.14]  | .254    | 1.000    |
| Credibility            | Any ad — no ads     | Extraversion            | −0.11 | [−0.33, 0.11]  | .316    | 1.000    |
|                        |                     | Agreeableness           | −0.20 | [−0.42, 0.02]  | .069    | .969     |
|                        |                     | Conscientiousness       | −0.07 | [−0.30, 0.17]  | .578    | 1.000    |
|                        |                     | Neuroticism             | −0.16 | [−0.33, 0.02]  | .079    | 1.000    |
|                        |                     | Openness                | −0.12 | [−0.31, 0.07]  | .227    | 1.000    |
|                        | Implicit — explicit | Extraversion            | −0.06 | [−0.30, 0.18]  | .615    | 1.000    |
|                        |                     | Agreeableness           | +0.13 | [−0.12, 0.37]  | .304    | 1.000    |
|                        |                     | Conscientiousness       | +0.02 | [−0.24, 0.28]  | .865    | 1.000    |
|                        |                     | Neuroticism             | −0.08 | [−0.28, 0.11]  | .418    | 1.000    |
|                        |                     | Openness                | −0.01 | [−0.23, 0.20]  | .918    | 1.000    |
|                        | Early — late        | Extraversion            | −0.31 | [−0.55, −0.07] | .012    | .178     |
|                        |                     | Agreeableness           | −0.05 | [−0.29, 0.20]  | .709    | 1.000    |
|                        |                     | Conscientiousness       | −0.13 | [−0.39, 0.13]  | .331    | 1.000    |
|                        |                     | Neuroticism             | −0.02 | [−0.22, 0.17]  | .819    | 1.000    |
|                        |                     | Openness                | −0.04 | [−0.26, 0.18]  | .715    | 1.000    |
| Perceived manipulation | Any ad — no ads     | Extraversion            | −0.06 | [−0.48, 0.37]  | .790    | 1.000    |
|                        |                     | Agreeableness           | +0.11 | [−0.32, 0.53]  | .622    | 1.000    |
|                        |                     | Conscientiousness       | +0.16 | [−0.29, 0.62]  | .475    | 1.000    |
|                        |                     | Neuroticism             | +0.14 | [−0.20, 0.48]  | .431    | 1.000    |
|                        |                     | Openness                | +0.31 | [−0.06, 0.68]  | .104    | 1.000    |
|                        | Implicit — explicit | Extraversion            | −0.25 | [−0.72, 0.22]  | .302    | 1.000    |
|                        |                     | Agreeableness           | −0.36 | [−0.83, 0.12]  | .144    | 1.000    |
|                        |                     | Conscientiousness       | −0.10 | [−0.61, 0.40]  | .691    | 1.000    |
|                        |                     | Neuroticism             | +0.03 | [−0.35, 0.41]  | .878    | 1.000    |
|                        |                     | Openness                | +0.26 | [−0.16, 0.68]  | .220    | 1.000    |
|                        | Early — late        | Extraversion            | −0.08 | [−0.55, 0.39]  | .743    | 1.000    |
|                        |                     | Agreeableness           | −0.15 | [−0.63, 0.32]  | .532    | 1.000    |
|                        |                     | Conscientiousness       | −0.08 | [−0.58, 0.43]  | .769    | 1.000    |
|                        |                     | Neuroticism             | −0.13 | [−0.51, 0.25]  | .494    | 1.000    |
|                        |                     | Openness                | −0.02 | [−0.44, 0.40]  | .927    | 1.000    |
| Notice                 | Any ad — no ads     | Extraversion            | −0.05 | [−0.50, 0.39]  | .813    | 1.000    |
|                        |                     | Agreeableness           | +0.09 | [−0.35, 0.54]  | .686    | 1.000    |
|                        |                     | Conscientiousness       | +0.42 | [−0.05, 0.89]  | .079    | 1.000    |
|                        |                     | Neuroticism             | +0.10 | [−0.25, 0.46]  | .568    | 1.000    |
|                        |                     | Openness <sup>142</sup> | +0.18 | [−0.21, 0.57]  | .375    | 1.000    |
|                        | Implicit — explicit | Extraversion            | +0.02 | [−0.48, 0.52]  | .932    | 1.000    |
|                        |                     | Agreeableness           | −0.27 | [−0.77, 0.22]  | .281    | 1.000    |
|                        |                     | Conscientiousness       | +0.29 | [−0.23, 0.82]  | .272    | 1.000    |
|                        |                     | Neuroticism             | +0.09 | [−0.30, 0.49]  | .641    | 1.000    |
|                        |                     | Openness                | +0.46 | [0.03, 0.90]   | .038    | .566     |
|                        | Early — late        | Extraversion            | −0.31 | [−0.81, 0.18]  | .216    | 1.000    |
|                        |                     | Agreeableness           | −0.41 | [−0.90, 0.09]  | .110    | 1.000    |
|                        |                     | Conscientiousness       | −0.24 | [−0.76, 0.29]  | .377    | 1.000    |
|                        |                     | Neuroticism             | +0.35 | [−0.05, 0.75]  | .084    | 1.000    |
|                        |                     | Openness                | +0.09 | [−0.34, 0.53]  | .673    | 1.000    |

DEMOGRAPHIC FACTORS. Table E.10 is the full demographic family: one factor at a time against the three planned contrast codes in the same random-intercept model, joint Wald on the contrast  $\times$  factor terms, Holm within outcome, with levels kept where at least five people fill them (doctoral merged into master's, "Other" education excluded, the two least familiar and the two rarest users merged into their neighbouring levels). The two-level coding column and the OLS column, an ordinary least squares regression of the person-level  $D_i$  on the factor, agree with the mixed model on every verdict. Under a condition-dummy parameterisation without correction the same data yield 18 raw  $p < .05$  terms out of 440, 14 of them on levels of two or three people; two survive Holm and rest on the same two participants.

**Table E.10:** Demographic moderation of planned contrasts. Cohort  $N = 54$ ; sex  $n = 51$ , education  $n = 49$ .

| Outcome                | Contrast            | Factor (levels)   | Spread | $\chi^2$ (df) | Raw $p$ | Holm $p$ | Two-level | OLS   |
|------------------------|---------------------|-------------------|--------|---------------|---------|----------|-----------|-------|
| Trust                  | Any ad — no ads     | Sex (2)           | 0.23   | 0.27 (1)      | .600    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.86   | 2.09 (2)      | .353    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.18   | 0.12 (1)      | .724    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 0.80   | 2.37 (3)      | .499    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.05   | 0.01 (1)      | .912    | 1.000    | 1.000     | 1.000 |
|                        | Implicit — explicit | Sex (2)           | 0.22   | 0.31 (1)      | .580    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.28   | 0.43 (2)      | .807    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.46   | 1.06 (1)      | .303    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 1.12   | 5.51 (3)      | .138    | 1.000    | .436      | 1.000 |
|                        |                     | Environment (2)   | 0.10   | 0.06 (1)      | .804    | 1.000    | 1.000     | 1.000 |
|                        | Early — late        | Sex (2)           | 0.25   | 0.41 (1)      | .524    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.27   | 0.35 (2)      | .841    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.02   | 0.00 (1)      | .968    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 0.40   | 0.55 (3)      | .909    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.46   | 1.37 (1)      | .242    | 1.000    | 1.000     | 1.000 |
| Credibility            | Any ad — no ads     | Sex (2)           | 0.11   | 0.16 (1)      | .693    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.45   | 1.51 (2)      | .470    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.29   | 0.81 (1)      | .368    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 0.44   | 0.99 (3)      | .805    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.28   | 0.94 (1)      | .334    | 1.000    | 1.000     | 1.000 |
|                        | Implicit — explicit | Sex (2)           | 0.18   | 0.51 (1)      | .474    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.20   | 0.59 (2)      | .746    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.29   | 1.01 (1)      | .316    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 0.33   | 1.05 (3)      | .790    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.10   | 0.15 (1)      | .703    | 1.000    | 1.000     | 1.000 |
|                        | Early — late        | Sex (2)           | 0.11   | 0.17 (1)      | .679    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.33   | 0.99 (2)      | .610    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.12   | 0.17 (1)      | .677    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 0.42   | 2.19 (3)      | .535    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.48   | 3.50 (1)      | .061    | .919     | .919      | .716  |
| Perceived manipulation | Any ad — no ads     | Sex (2)           | 0.46   | 0.72 (1)      | .396    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 1.25   | 2.77 (2)      | .250    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.12   | 0.04 (1)      | .848    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 1.22   | 3.45 (3)      | .327    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.45   | 0.65 (1)      | .421    | 1.000    | 1.000     | 1.000 |
|                        | Implicit — explicit | Sex (2)           | 0.41   | 0.71 (1)      | .399    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.90   | 1.81 (2)      | .405    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.82   | 2.10 (1)      | .147    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 0.95   | 1.73 (3)      | .629    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.58   | 1.34 (1)      | .247    | 1.000    | 1.000     | 1.000 |
|                        | Early — late        | Sex (2)           | 0.27   | 0.31 (1)      | .580    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.29   | 0.19 (2)      | .907    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.07   | 0.02 (1)      | .895    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 1.42   | 3.82 (3)      | .282    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.16   | 0.10 (1)      | .748    | 1.000    | 1.000     | 1.000 |
| Notice                 | Any ad — no ads     | Sex (2)           | 0.36   | 0.39 (1)      | .531    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.36   | 0.22 (2)      | .894    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.23   | 0.12 (1)      | .730    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 1.15   | 1.73 (3)      | .631    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.14   | 0.06 (1)      | .813    | 1.000    | 1.000     | 1.000 |
|                        | Implicit — explicit | Sex (2)           | 0.94   | 3.33 (1)      | .068    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 1.08   | 3.69 (2)      | .158    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.51   | 0.75 (1)      | .388    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 1.60   | 4.27 (3)      | .234    | 1.000    | 1.000     | 1.000 |
|                        |                     | Environment (2)   | 0.14   | 0.07 (1)      | .791    | 1.000    | 1.000     | 1.000 |
|                        | Early — late        | Sex (2)           | 0.52   | 1.01 (1)      | .316    | 1.000    | 1.000     | 1.000 |
|                        |                     | Education (3)     | 0.37   | 0.52 (2)      | .773    | 1.000    | 1.000     | 1.000 |
|                        |                     | Familiarity (2)   | 0.55   | 0.85 (1)      | .356    | 1.000    | 1.000     | 1.000 |
|                        |                     | Use frequency (4) | 1.66   | 5.67 (3)      | .129    | 1.000    | 1.000     | .876  |
|                        |                     | Environment (2)   | 0.07   | 0.02 (1)      | .895    | 1.000    | 1.000     | 1.000 |

## E.6 REALISED DESIGN AND ARM SPLIT

The Latin-square rotation of section 6.2 balances task and session position against condition in expectation; with  $N = 54$ , which is not a multiple of 25, exact balance is not possible. Table E.11 and Table E.12 give the realised counts. Neither table departs from uniformity ( $\chi^2(16) = 17.0, p = .38; \chi^2(16) = 17.4, p = .36$ ). The largest cell is the study-environment task under the implicit-late condition (17 of 54), the smallest the gardening task under the same condition (5). Neither the paired  $t$  of the planned contrasts nor the mixed-model check enters task or position as a term; both rest on this balance, and a task effect that happened to align with a condition would not be removed by them.

**Table E.11:** Realised task  $\times$  condition counts ( $N = 54$ , one row per participant-condition).

| Task               | $a_2^{\text{imp}}$ | $a_4^{\text{imp}}$ | $a_2^{\text{exp}}$ | $a_4^{\text{exp}}$ | $a^\emptyset$ | Total |
|--------------------|--------------------|--------------------|--------------------|--------------------|---------------|-------|
| Laptop on a budget | 13                 | 13                 | 11                 | 8                  | 9             | 54    |
| Fitness restart    | 10                 | 6                  | 13                 | 12                 | 13            | 54    |
| Gardening gift     | 11                 | 5                  | 10                 | 14                 | 14            | 54    |
| Pet decision       | 10                 | 13                 | 13                 | 10                 | 8             | 54    |
| Study environment  | 10                 | 17                 | 7                  | 10                 | 10            | 54    |
| Total              | 54                 | 54                 | 54                 | 54                 | 54            | 270   |

**Table E.12:** Realised session position  $\times$  condition counts ( $N = 54$ ).

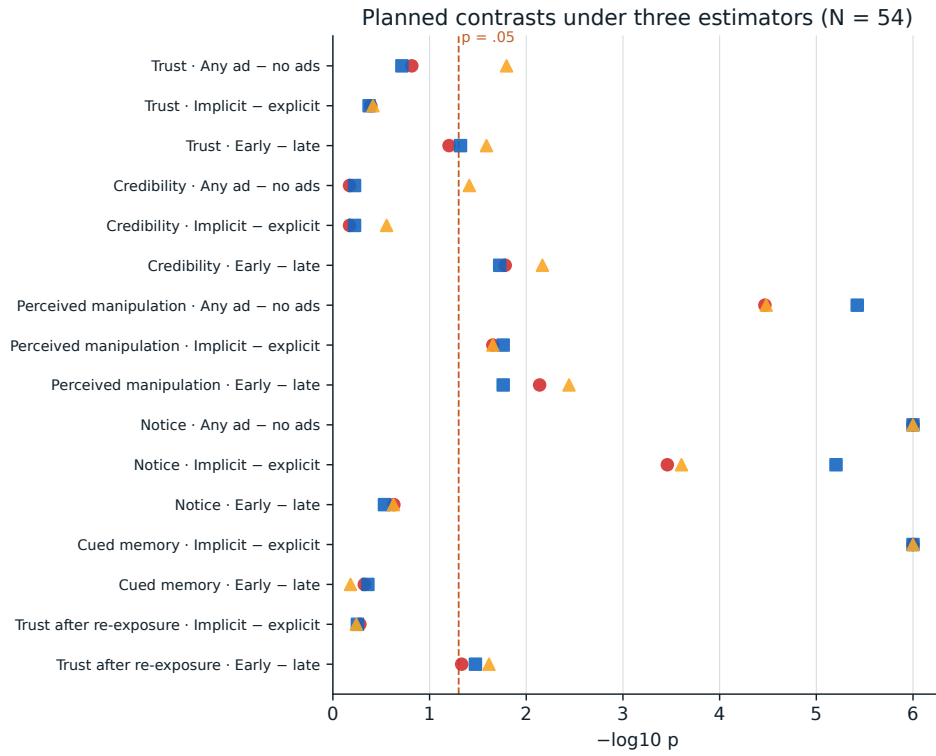
| Session | $a_2^{\text{imp}}$ | $a_4^{\text{imp}}$ | $a_2^{\text{exp}}$ | $a_4^{\text{exp}}$ | $a^\emptyset$ | Total |
|---------|--------------------|--------------------|--------------------|--------------------|---------------|-------|
| 1       | 11                 | 8                  | 12                 | 13                 | 10            | 54    |
| 2       | 14                 | 9                  | 9                  | 8                  | 14            | 54    |
| 3       | 11                 | 13                 | 7                  | 7                  | 16            | 54    |
| 4       | 11                 | 10                 | 12                 | 13                 | 8             | 54    |
| 5       | 7                  | 14                 | 14                 | 13                 | 6             | 54    |
| Total   | 54                 | 54                 | 54                 | 54                 | 54            | 270   |

The two arms are pooled throughout and arm is not a factor in any family. Table E.13 gives the any-advertisement minus no-advertisement contrast of the four primary outcomes within each arm. The sign and the order of the outcomes are the same in both arms: perceived manipulation and notice rise in each, trust and credibility do not move in either. The laboratory estimates are wider because  $n = 18$ . No between-arm difference in  $D$  reaches raw  $p < .05$  (Welch  $t$ ). Pooled values are those of Table 7.2.

**Table E.13:** Any advertisement minus no advertisement by arm: person-level  $D$ , 95%  $t$  interval,  $d_z$ , raw paired- $t$   $p$ . Uncorrected; descriptive.

| Outcome      | Arm        | $n$ | $D$   | 95% CI        | $d_z$ | raw $p$ |
|--------------|------------|-----|-------|---------------|-------|---------|
| Manipulation | laboratory | 18  | 1.57  | [0.52, 2.62]  | 0.74  | .006    |
|              | crowd      | 36  | 1.12  | [0.50, 1.74]  | 0.61  | < .001  |
| Notice       | laboratory | 18  | 2.17  | [1.06, 3.28]  | 0.97  | < .001  |
|              | crowd      | 36  | 2.03  | [1.36, 2.70]  | 1.02  | < .001  |
| Trust        | laboratory | 18  | -0.38 | [-1.24, 0.49] | -0.21 | .375    |
|              | crowd      | 36  | -0.33 | [-0.73, 0.08] | -0.27 | .113    |
| Credibility  | laboratory | 18  | -0.32 | [-0.98, 0.33] | -0.25 | .308    |
|              | crowd      | 36  | -0.05 | [-0.36, 0.26] | -0.05 | .754    |

## E.7 ESTIMATOR AND ASSUMPTION CHECKS



**Figure E.4:** Sixteen planned contrasts ( $N = 54$ ). Circles: paired- $t$  Holm  $p$ ; squares: LMM Holm  $p$ ; triangles: raw Wilcoxon  $p$ . Values below  $10^{-6}$  are plotted at 6.

The paired  $t$  is the pre-specified estimator; this section documents the assumption rather than switching on it. Shapiro–Wilk rejects on 7 of 16 planned  $D_i$  series. A percentile bootstrap interval of  $\bar{D}$  matches the  $t$  interval width to 3% on all sixteen and agrees on 15 of 16 Holm verdicts (exception: trust early minus late). Raw Wilcoxon agrees on 13 of 16 at the .05 line; the three exceptions are the trust any-ad and early-minus-late cells and credibility any ad minus no ads. The LMM agrees on 15 of 16 Holm verdicts, again except trust early minus late.

# F

## Additional Information on Genre Trajectories

section 7.5 reports the confirmatory family. This appendix holds the descriptive tables under both readings of Definition 1 and the context-window sweep.

### F.1 CORPUS AND DESCRIPTIVE TABLES

Messages have a median of 15 words (IQR 8–29) and a median response latency of 49 s (IQR 30–75). Arms are pooled throughout, which the variance checks below support.

**Table F.1:** Trajectory measures by condition, bare-utterance context ( $N = 54, 270$  conversations).  $M$  (SD); column headers give the ceiling.

| Condition       | $N_{\text{shift}}$ (/3) | $D$ (/4)    | $H_{\text{nats}}$ (/1.386) | $R$ (/4)    |
|-----------------|-------------------------|-------------|----------------------------|-------------|
| $a^{\emptyset}$ | 2.48 (0.77)             | 3.11 (0.74) | 1.055 (0.305)              | 1.52 (0.77) |
| Implicit early  | 2.56 (0.66)             | 3.17 (0.69) | 1.084 (0.277)              | 1.43 (0.63) |
| Explicit early  | 2.41 (0.71)             | 3.13 (0.75) | 1.064 (0.305)              | 1.57 (0.69) |
| Implicit late   | 2.46 (0.82)             | 3.00 (0.80) | 1.008 (0.347)              | 1.54 (0.82) |
| Explicit late   | 2.35 (0.78)             | 3.11 (0.69) | 1.057 (0.287)              | 1.63 (0.76) |
| All             | 2.45 (0.75)             | 3.10 (0.73) | 1.054 (0.304)              | 1.54 (0.74) |

**Table F.2:** Kruskal–Wallis on 270 conversations. Not a test of  $\delta^{(a)}$ . The participant ICCs are in the last row.

| Measure                                                          | Factor           | Levels | Kruskal $H$ | $p$ |
|------------------------------------------------------------------|------------------|--------|-------------|-----|
| $N_{\text{shift}}$                                               | Condition        | 5      | 2.89        | .58 |
| $N_{\text{shift}}$                                               | Task             | 5      | 4.61        | .33 |
| $N_{\text{shift}}$                                               | Arm              | 2      | 0.04        | .85 |
| $N_{\text{shift}}$                                               | Session position | 4      | 2.25        | .52 |
| $H$                                                              | Condition        | 5      | 1.16        | .88 |
| $R$                                                              | Condition        | 5      | 2.80        | .59 |
| Participant ICC: $N_{\text{shift}}$ 0.067, $H$ 0.111, $R$ 0.043. |                  |        |             |     |

TRANSITION STRUCTURE. Under the bare-utterance context, 120 of the 169 ordered pairs  $\tau_k = (\hat{g}_k, \hat{g}_{k+1})$  are observed, the diagonal ( $\delta_k = 0$ ) carries 0.183 of the mass, and  $\Pr(\delta_k = 1) = 0.817$ . The modal destination is general guidance and information (0.273) and the three heaviest edges are guidance to guidance (82), guidance to relationships (56), and relationships to guidance (36). Among the 89 cases with  $\delta_2^{(a)} = 1$ , the destinations  $\hat{g}_3$  are led by relationships (0.191), guidance (0.146), and other obscene or illegal (0.135); purchasable products receives 4 of the 89.

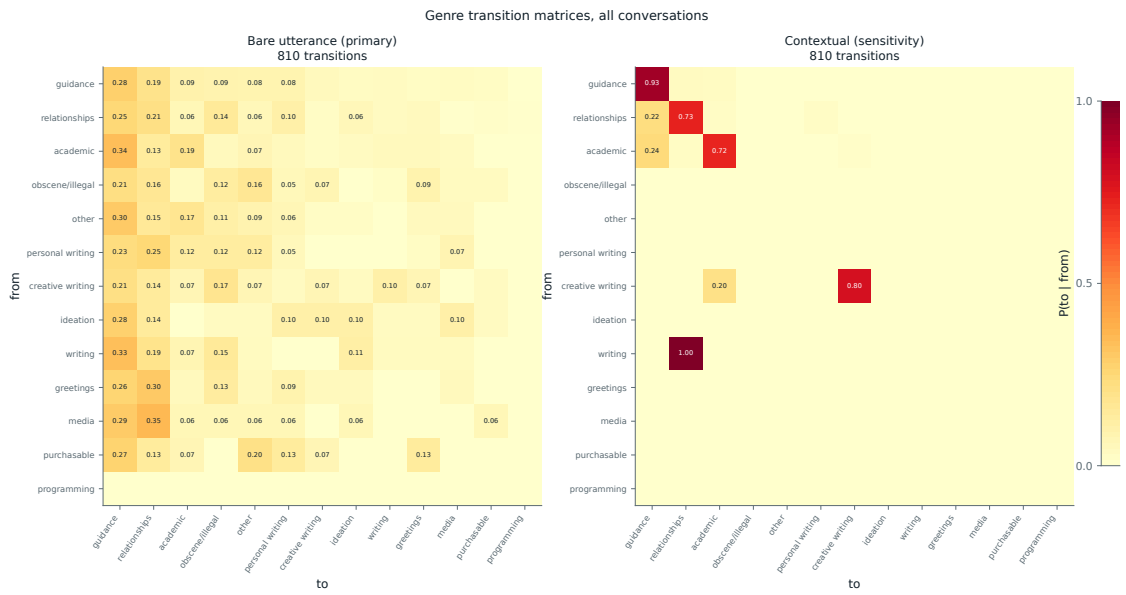
## F.2 THE TWO READINGS OF DEFINITION 1

Definition 1 applies  $f_{\text{genre}}$  to the utterance  $u_k$ . The deployed system instead classified a 510-token window holding the task prompt, the last three messages, and the current message, in that order. Both readings are carried in every table as rows of a `genre_source` column, and every analysis selects one. The bare reading agrees with the logged runtime label on 351 of 1,080 utterances (32.5%); the contextual reading reproduces it on 1,080 of 1,080, by construction.

The trade-off is between fidelity to the definition and fidelity to the deployed system, and the two readings are near-opposite failure modes rather than a spectrum. The bare reading is close to saturation: it uses all 13 classes, has  $\Pr(\delta_k = 1) = 0.817$ , and assigns a fallback class to a fifth of turn-4 utterances because late messages are short. The contextual reading is close to the floor: it uses 7 of 13 classes, has  $\Pr(\delta_k = 1) = 0.132$ , observes only 16 of 169 pairs  $\tau_k$ , leaves 70% of conversations with  $N_{\text{shift}} = 0$ , and carries 0.868 of its mass on the diagonal  $\delta_k = 0$ . The task prompt is what holds it there; Section F.3 isolates that. Task also appears to dominate under the contextual reading and not under the bare one, which is expected, because the task prompt sits inside the contextual classifier’s own input.

**Table F.3:** Bare-utterance and deployed-window readings on 1,080 utterances, with genre-aligned outcomes for 108 early-advertisement conversations.

| Quantity                                                     | Bare utterance (primary) | Contextual (deployed)     |
|--------------------------------------------------------------|--------------------------|---------------------------|
| Agreement with the runtime log                               | 351/1,080                | 1,080/1,080               |
| Genre classes used                                           | 13 of 13                 | 7 of 13                   |
| Mean $N_{\text{shift}}$ (of 3)                               | 2.45                     | 0.40                      |
| Mean diversity $D$ (of 4)                                    | 3.10                     | 1.33                      |
| Mean max persistence $R$ (of 4)                              | 1.54                     | 3.53                      |
| Shift rate                                                   | 0.817                    | 0.132                     |
| Diagonal transition mass                                     | 0.183                    | 0.868                     |
| Distinct genre pairs observed                                | 120 of 169               | 16 of 169                 |
| $\delta_2^{(a)}$ contrast, early pooled                      | +0.046 (Holm $p = .94$ ) | -0.019 (Holm $p = 1.00$ ) |
| $\delta_2^{(a)} = 0$ (no shift)                              | 19 of 108                | 94 of 108                 |
| $\delta_2^{(a)} = 1, \tilde{\delta}_2^{(a)} = 0$ (unrelated) | 80 of 108                | 12 of 108                 |
| $\tilde{\delta}_2^{(a)} = 1$ (aligned)                       | 9 of 108                 | 2 of 108                  |
| $g^{(a)} = \hat{g}_2$ already                                | 25 of 108                | 53 of 108                 |



**Figure F.1:** Row-normalised transition shares for the bare utterance and deployed window (810 steps each).

### F.3 SWEEP OVER CONTEXT WINDOWS

Because the deployed window is the stickier of the two readings, four further windows were classified with the same model to establish whether the confirmatory result depends on how context is supplied. The recipes were fixed before the tests were run: the current message alone (the primary reading), the deployed window, the deployed window with the task prompt removed, the previous user message plus the current one, the task prompt plus the current message, and the conversation’s opening message plus the current one. Task prompts were recovered for all 270 conversations. Holm is applied across the six early-pooled contrasts, which is the comparison a reader would otherwise select from.

No window produces a significant  $\delta_2^{(a)}$  contrast. The smallest unadjusted  $p$  is .096, for the previous-message window, where the estimate is negative ( $-0.139$ ); after Holm across the six it is .58.  $\tilde{\delta}_2^{(a)}$  remains at chance under every window ( $p$  between .79 and .97). The task prompt is what suppresses  $N_{\text{shift}}$ : removing it raises the mean from 0.40 to 0.94 while leaving  $\delta_2^{(a)}$  at  $+0.009$  ( $p = .90$ ).

**Table F.4:** Context-window sensitivity for early advertisements pooled versus  $a^0$  ( $N = 54$ ). Holm across six windows; “Sticky” is the share with  $N_{\text{shift}} = 0$ .

| Window    | Input                   | Classes | $N_{\text{shift}}$ | Sticky | $\delta_2^{(a)}$ | Holm $p$ |
|-----------|-------------------------|---------|--------------------|--------|------------------|----------|
| Bare      | current message         | 13      | 2.45               | 0.03   | +0.046           | 1.00     |
| Deployed  | task + last 3 + current | 7       | 0.40               | 0.70   | -0.019           | 1.00     |
| No task   | last 3 + current        | 12      | 0.94               | 0.38   | +0.009           | 1.00     |
| Previous  | previous + current      | 13      | 1.59               | 0.10   | -0.139           | 0.58     |
| Task only | task + current          | 5       | 0.50               | 0.68   | -0.019           | 1.00     |
| Opening   | first message + current | 12      | 1.28               | 0.37   | -0.056           | 1.00     |